

Supplementary methods

Experiment map

The current unified physical-time experiments are organized by scientific task: S13–S17 define the unified physical-time and physical-reference analyses; S22 provides the native-resolution policy-free task and S23 the British operating and trading comparisons; S1–S5 retain earlier catalogue and representation experiments as supplementary historical analyses. In particular, the earlier native-grid Greek experiment in S5 and the harmonized 30-min Greek experiment in S13 are distinct experiments. S7 and S11 describe scenario-cost studies, whereas S17 describes observed-price settlement. S6 and S12 now distinguish the earlier 120-region review, the v15 collection design, and the received 320-window v19 reference. These experimental cohorts retain their own sampling, normalization and evaluation units. Table S0 gives a compact map from each section to its data, statistical unit, scientific purpose and main-text result; this prevents catalogue, transfer, synthetic and cost populations from being interpreted as one evaluation cohort.

Table S0. Experiment map

Section	Data and period	Analysis unit	Scientific purpose	Main-text link	Status in this manuscript
S1. Archive coverage	Seven archives; primary 2014–2025 periods and external 2020–2021 periods	Turbine $\times$ 30-min row; complete 25-point shape	Define eligible records, chronological splits and quality flags	2.1; Methods 4.1	Complete
S2. Detector protocols	Seven archives; 17 fixed configurations on the harmonized grid	Turbine $\times$ detector interval	Construct the protocol-dependent event objects	2.1–2.2; Methods 4.2	Complete
S3. Matching and summaries	Same archives, detector pairs and IoU thresholds 0.3/0.5/0.7	One-to-one matched event pair	Measure structural survival together with left/right coverage	2.2; Methods 4.3	Complete
S4. Representation and perturbation	Pizhou training/holdout plus five-farm transfer; raw, PCA, statistics and angular representations	Event vector $\times$ partition	Quantify retained shape information and separate correspondence, order and attribute effects	2.2–2.3; Methods 4.4	Complete
S5. External representation transfer	Greek January–June 2020 and SDWPF 2020–2021	Configuration pair $\times$ turbine/batch	Test whether frozen or local representations transfer across archives	2.3	Complete; support strata retained
S6. Human-reviewed regions	Legacy 120-region cohort and v19 eight-archive packet (320 windows $\times$ 3 observers)	Window $\times$ observer; detector presence per window	Provide an external event-presence and morphology reference	Supplementary reference; Methods 4.6	Three observer exports complete; packet closed
S7. Forecasting and storage	Pizhou and Yandun chronological forecasts	Issue time $\times$ target interval $\times$ farm	Test event-aware forecasting and engineering cost scenarios	Supplementary cost context	Complete under stated scenario prices
S8. Controlled localization	Synthetic sequences with known episodes and fresh confirmation seeds	Sequence $\times$ injected episode	Measure event-level localization and delay under known truth	2.7; Methods 4.7	Complete
S9. Weather and post-event context	ERA5 panel, NOAA station comparison and Yandun sampling sensitivity	Time anchor $\times$ turbine; subsequent event window	Test external-process tracking and post-event association	2.4	Exploratory association; causal effect not estimated
S10. Table-reading convention	All result populations	Table row with an explicit unit and denominator	Keep structural, physical, forecast and cost estimands distinct	Methods 4.5	Complete
S11. Policy-capacity scenario	Jiangsu-informed engineering scenarios on Pizhou and Yandun	Farm $\times$ test interval $\times$ storage design	Quantify policy-informed capacity and price sensitivities	Supplementary cost context	Scenario study; not a realized settlement replay

Section	Data and period	Analysis unit	Scientific purpose	Main-text link	Status in this manuscript
S12. Independent collection design	v15 packet specification and annotation interface	Window $\times$ observer response	Record provenance and the distinction from the received v19 exports	Supplementary reference; Methods 4.6	Design archived; received responses reported in S6
S13. Protocol space and hierarchy	Seven archives on the v18 physical-time grid	Protocol tuple $\pi \times$ primitive/composite event	Formalize detection, matching, representation and evaluation layers	Introduction; Methods 4.1–4.3	Complete
S14. Common support and information	Pizhou-fitted representations evaluated on site/configuration pairs	Matched pair $\times$ conditioning stratum	Estimate conditional partition information and metric sensitivity	2.2–2.3	Complete
S15. Physical-process probes	Five primary farms with regional weather targets	Event $\times$ site $\times$ weather class	Test whether retained shape information tracks an independent process	2.3–2.4	Complete for the archived association analysis
S16. Hill LiDAR chronology	Hill of Towie T07/T11, 2026 instrument release	Event $\times$ instrument $\times$ seven-day block	Chronologically test calibrated physical-process discrimination	2.4	Complete; instrument and block scope reported
S17. Forecast-to-cost replay	Hill of Towie 2020 and GB Elexon half-hour prices	Farm $\times$ half-hour settlement interval	Convert high-ramp forecast errors into observed-price exposure and storage replay	Supplementary cost context	GB price replay; native Jiangsu rule test in S22
S18. Output and reproducibility	Versioned public repository and release artifacts	Artifact $\times$ protocol/result record	Bind code, tables, figures and validation records to reproducible paths	Data availability; Methods 4.8	Versioned GitHub repository release
S19. Decision-aligned forecast extension	Hill of Towie existing forecast replay; 1/2/4-h horizons	Issue time $\times$ target interval $\times$ candidate model	Test whether event information can improve error and priced exposure under a fixed validation budget	Supplementary cost context; Table S56	Exploratory extension; existing test calendar reused
S20. Metered correction	Hill 2020; 58 complete days	Half-hour $\times$ schedule $\times$ battery	Separate gross debit from signed settlement	Supplementary S23	Ex-post capability
S21. Overlap correspondence	Eight structural populations	Connected overlap component	Resolve splitting, merging and composite episodes	2.2; Methods 4.3	Complete
S22. Native policy-fee test	Pizhou 2023–2025; held-out late calendar	Native 15-min target $\times$ issue horizon	Connect historical shape and issued weather to rule-based charges	2.8; Methods 4.6	Complete; 2022-rule simulation
S23. British full-income tests	Hill 2020–2021; Elexon prices and vintages	Operating day or scheduled interval	Distinguish exposure, net operating gain and trading income	Discussion; S17–S20	All outcomes retained
S24. Polarity increments	Five ERA5 populations, two LiDAR instruments, SMARTEOLE	Paired event predictions and shared calendar blocks	Separate encoding recovery, scalar redundancy and noise sensitivity	2.4–2.5; Methods 4.5	Reviewer-triggered exploratory extension
S25. AI/human reference	Same 320 displayed windows; 3 human + 7 AI sources	Window $\times$ review source	Evaluate auxiliary reference agreement without conflating sources	Supplementary S6; Methods 4.6	Seven actual AI sessions complete
S26. Complete physical trajectories	Hill T11/T07 2026 and SMARTEOLE 2020	Unique power interval $\times$ measured wind curve	Locate the incremental value of temporal geometry in compound processes	2.6; Methods 4.5	Validation-locked models; exploratory chronological test

S1. Archive coverage and source eligibility

The study contains seven archives and 333 turbines. The primary catalogue contains Pizhou (33), Suining (14), Yandun (117), La Haute Borne (4) and Hill of Towie (21). The Greek monitoring archive adds ten turbines, and SDWPF adds 134. Table 1 in the main text identifies their experimental roles.

A full primary-catalogue boundary check covers 1,984,249 candidate intervals and 1,538,376 complete shapes. All event timestamps agree with their source half-hour indices. Every eligible shape, includ-

ing four samples before and after the event, remains within its assigned 60%/80% chronological period; the check finds zero boundary violations across 189 turbines.

Primary site	Half-hour grid rows	Usable rows
Pizhou	823,680	507,047
Suining	245,952	173,437
Yandun	1,005,720	417,727
La Haute Borne	140,160	108,298
Hill of Towie	368,949	288,832

The primary catalogue uses source-specific complete-count eligibility flags. Each event and its context lie within one valid run and one chronological period. The split boundaries occur at 60% and 80% of elapsed archive time; event contexts crossing a boundary are excluded, so no morphology vector spans two splits. Primary amplitude scaling uses nameplate capacity where recorded and early-period q99.5 elsewhere. The data provider identifies the q99.5 upper production level as rated power for the applicable farms; event metadata retain the empirical scaling basis. One-minute Pizhou power records are averaged within each 30-min bin. Ten-minute and 15-min records use the corresponding valid-observation mean. A bin enters the catalogue when the required observations for its complete context are present.

The forecasting analysis uses a distinct signed-observation rule. Every turbine in the fixed fleet supplies the required native observation count. Accepted mean power lies between -5% and 120% of the recorded nameplate or early-period q99.5 scale. This rule retains small negative measurements and preserves the calendar grid.

S2. Detector configurations

The seventeen primary configurations comprise threshold, financial-tail and mean-shift rules at one, two and four hours, three endpoint corridors, an adaptive corridor, SDA and three OpSDA configurations.

The fixed-change rules use a magnitude threshold of 0.20. The historical-tail rule uses the preceding 48 lagged differences, 5th and 95th percentiles, and a minimum magnitude of 0.10. The reference history ends before the current change. Same-direction overlapping detections are merged.

The fixed endpoint corridors use tolerances 0.025, 0.050 and 0.100. Their chord criterion closes a segment at the previous point when the next endpoint violates the constraint. Adaptive tolerance is fixed at segment start. It combines a 0.025 floor, scaled historical median absolute deviation and a local power-level term.

SDA uses the original boundary-crossing convention [1]. The conference OpSDA implementation merges eligible SDA segments by dynamic programming and a squared-length objective [2]. Door tolerance is 0.025 and amplitude threshold is 0.20.

S3. Shapes, matching and summary units

Each complete shape has $4 + 17 + 4$ coordinates: observed pre-event values, normalized event-time values, and observed post-event values. The vector is centered at event-start power and divided by its maximum absolute displacement. Duration, amplitude and other interval statistics remain separate fields.

Within each site, turbine and period, temporal IoU defines candidate correspondence. Greedy selection uses decreasing IoU, start separation and event identifiers. The primary event-matching threshold is 0.5. The additional support-overlap diagnostic tabulates the fraction of detector-occupancy comparisons exceeding 0.3, 0.5 and 0.7. Each event is assigned once per configuration pair. Counts refer separately to detector intervals, pair records and supported configuration comparisons. Pair-weighted NMI and ARI use matched-pair counts as weights. The machine-readable composition table compares events matched to at least one other configuration with events matched to none. It records amplitude, duration, direction, power range and pre-window mean power. Table S14 gives matched-count-weighted agreement and catalogue reach.

For event-level sensitivity, each configuration pair is rebuilt from interval endpoints at IoU cutoffs 0.3, 0.5 and 0.7. Greedy ties use start-time separation and event identifiers. Maximum-total-IoU assignment uses zero-reward dummy columns and negative reward for infeasible edges. ID-sorted matrices make repeated optimal assignments deterministic within the fixed solver implementation. All 1,465,997 greedy pairs at IoU 0.5 exactly reproduce the original event identifiers. Table S16 pools labels across turbines within each configuration pair, then reports equal-pair and matched-count-weighted summaries.

NMI uses arithmetic entropy normalization. ARI uses chance-adjusted pairwise partition agreement. Primary site summaries average configuration-pair scores equally. The learned-model site scores use within-site contingency tables. The cross-farm learned summary averages five farms equally and averages the three seeds. SDWPF uncertainty summaries retain seven computational batches within its single farm.

Table S1 contains the full primary representation summary. Table S2 gives the raw25 block-length sensitivity, including the number of occupied blocks and paired records.

The expanded composition analysis evaluates every detector pair and each of its two catalogue sides separately. It compares 526,276 eligible primary test events using duration, signed and absolute amplitude, power range, pre-window mean power, upward fraction and earlier turbine wind. Wind is the mean of four complete half-hour bins strictly preceding event start; duplicate source timestamps and incomplete wind bins are excluded. Detailed rows retain event counts, valid-feature counts and quartiles. Table S32 averages feature means over the same supported pair sides in both populations. This pair-specific comparison complements the historical union-of-any-match composition file.

The conditional agreement audit fits quartile boundaries for absolute amplitude, duration and pre-window mean power on the Pizhou training event IDs. Each statistic is then computed within a site, detector pair and covariate stratum. A matched pair enters a stratum when both event sides share its covariate code. One-side-constant partitions contribute zero NMI/ARI; both-constant partitions and strata with fewer than two pairs have undefined informative scores. Table S33 reports pair-weighted summaries over defined strata with at least thirty matches, along with the fraction of retained pairs represented by those scores. Temporal-match retention and left/right catalogue coverage have separate denominators. The joint-stratum analysis uses all four variables simultaneously. Its point summaries describe conditional partition agreement.

S4. Representation fitting and perturbation analysis

Primary representation fitting uses eligible events from the first 26 sorted Pizhou turbines. Seven turbines form the primary holdout. Sampling uses at most 2,500 events per configuration, seed 41, six PCA components and twenty K-means initializations.

The fixed representations are raw25, raw-PCA6, statistics9, GAF-PCA6 and signed-GAF-PCA6. Statistics9 includes duration, signed amplitude, range, total variation, maximum rate, chord residual, curvature and pre/post means. The fixed signed-GAF representation appends the 25 signed coordinates to the flattened 625-value field.

Raw-PCA6 is a compression control for the same 25-point path. Across the three current Pizhou source fits it retains 93.4% of standardized training variance; same-event test partition ARI against raw25 ranges from 0.985 to 0.998. The earlier 0.989 pooled v5 value describes a separate historical experiment. Current cross-protocol comparisons use the frozen v18 models and common supported configuration pairs.

The learned image encoders use two convolutional layers and a six-dimensional latent vector. The signed version adds an orientation channel. TCN uses dilated one-dimensional convolutions. Transformer uses positional features and two attention layers. All four reconstruct the ordered trajectory. Input and latent standardization use the corresponding Pizhou training sample. The primary archived models use fifteen training epochs, validation reconstruction selection and seeds 41, 42 and 43. A completed 100-epoch, three-seed rerun uses the same data, transformations, latent dimension and early-stopping rule. Its assembled metrics are reported as the training-budget sensitivity in Table S12 and the polarity-mechanism figure. Parameter counts are 11,167, 11,311, 4,567 and 15,559 for CNN-GAF, signed-GAF, TCN and Transformer, respectively.

Row permutation reallocates complete event shapes within site and turbine. Coordinate permutation changes order within each event. Conditional comparisons use direction and training-defined amplitude, duration and pre-event-power bins. The complementary perturbations distinguish correspondence, order and coarse event attributes.

Calendar uncertainty uses 2,000 resamples with shared weights across turbines in the same block. Seven-day blocks define the primary estimate, and three- and fourteen-day blocks provide sensitivity checks. The 2.5th and 97.5th percentiles define pointwise intervals. Ten further Pizhou turbine splits use seeds 20260912 through 20260921.

S5. External representation evaluation

The Greek archive contains ten turbines observed during January-June 2020 [3]. Its twelve base configurations operate on ten-minute samples with lags of two, four and eight points. The physical lags are twenty, forty and eighty minutes. Four source observations on each side provide forty minutes of context.

SDWPF contains 134 turbines observed during 2020-2021 [4]. The external pipeline aggregates power to thirty-minute bins and applies the twelve base configurations. Seven batches divide the turbine list for computation. The complete archive has 11,361,190 source records and 134 location entries.

Both external catalogues use a turbine-wide q99.5 amplitude scale. Pizhou supplies the raw25 standardizer and centroids. Saved learned encoders, input scaling and latent scaling provide the SDWPF representations. Learned clusterers use saved Pizhou training latent vectors. The Greek local baseline fits standardization and k=4 K-means on Greek training shapes only, then evaluates the same 62 Greek test configuration pairs. This comparison changes the training population while retaining the event correspondence table.

The TCN adaptation comparison uses one source checkpoint (seed 41), the same 26 source fitting turbines, and the same 17,357 Greek test pairs. A second copy of the checkpoint receives five reconstruction-training epochs on Greek training shapes. Each encoder fits latent scaling and k=4 prototypes on the identical Pizhou training sample. Table S21 reports both models using identical pooled, equal-pair and matched-count-weighted metrics. Greek test shapes are used for evaluation.

Greek configuration pairs are grouped by support: 2-4, 5-9, 10-29, 30-99 and at least 100 matches. Table S3 reports every support stratum. The larger-support median is 0.669. Table S13 contains SDWPF batch-seed results for two, four and six clusters.

S6. Human-reviewed regions and operating logs

Earlier focused-region review

One researcher labeled thirty regions from each of Pizhou, Yandun, Greece and SDWPF. The first display provided twelve hours of context with a highlighted central two hours. The researcher identified the highlighted region as the principal target. A second display showed four hours and retained the same two-hour target for 52 refinements.

The merged labels contain 35 downward, 28 sustained-low, 24 upward, 14 V-shaped, 13 inverted-V and six quiet regions. Source submissions are retained separately. The second round supplies the final category for its 52 regions. The review record provides a region category and its display scope.

Catalogue agreement marks a region positive when a same-turbine detector interval overlaps its highlighted period. Precision divides positive-label hits by all flagged regions. Recall divides positive-label hits by all presence-positive regions. F1 summarizes their harmonic balance. Table S10 gives sampling support and Table S11 gives all configuration-level results. The sampling population contains 114 positive and six quiet labels.

Source-clock strings are parsed individually because the review table contains both naive and UTC-tagged timestamps. Every region has a finite anchor before catalogue intersection. The selected population and its clock convention remain attached to the reported metrics.

Greek logs provide generator, run, stop, yaw, reset and oil-flow context. The mapping retains 63,163 records. Event-start proximity uses thirty-minute, one-hour and four-hour neighborhoods within each turbine. The review windows also contain 2,336 source-clock log rows.

Current eight-archive human reference

The v19 packet contains 320 four-hour windows, with 40 per archive from Pizhou, Suining, Yandun, La Haute Borne, Hill of Towie, Greece, SDWPF and SMARTEOLE. The rated region is the central two hours. Each of the first seven archives contributes 20 random and 20 detector-disagreement windows; the 40 SMARTEOLE windows come from its operating-period sample. Sampling source, detector outputs and other responses are hidden from each observer. The exported packet records presence, dominant morphology, assessability and an optional free-text note.

Three actual identifiers supplied 320 assessments each. R_147f4682 recorded 75 upward, 65 downward, 21 V, 22 inverted-V, 11 oscillatory, 75 quiet and 51 sustained-low regions (194 positive). R_2c332025 recorded 64 upward, 50 downward, 27 V, 21 inverted-V, 34 oscillatory, 104 quiet and 20 sustained-low regions (196 positive). R_29d247dd recorded 46 upward, 39 downward, 52 V, 64 inverted-V, 66 oscillatory, 42 quiet and 11 sustained-low regions (267 positive). All 960 responses were assessable. The exports supply region categories rather than start/end labels and are retained byte-for-byte. The earlier 120-region cohort remains separate.

Nominal Krippendorff's alpha was 0.528 for presence and 0.476 for morphology. The three-way coincidence calculation retains all available ratings per window; missing ratings would remain missing rather than being imputed. Shared calendar resampling preserves the observer set within each sampled window. Seven-day 95% alpha intervals were 0.455–0.599 and 0.426–0.523. Pairwise Cohen kappa and disagreements are retained in the machine-readable agreement record, without an automatically generated consensus label.

Source recovery links each displayed power sequence to its original site, turbine and relative clock. For 280 inherited windows, missing display metadata were recovered through the original private sampling manifest and exact sequence equality. The 40 SMARTEOLE sequences were checked against the prepared source records at the displayed rounding precision. The distributed packet remains unchanged for additional observers. Absolute geographic clock claims are unnecessary for this within-archive comparison.

Each of the seventeen fixed configurations predicts presence if at least one basic detector interval overlaps the target region. Precision is $TP/(TP+FP)$, recall is $TP/(TP+FN)$, and F1 is $2TP/(2TP+FP+FN)$. Turning recovery additionally requires a matching V or inverted-V composite with its turning point inside the rated target. Multiple detections in a region count once for presence. This unit differs from the boundary-resolved one-to-one event evaluation in S8.

Aggregate estimates condition on the selected windows, with site and sampling-arm results supplied separately. For uncertainty, 2,000 resamples draw calendar blocks within each site, giving all windows and turbines in a sampled block the same multiplicity. Block lengths are 3, 7 and 14 days; paired detector differences share each draw. The primary seven-day analysis contains 197 occupied farm-blocks. The intervals condition on these observers; agreement was calculated on all 320 windows with three ratings. The packet contains no preassigned label-validation/test split, and none of the detector thresholds was tuned to these ratings. Tables S44–S47 report all three observers and all detector results; Table S54 reports agreement and its block sensitivity.

The received v19 assessment set now contains three observers. The third export is included in the agreement record, detector metrics and Tables S44–S47; all three ratings remain separate and no consensus label is created.

S7. Forecasting and storage objectives

The forecasting experiment uses fixed fleets of 33 Pizhou turbines and 117 Yandun turbines. A 24-point half-hour history predicts the production interval ending one hour after issue. Each input interval is fully

observed by issue time.

The chronological training/validation/test window counts are 6,813/3,173/3,549 in Pizhou and 1,847/375/1,525 in Yandun. Each retained context and target lies within one period and one continuous valid run. Persistence, TCN and TimesNet share the target timestamps.

The learned forecasters use seeds 41, 42 and 43 and up to twenty minibatch epochs. Validation MSE selects the checkpoint. The primary event-weighted training multiplies the squared error by four for training changes above the training q95 magnitude. A sensitivity series uses factors 0, 1, 2, 4 and 8 under the same data split, model, optimizer and event threshold. Factor is the incremental coefficient in $1 + \text{factor} \times \text{ramp_indicator}$, giving actual ramp weights 1, 2, 3, 5 and 9. The same observed history supplies every training variant at prediction time.

The capacity grid includes zero storage and twenty-five positive combinations. Power fractions are 0.01, 0.02, 0.04, 0.08 and 0.16. Energy durations are 0.5, 1, 2, 4 and 8 h. Validation selects capacity for each model and cost scenario.

Annual capacity costs are 350 CNY/kW and 1,100 CNY/kWh. Throughput costs 25 CNY/MWh. The low, base and high shortfall/surplus penalties are 150/40, 300/80 and 600/160 CNY/MWh. Results are expressed per one-MW-normalized production scale.

Dispatch uses a 0.92 round-trip efficiency, zero initial stock and zero terminal inventory credit. The calendar includes missing intervals, during which stored energy is carried forward. Capacity cost covers the full test span. Delivery penalties accrue on the common observed targets. Table S7 reports forecast error and Table S8 reports the selected scenario costs.

The paired forecast analysis resamples seven-day calendar blocks 2,000 times. Pizhou has fifteen occupied blocks and Yandun has five. Each bootstrap draw is shared across the fixed model seeds. Positive reported error gains mean lower error under event weighting.

S8. Controlled episode localization

The generator produces 96-point sequences with smooth background variation and known finite episodes. It supplies 1,600 normal training sequences, 400 labeled validation sequences, 800 labeled test sequences and 800 higher-noise sensitivity sequences. Validation and test sets each contain equal numbers of positive and background sequences.

Episodes have amplitudes 0.08, 0.15, 0.30 or 0.50, positive or negative direction, and trapezoidal or smooth single-turn profiles. Rising/falling spans contain 2, 4, 8 or 16 points. Generator seeds are 9101, 9102, 9103 and 9104 for the four datasets.

TimesNet, KAN-AD, TCN-AE and positional-Transformer-AE reconstruct the normal training sequences [5, 6]. Each run receives thirty full minibatch epochs and 750 parameter updates. Normal validation reconstruction selects a checkpoint. The three initialization seeds are 41, 42 and 43.

The default decision threshold is the training reconstruction-score q99. Threshold-only calibration searches nine validation-score quantiles. Complete protocol calibration also searches low/high ratios 1.0 and 0.5, merge gaps zero and two points, and minimum durations one and three points. Validation event F1 selects the settings.

Greedy one-to-one matching compares fully labeled episodes at IoU 0.3. IoU 0.1 and 0.5 provide sensitivity checks. Table S9 records every seed and decision protocol. Paired intervals resample whole generated sequences 2,000 times, with the same draws across the fixed trained seeds.

A 100-epoch upper-budget rerun uses the same generated sequences, model configurations and three seeds. Validation-only calibration selects the threshold and interval-grouping parameters. Table S22 reports its default and calibrated test F1, precision and recall together with optimizer steps.

The subsequent model-size search evaluates three configurations per neural model, each with seeds 41, 42 and 43 and a maximum of thirty epochs. TimesNet uses d_model 8, 16 and 32; KAN-AD uses Fourier orders 8, 16 and 32. Their learning rates are 0.0005, 0.001 and 0.001 respectively. The sequence autoencoders use (width, latent size, learning rate) combinations (8, 8, 0.0005), (16, 16, 0.001) and (32, 16, 0.001). Adam, batch size 64 and five-epoch early stopping follow the original implementation. Mean

validation episode F1 across seeds selects each configuration. The analytic reference selects adjacent-mean windows of 2, 4, 8 or 16 points using validation F1; window 4 is selected. Every candidate uses the same threshold, hysteresis, merge-gap and minimum-duration search.

After selection, all checkpoints and output settings are frozen in the confirmation record. Generator seeds 2026091501 and 2026091502 then supply 1,600 fresh sequences each, half containing episodes. Innovation-noise standard deviations are 0.015 and 0.027; the autoregressive coefficient remains 0.65 and the event generator family is retained. The confirmation evaluates the frozen thresholds directly in both conditions. Tables S29–S30 report mean metrics, between-training-seed SD and 2,000 paired sequence-resampling intervals. The intervals condition on the fixed trained checkpoints; they quantify sequence-sampling variation.

Parameter counts use trainable model tensors and exclude positional and normalization buffers. CPU timing uses an AMD Ryzen 7 9700X under Ubuntu 24.04 / WSL2 with two torch threads. Five repeats follow one warm-up run of 800 sequences. Neural score batches contain 64 sequences; the analytic score uses vectorized NumPy cumulative sums. Table S31 reports normalization and score-generation time, with interval postprocessing excluded.

S9. Weather and post-event context

The regional ERA5 exposure is an absolute 100-m wind-speed change of at least 1.5 m/s over three hours [7, 8]. Weather timestamps use a fifteen-minute nearest-match tolerance. Outcomes record subsequent threshold-event starts within one, two or four hours.

The archived panel has 21,300 exposed and 84,761 unexposed records. The unadjusted risk differences are 0.0388, 0.0679 and 0.0863. NOAA station 580270 supplies a regional surface comparison with a ninety-minute match tolerance [9]. Tables S4 and S5 show the observed risks and shifted-exposure contrasts. Table S15 adds 2,000 calendar-block resamples for 3-, 7- and 14-day blocks. Each resampled block carries all its turbine records, preserving within-block cross-turbine dependence. The main seven-day analysis occupies 31 ERA5 blocks and 27 NOAA blocks.

Yandun sampling sensitivity uses 15-, 30- and 60-min arithmetic-mean grids from the same native records. A four-hour physical lag is 16, 8 and 4 rows respectively; a one-hour lag is 4, 2 and 1. Every bin between the two endpoints is required to be valid. Table S17 reports eligible anchors, threshold-positive anchors and their ratio, both on each grid and at hourly clock times. These are anchor counts before event merging.

The post-event analysis retains 96 eligible events, with 17 exposed and 79 unexposed. Its scaled power-recovery index divides post-end power change by the pre-event-to-end amplitude and clips the index to [-2,2]. The exposed-minus-unexposed contrast is -0.390, with a seven-day block interval [-1.429,0.363]. The residual-power contrast is 191.0 source units with interval [-368.5,645.1]. Table S6 retains both estimates. Their interval widths describe the uncertainty in the recovery contrast.

The adjusted binomial model encodes the outcome numerically as 1 for a subsequent event. Exact joins obtain power, wind speed and direction at $t-4$ h and power change from $t-5$ h to $t-4$ h, preceding the exposure window $[t-3 \text{ h}, t]$. Complete cases number 76,538, including 12,011 exposed and 64,527 unexposed records. The same cohort supplies the crude and standardized risk differences. Hour, month and turbine indicators accompany these covariates. Clustered covariance uses 31 seven-day blocks; delta-method intervals use a t critical value with 30 degrees of freedom. Coefficient tables also provide Benjamini–Hochberg adjusted q -values [10].

S10. Reading the result tables

Each table identifies its measurement unit. NMI and ARI describe cluster correspondence. Coverage describes how much candidate support enters matching. Human-region precision and recall describe catalogue agreement with the reviewed sample. Synthetic F1 describes episode localization. Forecast nMAE describes error relative to training power scale. Scenario cost prices the corresponding errors under the listed rates.

The five-farm comparison, Greek external result, SDWPF external result, reviewed regions, controlled episodes and forecasting targets remain separate evaluation populations. Their complementary measurements form the evidence chain developed in the main text.

S11. Policy-capacity storage experiment

The fixed baseline has battery power equal to 10% of wind-farm nameplate capacity and energy equal to two hours of battery power. The source is Jiangsu’s 2023 market-connection policy [11], applied here as a historical engineering scenario. The 2025 national tariff reform changes the role of compulsory storage [12]. The Jiangsu rule supplies a historical engineering scenario. Yandun is an analogous Xinjiang archive, and the same technical design serves as an engineering comparator in this study.

Pizhou nameplate capacity is 87.45 MW according to the supplied data description; the sum of the 117 Yandun metadata capacities is 200.5 MW. Forecasts are converted from their original training-scale normalization to MW before dispatch. Costs are reported per installed MW over 2,496 test-calendar hours at Pizhou and 914 hours at Yandun, with missing intervals retained. These units differ from the earlier normalized-reference-MW tables.

The policy-capacity study reuses the original MSE models and the models with ramp sample weight four. The full capacity grid contains a zero-storage option and power fractions 0.05, 0.10 and 0.20 at durations 1, 2 and 4 h. Online dispatch charges during positive delivery deviations and discharges during negative deviations, with SOC limits 10%–90%, initial SOC 50% and round-trip efficiency 92%. Candidate reserve fractions are 0, 0.25 and 0.5 of usable energy; severe deviations above 20% of farm capacity release the reserve. Selection uses validation costs only. Terminal inventory is valued at an assumed 300 CNY/MWh, with initial inventory valued identically.

Assumed upfront power/energy costs are 350 CNY/kW and 1,100 CNY/kWh. A ten-year life and 8% discount rate give the capital-recovery factor; fixed annual maintenance is 2% of upfront cost. This replaces the earlier convention that treated those amounts as annual capacity charges. Shortfall/surplus prices are 150/40, 300/80 and 600/160 CNY/MWh. Additional premiums of 0, 1,000 and 5,000 CNY/MWh apply only to remaining deviations above 20% of installed wind capacity. These prices define scenario sensitivities; comparison with individual farm settlements remains a separate validation task.

The unconstrained design and the design requiring at least 10% power and 2 h duration are selected separately on validation records. A sparse linear program solves a perfect-information cost bound at fixed 10%/2 h capacity. It uses the complete realized residual path, so its cost is a perfect-information lower bound. The online policy uses validation-selected reserves and delivery-time deviations. Positive/negative-deviation-specific charging bounds prevent simultaneous charging and discharging. SOC conservation, feasibility and the oracle-versus-online inequality are checked in tests. Tables S23–S25 report the capacity-price surface, selected designs and cost bounds.

S12. Independent multirater collection

This section records the earlier v15 collection design. The current submitted file was recognized by its internal v19 schema and packet identifier, despite its filename retaining the v15 prefix; its actual sampling and results are described in S6.

The v15 collection packet contains 200 real SCADA windows, with fifty from each of Pizhou, Yandun, Greece and SDWPF. Each site contributes twenty steady-screen, five rising-screen, five falling-screen, ten turning-screen and ten random-screen windows. These strata balance the sampling design; the human class distribution is determined by new responses. Four hours are displayed and only the central two hours are rated. Detector identities, sampling strata and earlier human answers are hidden in the interface.

Each participant uses an anonymous identifier and exports a versioned JSON file. Duplicate exports from the same identifier are merged by response timestamp; changed target regions are rejected. Cohen’s kappa is calculated on shared assessable windows for each rater pair, and nominal Krippendorff alpha uses units with at least two available ratings. Uncertain and data-quality responses are tabulated separately.

Constant-category cases return an undefined chance-corrected coefficient. Multirater coefficients will be calculated after independent responses are collected.

The collection interface is available at <https://longjingpy.github.io/wind-power-event-protocol-audit/annotation/>. It works offline and stores answers locally. The v15 definition records sustained low production as a state and assigns transitions to their dynamic categories, so its responses remain distinct from the original v9 review labels.

S13. v18 protocol space and event hierarchy

The v18 protocol is $\pi=(Q,D,M,F,C)$, where Q specifies the observation clock, sampling, quality and amplitude calibration; D constructs intervals; M establishes correspondence; F encodes trajectories; and C assigns structure. The main grid is 30 min, with 60-, 120- and 240-min detector horizons, a 24-h historical window and 120-min context on each side. All transformations and scales are fitted on the chronological training population. The primary structural estimand is evaluated on matched, encodable events; weather and cost estimands use their eligible calendar populations.

Primitive detector intervals and composite V or inverted-V intervals are separate event levels. A composite joins opposite legs within 30 min and 240 min total duration, retains both child event IDs and stores the observed internal turning index. Primitive candidates describe detector intervals and orientation; they are not assumed to be monotonic physical trajectories.

S14. v18 common-support matrix and conditional information

Terminology and support units

An event pair consists of one interval from each detector configuration. A supported configuration pair contains at least 100 matched events; informative summaries additionally require nonconstant partitions. Coverage uses unique eligible events on each side. A nat is the natural-log information unit: 1 nat = $1/\ln(2)$ bits. ARI is chance-adjusted partition agreement with 1 for identical partitions, 0 at the random-label expectation, and possible negative values.

For positive-duration intervals $I_a = [s_a, e_a)$ and $I_b = [s_b, e_b)$, let $J = \max(0, \min(e_a, e_b) - \max(s_a, s_b))$ and $U = (e_a - s_a) + (e_b - s_b) - J$. The intersection over union (IoU) is

$$\text{IoU}(I_a, I_b) = J/U, \quad U > 0.$$

The implementation rejects zero-duration and reversed intervals ($e \leq s$) before overlap calculation. A zero union is an invalid comparison and receives no score or edge. Empty catalogues contain no matches and their coverage is undefined; intervals touching only at an endpoint have $\text{IoU}=0$. For overlapping valid intervals, U equals the enclosing duration used by the implemented candidate-edge calculation. Matches share site, turbine, split and hierarchy. Greedy assignment sorts feasible edges by descending IoU with deterministic tie rules, using $\text{IoU} \geq 0.5$. If N_a and N_b are the eligible event counts and M is the number of one-to-one matches, left and right coverage are M/N_a and M/N_b . Primary summaries use supported, informative configuration pairs common to all six representations and three seeds, first taking the median across seeds within each pair and then across pairs. ARI [13] and normalized mutual information (NMI) quantify partition agreement. Constant partitions and pairs with fewer than 100 matches remain in separate support tables.

The common-support matrix intersects informative configuration pairs supported by raw25, statistics9, raw-PCA6, GAF-PCA6, signed-GAF-PCA6 and protected-polarity representations across seeds. Within a configuration pair, seed values are summarized by their median; site values are medians across the common intersection. The principal IoU threshold is 0.5; 0.3 and 0.7 are sensitivity values. Maximum-total-IoU assignment is an additional matching sensitivity. Pairs with fewer than 100 matches and constant partitions remain in the machine-readable output and are excluded from the primary informative median.

For matched partition labels L_a, L_b and conditioning stratum W , conditional mutual information is $I(L_a; L_b | W) = H(L_a | W) - H(L_a | L_b, W)$, using natural logarithms and empirical frequencies weighted by stratum size. The reported excess subtracts the mean of 100 within-stratum, within-turbine permutations. Individual descriptor analyses use training terciles; the joint analysis uses median splits of amplitude, duration and pre-window mean power together with both event directions. Cut points are fitted on Pizhou training events. Each stratum contains at least 30 matched pairs. Calendar uncertainty is reported separately from the permutation reference.

An additional metric analysis reuses the Pizhou-frozen K-means models with $k=4$ and seed 41 for raw25, GAF-PCA6 and protected-polarity GAF. Adjusted mutual information (AMI), Fowlkes–Mallows index and variation of information (VI) complement ARI and NMI. VI equals $H(L_a) + H(L_b) - 2I(L_a; L_b)$, in nats; lower values indicate closer partitions. All five metrics use identical basic-event test matches. Summaries intersect configuration pairs with at least 100 matches and nonconstant labels in all three representations, and then take the median over configurations. This single-seed, three-representation sensitivity has a different common-support intersection from the primary three-seed, six-representation analysis. Table S48 preserves that distinction.

S15. v18 representation and physical-process probes

The primary analysis fixes K-means at $k=4$, with $k=2$ and $k=6$ sensitivities shown explicitly in Table S57. All three clustering algorithms retain their original training budgets. For the finite signed unit-domain vector x , set $c_i = \sqrt{1 - x_i^2} \geq 0$. The Gramian angular summation field (GASF), the GAF variant used here, is $G = xx^T - cc^T$ [14], equivalent to $G_{ij} = \cos(\arccos x_i + \arccos x_j)$. For an exact, uncompressed full field, its diagonal yields $|x_i| = \sqrt{(G_{ii} + 1)/2}$, and $G + cc^T = xx^T$. Choose an anchor r with $|x_r| > 0$; the sign bit $b = \text{sgn}(x_r)$ gives $x_r = b\sqrt{(G_{rr} + 1)/2}$ and $x_i = (G + cc^T)_{ir}/x_r$. The deterministic largest-magnitude anchor, with the first index breaking ties, is recoverable from the diagonal. The all-zero path is already determined, although constant supports are excluded by preprocessing. This recovers the normalized trajectory; restoring absolute power also requires its offset and scale. PCA retains only part of the field, so GAF/PCA5 plus polarity is an empirical six-dimensional representation, not an exact-inversion guarantee. It preserves the anchor sign explicitly and is evaluated on held-out clustering and physical targets.

The protected polarity coordinate contributes at most about 0.225% of the six-dimensional PCA variance in the primary fit. It therefore changes the physical-readout interface more strongly than the Euclidean cluster geometry. This is the reason the structural ARI of GAF+polarity can remain close to ordinary GAF while its external directional AUROC changes substantially.

Hourly ERA5 wind, temperature, pressure and related surface fields provide regional context for five geographically aligned archives [7, 8]. The event-level physical target is $\Delta u = u_{100}(e) - u_{100}(s)$, classified as decrease for $\Delta u \leq -1.5 \text{ m s}^{-1}$, increase for $\Delta u \geq 1.5 \text{ m s}^{-1}$, and small net change otherwise; event duration satisfies $0 < e - s \leq 4 \text{ h}$. The $\pm 1.5 \text{ m s}^{-1}$ cutoffs are study-defined symmetric contrast levels, fixed for the regional probe before the later LiDAR transfer. They provide a common operational definition of a substantial endpoint wind-speed tendency, rather than a turbine cut-in/cut-out boundary or a universal meteorological ramp threshold. The four-hour limit matches the longest principal detector horizon and the composite-event window used in this study, focusing the probe on intraday dynamics. It does not require every wind change to span four hours: the same endpoint contrast may occur over a shorter event. A V-shaped excursion may return to small net change while retaining substantial internal variation, which is evaluated through event morphology and the overlap analysis. NOAA Integrated Surface Database (ISD) observations are archived with variable-level quality codes and precipitation accumulation duration [9]; independent LiDAR supplies the local-wind test.

S16. v18 Hill LiDAR confirmation and chronology

The Hill of Towie v2.1.0 release supplies 2026 Wind10 aggregates. T11 uses ZX300 unit 2428 horizontal speed near 58 m; T07 uses the available ZXTM unit 5060 fit-derived hub-height speed at 208 m. Posi-

tive packets and wind values in the physical range define valid measurements. Missing intervals remain missing. Endpoint clock offsets of -10 , 0 and $+10$ min are reported. Frozen Hill-2020 physical heads and Pizhou representation transforms are applied before reading the 2026 scores.

A separate chronological calibration fits multinomial logistic calibration to source log probabilities. Calibration training ends before 14 March 2026, validation ends before 7 April and later events form the test period. Regularization C is selected from 0.1, 1 and 10 on validation. The field test reports AUROC, Brier score, signed log loss and seven-day block intervals. It is an external chronological probability test and does not change the frozen structural matrix.

S17. v18 forecast-to-cost and storage replay

Elxon prices specify observed half-hour imbalance settlement [15]. Here S_t is the experimental scheduled power position in MW for delivery interval t ; its definition depends on the comparison. In the forecast-exposure experiment, $S_t = \hat{P}_{t|o}^{(m)}$ is model m 's prediction issued at o . In the common-commitment storage experiment, every information arm shares the one-hour persistence schedule from the last completed observation; missing forecasts command zero desired correction while preserving inventory restoration. In the ex-post capability experiment, each model retains its own issued schedule across battery sizes, with persistence used where that model's forecast is unavailable. Thus schedules are fixed across capacities within a model, and shared across models only in the common-commitment arm. For metered delivery P_t , interval duration Δt in hours, and buy/sell prices p_t^b, p_t^s in GBP MWh -1 , the signed settlement cost is

$$C_t = \max(S_t - P_t, 0)\Delta t p_t^b - \max(P_t - S_t, 0)\Delta t p_t^s.$$

Hill aggregate power uses 21 turbines and a 48.3-MW reference capacity. Six information variants are persistence, power plus calendar, observed wind, historical event structure and two three-expert ramp mixtures. Event features become available at event end plus 150 min. Forecast horizons are 1, 2 and 4 h and target a 30-min interval. Elxon half-hour buy and sell prices score signed energy deviations. Gross debit is the positive part of the interval cashflow; net cashflow retains both deficit purchases and surplus settlement.

The common-commitment storage arm holds the persistence schedule fixed across information variants when an issued forecast is available; unavailable-forecast intervals use the documented zero-correction fallback. Battery power is 0%, 5%, 10% or 20% of fleet capacity, with two-hour energy capacity, 10–90% state-of-charge bounds, 50% initial and terminal inventory, and 0.92 charge/discharge efficiencies. The issued-forecast controller uses current inventory and a reachable terminal-inventory band; it does not use future power or future prices. Throughput sensitivities are 0, 10 and 30 GBP MWh -1 . Validation selects capacity before the test replay.

The v20 Jiangsu duplicated-slot screen is a historical scenario recorded under `outputs/protocol\_benchmark\_v20/economics/jiangsu\_counterfactual/`. It applies a medium/short-term allowance to simulated slots and does not evaluate the native ultra-short-term accuracy component. S22 supersedes it for that question using actual quarter-hour points and the 97%/87% rule. The French transfer design still requires historical RTE prices and a specified responsible-party perimeter. Table S55 separates these policy components.

S18. Output and reproducibility map

The v18 event engine and unit-explicit economic functions are in `src/wind\_events/`. Structural, physical, LiDAR, threshold, weather, forecast and storage outputs are under `outputs/protocol\_benchmark\_v18/`. Independent SVG, PDF, PNG and data-table sources are under `manuscript/figures\_v18/`. The static support-aware benchmark index is under `outputs/protocol\_benchmark\_v18/leaderboard/`. The package wheel, isolated demo and unit tests are recorded under `outputs/protocol\_benchmark\_v18/package/`. The

complete v18 input, transformation and evaluation record is `docs/V18\_CURRENT\_EXECUTION.md`. The v19 manifest records the multirater and SMARTEOLE stages. Current v21/v22 representation, sampling and economic tables remain under their versioned output folders. The v23 extension adds `outputs/protocol\_benchmark\_v23/cluster\_count\_common\_support.csv` and the `many\_to\_many/` directory: `configuration_pairs.csv`, `site_summary.csv`, `verification.json` and `coverage_aggregation_reconciliation.csv`. The last table verifies the equal-configuration versus pooled-coverage distinction against the original one-to-one counts. Supplementary Tables S57–S58 and Fig. S1 summarize these additions. The v24 native economic task adds Tables S59–S64, S22–S23 methods and the model-input replication bundle. The public v0.7.0 release mirrors aggregate outputs under `results/`, contains the current manuscript and editable Fig. 1, and links the de-identified processed-data release. The detailed local component ledger retains internal event clocks; public tables use aggregate counts. The SMARTEOLE support diagnostic remains `outputs/protocol\_benchmark\_v19/smarteole/diagnostic/report.json`, with the supported-row median, all-row and pair-weighted summaries.

S19. Decision-aligned forecast extension

The extension fits residual power targets (future 30-min power minus issue-time power) using two feature arms: past weather and past weather plus frozen historical raw25 event features. The fixed candidate matrix crosses absolute- and squared-error losses, 7/15 leaf budgets, ramp weights 1/4 and persistence blends 0/0.25/0.5/0.75/1, giving 240 candidate predictions across 1-, 2- and 4-h horizons. Validation selects the primary nMAE candidate and a separate gross-debit candidate; no future prices or target power enter model actions. The test period is the previously used calendar and is labelled exploratory. Paired 3/7/14-day farm-block resampling supplies uncertainty. The 2-h event-aware selected model reduces gross debit by 16.51% relative to the legacy historical-event ramp mixture, with a seven-day interval of 11.39–22.31%; its nMAE reduction is 0.79 percentage points (0.40–1.22). Complete candidate rows, selections and effect intervals are under `outputs/protocol\_benchmark\_v20/economics/`.

S20. Full settlement and online metered correction

The v22 economic extension is a separate 58-complete-day capability arm after the common-commitment forecast-only arm. It writes the battery action back to delivered power before applying the observed Elexon single-price settlement formula. Gross debit is the positive part of signed settlement cost, including negative-price periods; it is not the absolute sum of purchase and sale legs. The forecast-only arm keeps a persistence commitment, uses the issued forecast, current state of charge and a daily terminal-inventory constraint, and reports the capacity frontier for 0%, 5%, 10% and 20% of the 48.3-MW farm capacity. The separate ex-post capability arm uses the current half-hour power measurement in `planned_inventory_step`, includes settlement of the restoration actions required to return terminal inventory, and reports gross and signed settlement separately under the same 10–90% state-of-charge bounds. The buy-side and sell-side price columns are both present and identical in this archived calendar. The metered balance allows negative delivery when charging exceeds wind generation; this represents grid import. Extra export headroom and grid charging are assumed engineering permissions, and terminal-restoration actions are settled. It evaluates 1-, 2- and 4-h schedules and retains 3-, 7- and 14-day block intervals. At the two-hour horizon, the 20% battery paired with the selected weather-plus-event schedule reduces test gross debit from GBP 223,345 to GBP 162,784 (27.1%; seven-day block interval 23.95–30.73%) over 58 complete days; the accompanying signed settlement cashflow is GBP 57,195 before and GBP 71,075 after correction. Machine-readable outputs are `outputs/protocol\_benchmark\_v22/economics/capability\_daily.csv`, `capability_frontier.csv` and `capability_intervals.csv`.

S21. Many-to-many correspondence and hierarchy

The many-to-many extension builds a bipartite graph for each site, turbine, test split and configuration pair. All edges satisfying $\text{IoU} \geq 0.5$ are retained, with 0.3/0.7 sensitivities. Connected components classify one-to-one, one-to-many, many-to-one and many-to-many correspondence. A primitive-only arm keeps the original event population; a hierarchy-inclusive arm allows primitive intervals to correspond to composite V/inverted-V episodes. Coverage counts unique connected nodes on each side, never edge multiplicity. Each component receives one vote when comparing label composition: if p_a and p_b are its four-class frequency vectors, composition overlap is $\sum_j \min(p_{aj}, p_{bj})$. This descriptive score preserves mixture proportions, while within-component temporal order is represented by the stored intervals and turning points. It is distinct from ARI. A separate containment sensitivity uses intersection/minimum duration ≥ 0.5 , allowing a short leg to belong to a longer episode. Components longer than four hours remain explicitly counted as linked measurement supports. Shared seven-day calendar blocks anchor each component at its earliest start; 2,000 draws quantify conditional temporal variation. Per-event predictions use the frozen Pizhou raw25 k=4 seed-41 model.

Configuration names fix the left/right order lexicographically. A supported component pair contains at least 100 connected components; this count is descriptive and does not filter the all-configuration site summary. Site composition scores and topology proportions weight components equally across all 136 configuration pairs. Pooled coverage divides total unique connected nodes by total eligible nodes across those comparisons; an event can contribute once within each comparison. The main one-to-one coverage table instead averages the 136 configuration-specific fractions equally. An exact count reconciliation confirms identical primitive populations and one-to-one matches under both summaries (coverage_aggregation_reconciliation.csv).

S22. Native-resolution rule-based accuracy-charge experiment

Data, operational clock and weather inputs

The v24 cost experiment is a new task on the existing Pizhou archive. The two source batches contain 33 turbines and overlap by 90,720 minute records per turbine; overlapping power values were checked before merging. The provider-supplied capacity note reports 87.45 MW. Source times use Asia/Shanghai and are converted to UTC internally. The aggregate grid contains 49,920 quarter-hour timestamps and 48,206 complete target points. Target power is measured at the quarter-hour timestamp; input features use samples at least one minute before issue. Neither target values nor quarter-hour inputs are replicated from the main half-hour catalogue. Daily and monthly eligibility counts accompany the results. The chronological split ends training at 2024-07-08 16:00 UTC and validation at 2024-10-20 16:00 UTC. Test targets extend through 2025-02-01 15:45 UTC. Fifteen-minute and four-hour tests contain 9,450 and 9,345 eligible targets; models within a horizon share the same targets.

Causal prefixes comprise nine observations spanning two hours for the 15-min forecast and seventeen observations spanning four hours for the four-hour forecast. Interpolation produces 25 coordinates. These prefixes are wholly historical, in contrast to the event-centred 4+17+4 shapes used in structural and physical analyses. Shape encoders and standardizers are fitted on training prefixes, with a maximum of 10,000 training samples for representation fitting. Scalar descriptors include available power, the preceding 15-min mean, wind, range, total variation and calendar fields. The five feature arms are scalar only, raw25, raw/PCA6, GAF/PCA6 and GAF/PCA5 plus a protected sign coordinate; all representation arms also receive the common scalar descriptors.

Weather inputs use the Open-Meteo Previous Runs API, fixed previous_day1 forecasts from JMA GSM and GFS Global [16]. JMA 10-m wind covers the full September 2023–February 2025 study window (12,504 hourly values); GFS 100-m wind begins in February 2024 (8,466 values). Wind speed/direction are converted to u/v before time interpolation; missing hourly support is not bridged. The queried administrative proxy is 34.3403 degrees N, 118.0068 degrees E. The nominal fixed-lead vintages precede issue by at least 19 h 15 min, allowing for the upper hourly endpoint used to interpolate

a quarter-hour target. These are archived forecast lead times rather than station-level receipt logs. Consecutive valid times may originate from different numerical-model cycles. Three arms use no numerical weather, JMA, or JMA plus GFS. Their test and validation targets are identical; GFS availability shortens its training support. Accordingly, weather-arm differences include both information and source-coverage effects. Source response files and their availability record are archived with the experiment.

Prediction, selection and policy calculation

The probability model is an ExtraTrees classifier with 150 trees, maximum depth 16, seed 41 and minimum leaf sizes 8/32. Normalized power is binned in 0.01-capacity increments over the implemented 0–1.20 range. Candidate outputs are conditional mean, conditional median and the centre maximizing predictive mass in the admissible error band. Forecasts are blended with persistence using weights 0/0.25/0.5/0.75/1 and clipped to 0–1 nameplate capacity. Validation ranks failed-point count, then nMAE, then the smaller model weight. Five representations, two leaf sizes, three actions, five blends and two horizons produce 300 candidate rows per weather arm and 900 total correlated scoring rows. The primary and scalar-control selections across weather arms are saved in `pipeline_selection_before_weather_test.json` before the new weather test scores are read. The already examined no-weather task and the use of an existing research archive are retained in the experiment’s exploratory identity. The nMAE-selected reference is a separate selection rule on the same candidate pool.

For a predictive distribution q_j over normalized power-bin centres y_j , the band action was $a^* = \arg \max_a \sum_j q_j \mathbf{1}\{|y_j - a| \leq \tau\}$, evaluated over the implemented candidate values a at normalized tolerance τ . This action targets the expected number of failed accuracy points. The primary four-hour selection uses a median action, JMA/GFS inputs, protected-polarity angular coordinates, minimum leaf size 32 and model blend 0.75. The 15-min selection uses raw/PCA6, no numerical weather, minimum leaf size 8, median action and blend 0.75. The separate same-model action ablation uses no-weather raw25, leaf size 32 and blend 0.75; only conditional mean versus band action changes.

The policy source is the official Jiangsu 2022 No. 53, Article 44(II), printed pages 19–20 [17]. Its ultra-short-term point accuracy is $100(1 - |\text{actual-forecast}|/\text{capacity})$; the qualification thresholds are 97% at 15 min and 87% at four hours. A failed point costs CNY 4 per 10 MW, or CNY 34.98 at 87.45 MW. Equality at the threshold qualifies. The medium/short-term 2% monthly allowance is not used for this distinct accuracy component. Costs aggregate the observed eligible points; exemptions, reporting failures and contractual purchase/sale income are outside this component. This is a 2022-rule simulation on the observed power calendar, not a reconstruction of a contemporaneous invoice.

Paired nonoverlapping calendar blocks are anchored to UTC epoch boundaries, with seven days primary and three/fourteen days as sensitivities. Two thousand multinomial block resamples preserve the model/baseline pair. The test occupies 35, 16 and 9 blocks, respectively. Intervals are percentile intervals of the relative fee reduction. They quantify temporal variation conditional on the selected training models; seeds 42 and 43 separately refit classifiers with the transformations and selected settings fixed. Sensitivity rescales the assumed capacity by 0.95/1/1.05 while preserving MW predictions. Discrete point charges can yield identical totals for distinct forecasts: the table of equal-fee pairs reports prediction equality and differing exceedance timestamps rather than interpreting a fee tie as duplicated data.

Reproduction and relation to earlier cost experiments

Machine-readable results are under `outputs/protocol_benchmark_v24/economics`. Primary tables, all 3/7/14-day intervals, model-seed checks, ramp attribution and selection-objective comparisons are retained. The economic replication bundle includes de-identified model-input arrays, selected estimators, environment versions and six exact-output reproduction cases. It reproduces these model-level tests from transformed inputs; the private original-clock minute archive remains a distinct upstream preparation stage. The v20 duplicated-slot screen is superseded for the ultra-short-term question by this native point-level experiment. The historical screen remains a record of an earlier scenario, not evidence for the Article 44(II) charges.

S23. British economic comparisons and the scope of monetary benefit

Preserved observed-price and metered-correction results

The earlier main-text British forecast and storage results are retained here alongside their methods (S17 and S19–S20) and full tables (S51–S52 and S56). The original 2020 forecasting experiment evaluates persistence, power/calendar, completed turbine wind, historical raw25-cluster inputs and two ramp-mixture variants. At two hours, the matched historical-event addition changes nMAE from 14.69% to 14.47% and gross imbalance debits from GBP 292,263 to GBP 279,178. The later fixed-matrix extension reduces debits to GBP 233,084 on that reused test period. These are gross-exposure comparisons; signed settlement includes the associated credits. Figures S2–S4 retain the original ramp-concentration, forecast and storage views.

The common-commitment forecast-only controller selects zero additional storage on validation for every information arm. The separate 58-day, ex-post metered-correction arm reduces two-hour event-schedule gross debits from GBP 223,345 to GBP 162,784, while credits decrease from GBP 166,150 to GBP 91,709 and signed settlement cost rises from GBP 57,195 to GBP 71,075. This result concerns the stated controller, commitment and settlement service. It is not an estimate of whether a farm should own storage for all possible services. It remains part of the evidence rather than being reclassified as a net-profit gain.

Forecast-only price-driven storage and complete trading income

The v24 wind-storage controller optimizes forecast operating revenue over a receding horizon. It can charge only from wind, respects an assumed 48.3-MW export limit, uses charge/discharge efficiencies of 0.92 and keeps SOC within 10–90%. Initial and terminal daily SOC are both 10%; inventory restoration is settled. Battery power is 0/5/10/20% of farm capacity with two-hour energy capacity. An explicit GBP 10 per MWh AC-throughput charge represents wear. Past price features respect a gate of max(publication time, interval end) plus five minutes; archived national WINDFOR vintages respect publication time plus five minutes. The latter is a national generation forecast rather than a local weather observation. Training ends before August 2020, validation uses the remaining specified 2020 calendar, and frozen settings are applied to 2021 January–June. Ninety-one complete eligible days enter scoring. Realized half-hour wind constrains physical wind-only charging and export after planning; future realized power and prices do not enter the planner.

For a battery power equal to 10% of farm capacity, the event-information arm gains GBP 5,986.77 relative to no storage after the assumed wear charge; the market-information arm gains GBP 7,168.89 and the weather arm GBP 6,413.51. At 20%, the corresponding gains are GBP 8,763.00, GBP 12,287.43 and GBP 11,503.57. These demonstrate operating gains for the tested assets; the event arm is not the best information arm. Capital costs and contractual service payments are separate investment quantities. The full per-day outcomes, capacity frontier and block intervals accompany the package.

A separate complete-income experiment includes market-index reference revenue, signed imbalance settlement and an assumed GBP 0.5/MWh scheduled trading fee. Forecasts and bounded schedule adjustments are selected on 2021 April–June validation, with July–December reserved for this test. The observed market index is an execution-price reference rather than a transaction record. Test revenue differences for the event arm relative to passive persistence are GBP -19,829.64, -21,721.27 and -26,310.43 at 1/2/4 h; weather-only differences are GBP -16,311.56, -22,117.41 and -25,991.07. These complete-income results prevent the gross-debit improvements from being interpreted as a general trading-profit advantage. They concern a different task, jurisdiction, period and objective from the positive Jiangsu accuracy-charge simulation.

S24. Scalar-controlled polarity increments and noise sensitivity

The v25 mechanism extension was designed after the reviewer identified a distinction between restoring angular direction and adding information beyond existing scalar covariates. It is exploratory on the ex-

isting populations. The four factorial cells are S, S+G5, S+b and S+G5+b, where S contains endpoint direction, signed endpoint change, duration, pre-window mean power and pre-window mean. G5 is the first five coordinates of the same frozen Pizhou six-component GAF/PCA transform; b is the sign at the largest-absolute-value coordinate. Thus the conditional bit contrast holds angular coordinates fixed. G6, pre-PCA signed-channel GAF/PCA6 and raw25 provide dimension, compression-order and trajectory controls.

The complete thirteen arms are scalar, scalar_g5, scalar_bit, scalar_g5_bit, scalar_g6, scalar_signed6, scalar_raw25, scalar_g5_random, scalar_g5_reversed, scalar_g5_endpoint, scalar_g5_pre, scalar_g5_post and scalar_g5_internal. Random signs are independent Bernoulli signs with seed 250041 and use separate events across source and target populations. Consistent reversal negates b during both fitting and evaluation, preserving information. Test-only corruption independently reverses 25%, 50% or 100% of the true signs after fitting, with seed 404. Alternative signs come from event-end coordinate 20, pre-context coordinate 0, post-context coordinate 24, or the largest absolute value among event-interior coordinates 4–20. The pre-PCA signed-channel control appends the original path before standardization and compression; it differs from reserving a sign coordinate after compression.

Each of the five regional ERA5 sites reuses its archived training-event identifiers and matched, weather-valid validation/test events. Training counts are 9,732 Pizhou, 9,648 Suining, 8,606 Yandun, 9,562 La Haute Borne and 10,000 Hill. A histogram gradient-boosted classifier uses 200 iterations, learning rate 0.05, L2 regularization 1, seed 41, no early stopping and candidate leaf budgets 7/15. Validation log loss selects the budget; scalar temperature in [0.5,5] is fitted on the same validation period. The source representation remains Pizhou-fitted. SMARTEOLE uses Pizhou source heads without target fitting on the exact stored WindCube-eligible population. Hill’s heads supply the two LiDAR instruments, followed by multinomial-logistic calibration with C in {0.1,1,10}; the original 14 March/7 April boundaries and context restrictions are preserved. There are 65 selected source heads (130 candidate fits) and 104 feature-arm/population evaluations across eight populations.

Direction AUROC conditions on observed increase/decrease; three-class log loss and summed Brier score also retain small-net-change outcomes. Event identifiers and target times are identical across all feature arms within each population. Shared UTC-anchored 3/7/14-day blocks, with 2,000 fixed-seed resamples, preserve model pairing and all same-farm turbines. A weighted positive-negative concordance calculation gives exact block-resampled AUROC, including half credit for tied scores; unit tests compare it with explicit weighted AUROC. The LiDAR final periods each contain four seven-day blocks. Draws are not additional independent weather periods. Confidence intervals quantify temporal sampling conditional on these fitted models and are not multiplicity-adjusted simultaneous bounds. Full scores and paired contrasts are archived, rather than selecting models by significance.

Anchor stability and measurement-noise experiments

Valid nonconstant shapes satisfy $\max|x|=1$; the pre-normalization scale $a=\text{shape_normalizer}$ measures the original maximum deviation in training-production-scale units. Zero a is excluded before encoding. For additive representation perturbation e, define the opposing-extrema margin $m=|\max(x)+\min(x)|$. If $m>2\|e\|_\infty$, the dominant positive/negative sign cannot exchange under that perturbation. When noise is added to samples before subtracting the noisy start value, e includes that subtraction; the experiment implements it explicitly. This explains why opposing extrema close to equal magnitude, rather than a small normalized anchor, are the important sign-instability case.

Relative-noise tests use independent Gaussian samples with standard deviation 0, 0.01, 0.05, 0.10, 0.20 or 0.40 times a and seeds 101/202/303. After subtracting the perturbed event-start sample, paths are renormalized and encoded with frozen transforms and heads. Scalar descriptors stay fixed, so this isolates the representation channel rather than re-running detection on noisy SCADA. Pizhou, Hill, both LiDAR instruments and SMARTEOLE are included. At relative noise 0.05, mean anchor-flip fractions over the three seeds range from 0.0097 to 0.0175 across these populations. All probability scores and paired differences are retained across noise levels.

A complementary absolute-noise test applies standard deviations 0.001/0.005/0.01 in the original

training-production-scale units before normalization, on Pizhou and both LiDAR instruments. The effective normalized noise is σ/a . Results are stratified by $a < 0.05$, $0.05 \leq a < 0.2$, $a \geq 0.2$, and opposing-extrema margin below/above 0.1. Empty strata remain explicitly empty; notably the final T11 sample contains no $a < 0.05$ events. These checks distinguish algorithmic instability near zero scale from the actual amplitude support of the evaluated populations.

S25. Seven AI review sessions and source-separated reference agreement

The three received human assessments remain unchanged and are described in S6. Seven additional independent AI review sessions each assessed all 320 windows, producing 2,240 model-generated judgments. Together there are ten review configurations and 3,200 judgments, not ten human respondents or 3,200 independent events. The final seven-session roster was fixed before calculating expanded-panel agreement, and every completed session was retained. Session identifiers A01–A07 preserve this analysis roster.

Each clean-context session received only the same nine normalized-power samples at -120,-90,-60,-30,0,30,60,90,120 min and the rubric below, with a different fixed random presentation order. The central five samples define the rated two-hour region; the outer samples provide context. No site, detector vote, human answer or other AI response was supplied. Reviewers produced labels by judging each curve; code was permitted only for reading input, writing the answers and checking schema/coverage, not for generating labels with a threshold classifier. Each response records morphology and H/M/L confidence. The source type is AI, with its actual task-context identifier stored separately from anonymous human identifiers.

The review sessions used Codex’s inherited model configuration on 20 September 2026. The collaboration interface did not expose a verified effective model version, temperature, top-p or reasoning setting; these are recorded as unavailable rather than inferred. The sessions therefore represent repeated model-based reviews under a common interface and rubric, not seven independently developed model families. The human observers retain their original separate provenance. No requests failed, no missing answers were imputed, and all seven outputs contained 320 unique valid window identifiers.

The task rubric was: judge recognizable upward, downward, V-shaped, inverted-V or repeated dynamic changes in the central two hours; a return to the initial level can still contain an event. Persistent low power is a state, whereas entering or leaving it can be dynamic. Ordinary small jitter need not be an event. Use uncertain for unsupported judgments and bad_data for invalid observations; infer neither sub-sample boundaries nor weather/fault causes. No numerical event threshold was added to the original human rubric. Confidence records the morphology judgment. The complete per-session task text, nine-point inputs, output schema and raw AI labels are archived with the reference results.

For category alignment, the original UI names valley/peak/data_issue are mapped bijectively to v_shape/inverted_v/bad_data; original strings are retained. This harmonizes names without changing judgments. Dynamic classes map to presence=yes; quiet/low_state map to no. Uncertain or bad-data responses would remain missing for nominal-alpha calculations; all received judgments were assessable. Human-only, AI-only and mixed panels are reported separately with 3/7/14-day resampling within farm, preserving every observer in each sampled window. The common seven-day analysis contains 197 occupied farm-blocks.

AI-only presence/morphology alpha is 0.958/0.958; mixed-panel alpha is 0.707/0.741. The original human values remain 0.528/0.476. Source-pair comparisons distinguish human–human, human–AI and AI–AI agreement. In the ten-configuration panel, only 3 of 45 observer pairs are human–human, while 21 are AI–AI and 21 are human–AI. Consequently a higher mixed-panel alpha measures agreement in a different, model-heavy panel; it is neither evidence of additional human recruitment nor a measure of annotation accuracy. The reference remains auxiliary to the externally measured physical outcomes and known-support synthetic localization task.

S26. Complete physical-process reconstruction

Population, independent targets and chronology

This extension asks when ordered power-event information improves recovery of a measured wind process beyond scalar descriptions. It uses the T11 and T07 LiDAR references described in S16 and the SMARTEOLE WindCube 80-m series. It retains power-defined primitive and compound intervals lasting 60–240 min, then collapses identical turbine/start/end intervals across detector configurations. A retained interval has a compound flag if at least one source definition contains opposite legs; the complementary primitive-only stratum contains intervals without this flag. Neither flag uses wind outcomes. This unique-interval population is separate from the earlier endpoint-probability population and its repeated configuration rows.

At least seven consecutive valid 10-min wind observations, including both endpoints, are required. WindCube 10-min means require at least eight valid minute values. Missing internal bins exclude an interval rather than being interpolated across. Native power inputs at SMARTEOLE require ten valid minute observations per 10-min bin. The Hill timestamp convention follows S16. Event supports and the entire power context, from 120 min before start through 150 min after end, remain within one split. Hill training ends at 00:00 UTC on 14 March 2026 and validation at 00:00 UTC on 7 April; SMARTEOLE boundaries are 01:48 UTC on 16 April 2020 and 12:54 UTC on 5 May. Table S75 gives retained counts. Test calendars have appeared in preceding tasks; the new reconstruction analysis is exploratory.

For phase $r_j = j/16$, the wind target is $Y_j = u(s + r_j(e - s)) - u(s)$, with $j=0, \dots, 16$. Linear interpolation occurs only within consecutive valid wind support. All trajectories start at zero. Full-trajectory error averages squared differences over $j=1, \dots, 16$ and over events before taking the square root. Geometry error applies the same calculation after subtracting each measured or predicted curve's own endpoint chord, $R_j = Y_j - r_j Y_{16}$. Thus this diagnostic asks whether intermediate evolution is recovered after removing endpoint tendency; it is not an error in absolute wind speed. The 10/30-min power tracks retain identical event identifiers, wind targets and splits. All scalar and temporal arms within a track use that track's power support.

Scalar controls and representation matrix

C5 comprises endpoint direction, signed endpoint change, duration, pre-window mean power and pre-window mean. C19 adds range, total variation, maximum rate, maximum chord residual, absolute curvature, post-window mean, shape normalization scale, minimum and maximum phases, internal mean and standard deviation, positive and negative chord extremes, and chord-extreme phase. C27 adds eight summaries of the cube root of nonnegative power: starting and ending level, internal mean, standard deviation and range, pre/post means and mean absolute internal difference. These are input transformations; independent instruments remain the wind target.

The basic matrix contains C5, C19 and ten C19-plus-representation arms: raw25, raw/PCA6, raw/PCA12, GASF/PCA6, GASF/PCA5 plus sign, GADF/PCA6, dual angular fields, energy-balanced signed channels, and two angular/phase combinations. The Gramian angular difference field (GADF) follows the angular construction of Wang and Oates [14]. With the same signed unit-domain x and $c_i = \sqrt{1 - x_i^2}$, $D = cx^T - xc^T$ and $D(-x) = -D(x)$. Because the event-start coordinate $x_4 = 0$, the uncompressed row $D_{4j} = x_j$. This is an exact full-field identity; compressed-coordinate performance is empirical. Dual fields combine GASF and GADF, each divided by its training root-mean-square vector energy. The signed-channel arm similarly balances GASF and raw coordinates. The phase arms combine three angular PCs with three orthonormal type-II discrete-cosine coordinates from the seventeen internal samples, or six of each. All standardization, energy factors and principal components are fitted on the corresponding training partition.

The scale-restoration matrix contains nine arms: C19, C27, C27 plus normalized raw25, internal level changes at seventeen positions, cube-root changes at seventeen or twenty-five positions, six cube-root principal components, amplitude-scaled GADF/PCA6, and amplitude-scaled twelve-coordinate phase en-

coding. Restored levels are $p_j = p_s + ax_j$; the selected level-change arm supplies the seventeen internal values $p_j - p_s$ alongside S27. Power remains in the fitted production-scale units used by the catalogue. A final residual matrix compares S27, internal level changes, internal cube-root changes, cube-root/PCA6 and scaled phase12. Repeated controls across branches are explicit common comparators, not independent experimental replications.

Decoder selection and residual control

Every arm compares eight decoder configurations: standardized ridge regression with penalty 0.1, 1, 10 or 100, and 192-tree extremely randomized ensembles with minimum leaf size 4 or 16 crossed with depth 12 or unrestricted depth. Tree seed is 41, all features are eligible at a split, and two CPU threads are used. Validation full-trajectory mean squared error selects the configuration. Residual fitting first generates out-of-fold scalar predictions over up to five contiguous groups of seven-day training blocks. Fitting observations whose complete contextual support overlaps a held fold are excluded. Residual models receive the same eight decoders and correction weights 0, 0.25, 0.5 and 1; the scalar-only correction has the identical budget.

The final validation record compares all completed scalar-only controls and all temporal candidates, within native 10-min support and across 10/30-min support. Both selections choose native 10-min internal level changes in each population. T11 and T07 use the S27 residual comparator; SMARTEOLE uses the S19 comparator. The selected trajectory models use ridge penalties 100, 10 and 100 for T11, T07 and SMARTEOLE, respectively. This selection was frozen before opening any v26 test result. Validation scoring rows, all selected-arm test scores and all pairwise comparisons remain in the release; previously inspected endpoint tasks are recorded separately.

Paired uncertainty and mechanistic diagnostics

The relative reduction is $100[1 - \text{RMSE}_{\text{trajectory}} / \text{RMSE}_{\text{scalar}}]$. The shared calendar-block bootstrap draws 2,000 paired replicates, with seven days primary and three/fourteen days as sensitivities. Each native test population occupies four seven-day blocks; all turbines sharing the SMARTEOLE wind record receive the same draw. T11 and T07 are instrument-specific analyses at one farm, rather than independent climate replications. Tables S76–S77 retain full-population and hierarchy contrasts. Empty-support resamples are excluded and counted: the seven-day SMARTEOLE compound contrast has 1,990 valid draws.

For duration standardization, each exact-duration stratum present in both power-defined groups receives mass equal to the smaller group count. An event in group g then has weight $\min(n_{\text{compound}}, n_{\text{primitive}}) / n_g$. These fixed, outcome-blind weights are combined with each shared calendar draw. The resulting compound-minus-primitive difference is descriptive effect modification, not an intervention on event topology (Table S78).

Two frozen-decoder diagnostics preserve every scalar descriptor and each event’s endpoints. One shuffles interior temporal coordinates within the event. The other reassigns endpoint-chord residuals between events within direction, amplitude and duration strata. Seeds 101, 202 and 303 retain all realized controls. These perturbations evaluate reliance on ordered correspondence and can move inputs outside the training distribution; they are mechanistic checks, not independent natural experiments (Table S79).

For auxiliary process recognition, the maximum measured drawup and drawdown define four classes: weak, up, down and return. A return requires both changes to reach the study threshold of 1.5 m/s; it includes V, inverted-V and multistage dynamics. Applying this same rule to the predicted curve yields a derived class. Table S80 reports precision/recall/F1 with an always-return comparator; ranking uses the smaller of predicted maximum drawup and drawdown. These scores describe curve-derived process recovery, separate from a tuned classification benchmark. All model comparisons and 3/7/14-day intervals are supplied with the result tables.

Reproduction and result map

The data interface is prepared by `prepare_process_v26.py` and `prepare_native_process_v26.py`; `physical_process_v26.py` and `physical_calibration_v26.py` perform validation and locked evaluation. The final selection is `final_validation_lock.json`. Summary, hierarchy, duration-standardized and perturbation tables are produced by `summarize_process_v26.py`, `process_hierarchy_v26.py`, `process_duration_control_v26.py` and `process_controls_v26.py`. Outputs reside under `outputs/protocol_benchmark_v26`. The six prepared tracks passed identical-event, target, split, finite-value and saved-prediction checks; targeted tests cover angular identities and continuous wind targets. Source calendars and the differing experiment denominators remain explicit.

Supplementary result tables

Table S1. Primary representation comparison (k = 4)

Site/group	Representation	NMI	ARI	Agreement
Hill of Towie	event row permutation	0.024	-0.007	0.366
La Haute Borne	event row permutation	0.067	0.023	0.402
Pizhou	event row permutation	0.055	0.025	0.367
pizhou_seen	event row permutation	0.048	0.022	0.370
pizhou_unseen	event row permutation	0.102	0.051	0.364
Suining	event row permutation	0.037	-0.011	0.364
Yandun	event row permutation	0.026	0.006	0.340
Hill of Towie	gaf pca6	0.272	0.242	0.567
La Haute Borne	gaf pca6	0.301	0.238	0.574
Pizhou	gaf pca6	0.273	0.252	0.556
pizhou_seen	gaf pca6	0.275	0.248	0.560
pizhou_unseen	gaf pca6	0.313	0.268	0.568
Suining	gaf pca6	0.301	0.259	0.580
Yandun	gaf pca6	0.269	0.254	0.563
Hill of Towie	gaf signed pca6	0.276	0.245	0.568
La Haute Borne	gaf signed pca6	0.289	0.234	0.575
Pizhou	gaf signed pca6	0.275	0.255	0.558
pizhou_seen	gaf signed pca6	0.277	0.251	0.562
pizhou_unseen	gaf signed pca6	0.316	0.274	0.570
Suining	gaf signed pca6	0.302	0.261	0.583
Yandun	gaf signed pca6	0.271	0.259	0.567
Hill of Towie	raw25	0.601	0.622	0.736
La Haute Borne	raw25	0.698	0.687	0.776
Pizhou	raw25	0.643	0.664	0.739
pizhou_seen	raw25	0.651	0.674	0.747
pizhou_unseen	raw25	0.682	0.684	0.751
Suining	raw25	0.615	0.630	0.731
Yandun	raw25	0.590	0.626	0.717
Hill of Towie	raw pca6	0.597	0.618	0.734
La Haute Borne	raw pca6	0.696	0.686	0.776
Pizhou	raw pca6	0.639	0.660	0.736
pizhou_seen	raw pca6	0.648	0.669	0.744
pizhou_unseen	raw pca6	0.677	0.679	0.749
Suining	raw pca6	0.614	0.629	0.729
Yandun	raw pca6	0.588	0.624	0.716
Hill of Towie	statistics9	0.474	0.453	0.742
La Haute Borne	statistics9	0.475	0.517	0.819
Pizhou	statistics9	0.518	0.500	0.742
pizhou_seen	statistics9	0.514	0.493	0.746
pizhou_unseen	statistics9	0.537	0.509	0.726
Suining	statistics9	0.527	0.517	0.768
Yandun	statistics9	0.473	0.482	0.743
Hill of Towie	within event time permutation	0.489	0.594	0.731
La Haute Borne	within event time permutation	0.601	0.660	0.749
Pizhou	within event time permutation	0.532	0.644	0.719
pizhou_seen	within event time permutation	0.536	0.648	0.727

Site/group	Representation	NMI	ARI	Agreement
pizhou_unseen	within event time permutation	0.572	0.651	0.725
Suining	within event time permutation	0.513	0.612	0.732
Yandun	within event time permutation	0.438	0.580	0.687

Table S2. Raw25 block-length sensitivity

Site/group	Days	Occupied blocks	Paired records	NMI [95% interval]
Hill of Towie	3	26	243684	0.601 [0.589, 0.628]
Hill of Towie	7	12	243684	0.601 [0.592, 0.630]
Hill of Towie	14	6	243684	0.601 [0.591, 0.639]
La Haute Borne	3	45	34042	0.698 [0.690, 0.742]
La Haute Borne	7	21	34042	0.698 [0.691, 0.744]
La Haute Borne	14	12	34042	0.698 [0.696, 0.748]
Pizhou	3	34	274570	0.643 [0.633, 0.676]
Pizhou	7	16	274570	0.643 [0.631, 0.676]
Pizhou	14	9	274570	0.643 [0.636, 0.667]
Suining	3	24	102248	0.615 [0.615, 0.664]
Suining	7	11	102248	0.615 [0.613, 0.663]
Suining	14	6	102248	0.615 [0.613, 0.671]
Yandun	3	15	811453	0.590 [0.567, 0.625]
Yandun	7	7	811453	0.590 [0.574, 0.620]
Yandun	14	4	811453	0.590 [0.576, 0.624]
pizhou_seen	3	34	214524	0.651 [0.644, 0.684]
pizhou_seen	7	16	214524	0.651 [0.647, 0.683]
pizhou_seen	14	9	214524	0.651 [0.651, 0.677]
pizhou_unseen	3	34	60046	0.682 [0.672, 0.725]
pizhou_unseen	7	16	60046	0.682 [0.674, 0.728]
pizhou_unseen	14	9	60046	0.682 [0.676, 0.723]

Table S3. Greek configuration-pair support strata

Matches per pair	Configuration pairs	Median NMI	Minimum	Maximum
2-4	1	1.000	1.000	1.000
5-9	3	0.858	0.643	1.000
10-29	7	0.724	0.136	0.865
30-99	9	0.710	0.526	0.817
>=100	42	0.669	0.385	0.961

Table S4. Descriptive weather contrasts

Source	Horizon (h)	Exposed n	Control n	Risk exposed	Risk control	Difference
ERA5	1	21300	84761	0.230	0.191	0.039
ERA5	2	21300	84761	0.398	0.331	0.068
ERA5	4	21300	84761	0.572	0.486	0.086
NOAA proxy	1	41163	19668	0.201	0.202	-0.001
NOAA proxy	2	41163	19668	0.350	0.342	0.009
NOAA proxy	4	41163	19668	0.514	0.492	0.022

Table S5. Archived time-displacement diagnostics (4 h outcome)

Displacement	Exposed n	Control n	Risk difference
shift-24h	22083	83978	-0.007
shift-6h	21603	84458	0.050
shift+6h	20159	85902	0.038

Displacement	Exposed n	Control n	Risk difference
shift+24h	21409	84652	-0.004

Table S6. Scaled post-event power index and residual-power contrasts

Measure	Events	Exposed minus control	95% block interval
Absolute power response	96	52.959	[-347.182, 453.980]
Scaled post-event power index	96	-0.390	[-1.429, 0.363]
Residual power	96	191.034	[-368.477, 645.136]

Table S7. Forecasts on common chronological targets

Site	Model	Training	Subset	n	nMAE (%)
pizhou	persistence	none	all	3549	7.754
pizhou	persistence	none	training threshold ramp	189	37.730
pizhou	tcn	event weighted	all	3549	8.413
pizhou	tcn	event weighted	training threshold ramp	189	33.186
pizhou	tcn	mse	all	3549	7.738
pizhou	tcn	mse	training threshold ramp	189	35.449
pizhou	timesnet	event weighted	all	3549	8.502
pizhou	timesnet	event weighted	training threshold ramp	189	35.507
pizhou	timesnet	mse	all	3549	7.771
pizhou	timesnet	mse	training threshold ramp	189	36.155
yandun	persistence	none	all	1525	7.424
yandun	persistence	none	training threshold ramp	78	38.868
yandun	tcn	event weighted	all	1525	9.724
yandun	tcn	event weighted	training threshold ramp	78	33.833
yandun	tcn	mse	all	1525	8.364
yandun	tcn	mse	training threshold ramp	78	36.515
yandun	timesnet	event weighted	all	1525	8.857
yandun	timesnet	event weighted	training threshold ramp	78	37.074
yandun	timesnet	mse	all	1525	7.735
yandun	timesnet	mse	training threshold ramp	78	36.698

Table S8. Storage costs under the legacy annual-charge assumptions

Site	Model	Training	Scenario	P (MW/MW)	E (MWh/MW)	Cost (CNY per norm. MW)
pizhou	persistence	none	base	0.000	0.000	25837.708
pizhou	persistence	none	high	0.000	0.000	51675.415
pizhou	persistence	none	low	0.000	0.000	12918.854
pizhou	tcn	event weighted	base	0.000	0.000	29630.118
pizhou	tcn	event weighted	high	0.000	0.000	59260.236
pizhou	tcn	event weighted	low	0.000	0.000	14815.059
pizhou	tcn	mse	base	0.000	0.000	25098.126
pizhou	tcn	mse	high	0.000	0.000	50196.252
pizhou	tcn	mse	low	0.000	0.000	12549.063
pizhou	timesnet	event weighted	base	0.000	0.000	26388.821
pizhou	timesnet	event weighted	high	0.000	0.000	52777.642
pizhou	timesnet	event weighted	low	0.000	0.000	13194.410
pizhou	timesnet	mse	base	0.000	0.000	24212.394
pizhou	timesnet	mse	high	0.000	0.000	48424.787

Site	Model	Training	Scenario	P (MW/MW)	E (MWh/MW)	Cost (CNY per norm. MW)
pizhou	timesnet	mse	low	0.000	0.000	12106.197
yandun	persistence	none	base	0.000	0.000	10751.779
yandun	persistence	none	high	0.000	0.000	21503.559
yandun	persistence	none	low	0.000	0.000	5375.890
yandun	tcn	event weighted	base	0.000	0.000	16643.238
yandun	tcn	event weighted	high	0.000	0.000	33286.477
yandun	tcn	event weighted	low	0.000	0.000	8321.619
yandun	tcn	mse	base	0.000	0.000	12661.780
yandun	tcn	mse	high	0.000	0.000	25323.560
yandun	tcn	mse	low	0.000	0.000	6330.890
yandun	timesnet	event weighted	base	0.000	0.000	12633.754
yandun	timesnet	event weighted	high	0.000	0.000	25267.507
yandun	timesnet	event weighted	low	0.000	0.000	6316.877
yandun	timesnet	mse	base	0.000	0.000	10648.952
yandun	timesnet	mse	high	0.000	0.000	21297.904
yandun	timesnet	mse	low	0.000	0.000	5324.476

Table S9. Controlled synthetic event detection (IoU $\geq$ 0.3)

Model	Seed	Policy	Precision	Recall	F1
rate_rule	0	default	0.056	0.210	0.088
rate_rule	0	threshold only	0.099	0.175	0.126
rate_rule	0	full protocol	0.439	0.435	0.437
mean_rule	0	default	0.255	0.752	0.381
mean_rule	0	threshold only	0.364	0.568	0.444
mean_rule	0	full protocol	0.742	0.740	0.741
tcn_ae	41	default	0.118	0.802	0.206
tcn_ae	41	threshold only	0.582	0.383	0.462
tcn_ae	41	full protocol	0.575	0.667	0.618
tcn_ae	42	default	0.118	0.810	0.206
tcn_ae	42	threshold only	0.557	0.365	0.441
tcn_ae	42	full protocol	0.584	0.657	0.619
tcn_ae	43	default	0.117	0.807	0.204
tcn_ae	43	threshold only	0.578	0.372	0.453
tcn_ae	43	full protocol	0.567	0.662	0.611
transformer_ae	41	default	0.064	0.640	0.116
transformer_ae	41	threshold only	0.398	0.330	0.361
transformer_ae	41	full protocol	0.833	0.425	0.563
transformer_ae	42	default	0.059	0.652	0.109
transformer_ae	42	threshold only	0.434	0.328	0.373
transformer_ae	42	full protocol	0.879	0.417	0.566
transformer_ae	43	default	0.060	0.645	0.110
transformer_ae	43	threshold only	0.379	0.310	0.341
transformer_ae	43	full protocol	0.828	0.432	0.568
timesnet	41	default	0.172	0.580	0.266
timesnet	41	threshold only	0.536	0.448	0.488
timesnet	41	full protocol	0.685	0.522	0.593
timesnet	42	default	0.195	0.610	0.296
timesnet	42	threshold only	0.522	0.438	0.476
timesnet	42	full protocol	0.729	0.537	0.619
timesnet	43	default	0.144	0.537	0.227
timesnet	43	threshold only	0.689	0.260	0.377
timesnet	43	full protocol	0.811	0.495	0.615
kanad	41	default	0.149	0.680	0.244
kanad	41	threshold only	0.464	0.305	0.368
kanad	41	full protocol	0.718	0.445	0.549
kanad	42	default	0.196	0.735	0.309
kanad	42	threshold only	0.455	0.455	0.455
kanad	42	full protocol	0.683	0.677	0.680
kanad	43	default	0.244	0.710	0.363
kanad	43	threshold only	0.668	0.477	0.557
kanad	43	full protocol	0.665	0.730	0.696

Table S10. Human-review sampling support

Site	Sampling stratum	Windows
pizhou	downward	6
pizhou	upward	6
pizhou	low output	6
pizhou	screen disagreement	6
pizhou	random control	6
yandun	downward	6
yandun	upward	6
yandun	low output	6
yandun	screen disagreement	6
yandun	random control	6
greece	downward	6
greece	upward	6
greece	low output	6
greece	screen disagreement	6
greece	random control	6
sdwpf	downward	6
sdwpf	upward	6
sdwpf	low output	6
sdwpf	screen disagreement	6
sdwpf	random control	6

Table S11. Human-labelled centre-region detector agreement

Site	Detector	Windows	TP	FP	FN	Precision	Recall	F1
greece	Corridor 0.025	30	8	1	21	0.889	0.276	0.421
greece	Corridor 0.050	30	13	1	16	0.929	0.448	0.605
greece	Corridor 0.100	30	14	1	15	0.933	0.483	0.636
greece	Tail 20 min	30	15	1	14	0.938	0.517	0.667
greece	Tail 40 min	30	13	1	16	0.929	0.448	0.605
greece	Tail 80 min	30	19	1	10	0.950	0.655	0.776
greece	Mean shift 20 min	30	8	1	21	0.889	0.276	0.421
greece	Mean shift 40 min	30	14	1	15	0.933	0.483	0.636
greece	Mean shift 80 min	30	15	1	14	0.938	0.517	0.667
greece	Threshold 20 min	30	13	1	16	0.929	0.448	0.605
greece	Threshold 40 min	30	16	1	13	0.941	0.552	0.696
greece	Threshold 80 min	30	19	1	10	0.950	0.655	0.776
pizhou	Adaptive corridor	30	20	0	8	1.000	0.714	0.833
pizhou	Corridor 0.025	30	17	0	11	1.000	0.607	0.756
pizhou	Corridor 0.050	30	19	0	9	1.000	0.679	0.809
pizhou	Corridor 0.100	30	20	0	8	1.000	0.714	0.833
pizhou	Tail 1h	30	3	0	25	1.000	0.107	0.194
pizhou	Tail 2h	30	3	0	25	1.000	0.107	0.194
pizhou	Tail 4h	30	2	0	26	1.000	0.071	0.133
pizhou	Mean shift 1h	30	17	1	11	0.944	0.607	0.739
pizhou	Mean shift 2h	30	18	2	10	0.900	0.643	0.750
pizhou	Mean shift 4h	30	18	1	10	0.947	0.643	0.766
pizhou	OpSDA 1h 0.025	30	12	1	16	0.923	0.429	0.585
pizhou	OpSDA 2h 0.025	30	16	1	12	0.941	0.571	0.711
pizhou	OpSDA 4h 0.025	30	18	1	10	0.947	0.643	0.766
pizhou	SDA 0.025	30	16	1	12	0.941	0.571	0.711
pizhou	Threshold 1h	30	20	1	8	0.952	0.714	0.816
pizhou	Threshold 2h	30	21	2	7	0.913	0.750	0.824
pizhou	Threshold 4h	30	20	2	8	0.909	0.714	0.800
sdwpf	Corridor 0.025	30	6	0	23	1.000	0.207	0.343
sdwpf	Corridor 0.050	30	12	0	17	1.000	0.414	0.585
sdwpf	Corridor 0.100	30	16	0	13	1.000	0.552	0.711
sdwpf	Tail 1h	30	14	0	15	1.000	0.483	0.651
sdwpf	Tail 2h	30	14	0	15	1.000	0.483	0.651

Site	Detector	Windows	TP	FP	FN	Precision	Recall	F1
sdwpf	Tail 4h	30	18	0	11	1.000	0.621	0.766
sdwpf	Mean shift 1h	30	18	0	11	1.000	0.621	0.766
sdwpf	Mean shift 2h	30	21	0	8	1.000	0.724	0.840
sdwpf	Mean shift 4h	30	20	0	9	1.000	0.690	0.816
sdwpf	Threshold 1h	30	19	0	10	1.000	0.655	0.792
sdwpf	Threshold 2h	30	20	0	9	1.000	0.690	0.816
sdwpf	Threshold 4h	30	24	0	5	1.000	0.828	0.906
yandun	Adaptive corridor	30	17	2	11	0.895	0.607	0.723
yandun	Corridor 0.025	30	16	2	12	0.889	0.571	0.696
yandun	Corridor 0.050	30	17	2	11	0.895	0.607	0.723
yandun	Corridor 0.100	30	17	2	11	0.895	0.607	0.723
yandun	Tail 1h	30	14	2	14	0.875	0.500	0.636
yandun	Tail 2h	30	16	2	12	0.889	0.571	0.696
yandun	Tail 4h	30	19	2	9	0.905	0.679	0.776
yandun	Mean shift 1h	30	18	2	10	0.900	0.643	0.750
yandun	Mean shift 2h	30	22	2	6	0.917	0.786	0.846
yandun	Mean shift 4h	30	25	2	3	0.926	0.893	0.909
yandun	OpSDA 1h 0.025	30	11	2	17	0.846	0.393	0.537
yandun	OpSDA 2h 0.025	30	15	2	13	0.882	0.536	0.667
yandun	OpSDA 4h 0.025	30	19	2	9	0.905	0.679	0.776
yandun	SDA 0.025	30	17	2	11	0.895	0.607	0.723
yandun	Threshold 1h	30	18	2	10	0.900	0.643	0.750
yandun	Threshold 2h	30	21	2	7	0.913	0.750	0.824
yandun	Threshold 4h	30	25	2	3	0.926	0.893	0.909

Table S12. Learned representations by farm, averaged over three seeds (k = 4)

Representation Farm		NMI	ARI
cnn gaf	Hill of Towie	0.199	0.305
cnn gaf	La Haute Borne	0.183	0.280
cnn gaf	Pizhou	0.193	0.289
cnn gaf	Suining	0.194	0.295
cnn gaf	Yandun	0.210	0.308
cnn signed gaf	Hill of Towie	0.695	0.807
cnn signed gaf	La Haute Borne	0.739	0.836
cnn signed gaf	Pizhou	0.732	0.832
cnn signed gaf	Suining	0.713	0.815
cnn signed gaf	Yandun	0.667	0.786
tcn	Hill of Towie	0.719	0.815
tcn	La Haute Borne	0.763	0.840
tcn	Pizhou	0.754	0.840
tcn	Suining	0.744	0.834
tcn	Yandun	0.689	0.787
transformer	Hill of Towie	0.691	0.756
transformer	La Haute Borne	0.729	0.774
transformer	Pizhou	0.722	0.781
transformer	Suining	0.705	0.767
transformer	Yandun	0.659	0.745

Table S13. SDWPF learned-model transfer across seven batches and three seeds

Representation	k	Runs	Mean NMI	SD NMI	Mean ARI
tcn	2	21	0.856	0.043	0.918
tcn	4	21	0.626	0.076	0.693
tcn	6	21	0.520	0.006	0.493
transformer	2	21	0.848	0.041	0.913
transformer	4	21	0.589	0.049	0.651
transformer	6	21	0.521	0.009	0.516

Table S14. Matched-count-weighted agreement and detector-occupancy overlap

Site	Weighted NMI	Weighted ARI	Pairs	Left cov.	Right cov.	IoU ≥ 0.3	IoU ≥ 0.5	IoU ≥ 0.7
Hill of Towie	0.684	0.738	243684	0.384	0.352	0.754	0.218	0.014
La Haute Borne	0.754	0.781	34042	0.356	0.340	0.445	0.071	0.000
Pizhou	0.722	0.782	274570	0.375	0.326	0.783	0.260	0.043
Suining	0.705	0.765	102248	0.381	0.348	0.784	0.240	0.029
Yandun	0.671	0.727	811453	0.370	0.368	0.548	0.261	0.080

Table S15. Observational weather contrasts with shared calendar-block intervals

Source	Horizon (h)	Block days	Blocks	Risk difference	95% interval
ERA5	1	3	73	0.039	[0.016, 0.062]
ERA5	1	7	31	0.039	[0.014, 0.064]
ERA5	1	14	16	0.039	[0.014, 0.061]
ERA5	2	3	73	0.068	[0.036, 0.100]
ERA5	2	7	31	0.068	[0.029, 0.107]
ERA5	2	14	16	0.068	[0.033, 0.099]
ERA5	4	3	73	0.086	[0.045, 0.124]
ERA5	4	7	31	0.086	[0.040, 0.134]
ERA5	4	14	16	0.086	[0.041, 0.125]
NOAA	1	3	59	-0.001	[-0.028, 0.026]
NOAA	1	7	27	-0.001	[-0.030, 0.027]
NOAA	1	14	14	-0.001	[-0.027, 0.026]
NOAA	2	3	59	0.009	[-0.034, 0.050]
NOAA	2	7	27	0.009	[-0.037, 0.054]
NOAA	2	14	14	0.009	[-0.037, 0.053]
NOAA	4	3	59	0.022	[-0.029, 0.072]
NOAA	4	7	27	0.022	[-0.038, 0.080]
NOAA	4	14	14	0.022	[-0.033, 0.077]

Table S16. Event-level IoU and matching-strategy sensitivity

Site	IoU	Matcher	Pairs	NMI	ARI	W. NMI	W. ARI	Left cov.	Right cov.
Hill of Towie	0.3	Greedy	307826	0.533	0.567	0.629	0.682	0.468	0.437
Hill of Towie	0.3	Max. IoU	308389	0.526	0.561	0.620	0.676	0.469	0.438
Hill of Towie	0.5	Greedy	243684	0.601	0.622	0.684	0.738	0.353	0.324
Hill of Towie	0.5	Max. IoU	243833	0.595	0.617	0.677	0.734	0.353	0.324
Hill of Towie	0.7	Greedy	111656	0.763	0.747	0.892	0.908	0.164	0.133
Hill of Towie	0.7	Max. IoU	111656	0.763	0.747	0.892	0.908	0.164	0.133
La Haute Borne	0.3	Greedy	43890	0.638	0.632	0.707	0.728	0.430	0.416
La Haute Borne	0.3	Max. IoU	43921	0.638	0.637	0.705	0.728	0.430	0.416
La Haute Borne	0.5	Greedy	34042	0.698	0.687	0.754	0.781	0.327	0.312
La Haute Borne	0.5	Max. IoU	34055	0.697	0.688	0.753	0.781	0.327	0.312
La Haute Borne	0.7	Greedy	16527	0.858	0.849	0.899	0.914	0.162	0.139
La Haute Borne	0.7	Max. IoU	16527	0.858	0.849	0.899	0.914	0.162	0.139
Pizhou	0.3	Greedy	348190	0.582	0.615	0.672	0.731	0.473	0.419
Pizhou	0.3	Max. IoU	348609	0.578	0.612	0.667	0.729	0.474	0.419
Pizhou	0.5	Greedy	274570	0.643	0.664	0.722	0.782	0.350	0.305
Pizhou	0.5	Max. IoU	274716	0.639	0.662	0.718	0.780	0.350	0.305
Pizhou	0.7	Greedy	125508	0.814	0.807	0.892	0.914	0.161	0.124
Pizhou	0.7	Max. IoU	125508	0.814	0.807	0.892	0.914	0.161	0.124
Suining	0.3	Greedy	128928	0.548	0.583	0.650	0.712	0.462	0.430
Suining	0.3	Max. IoU	129077	0.542	0.577	0.645	0.707	0.463	0.430
Suining	0.5	Greedy	102248	0.615	0.630	0.705	0.765	0.350	0.320
Suining	0.5	Max. IoU	102293	0.612	0.628	0.700	0.762	0.350	0.320
Suining	0.7	Greedy	47316	0.794	0.798	0.899	0.918	0.163	0.132

Site	IoU	Matcher	Pairs	NMI	ARI	W. NMI	W. ARI	Left cov.	Right cov.
Suining	0.7	Max. IoU	47316	0.794	0.798	0.899	0.917	0.163	0.132
Yandun	0.3	Greedy	1082770	0.522	0.570	0.610	0.662	0.479	0.484
Yandun	0.3	Max. IoU	1085302	0.512	0.560	0.598	0.652	0.481	0.485
Yandun	0.5	Greedy	811453	0.590	0.626	0.671	0.727	0.351	0.349
Yandun	0.5	Max. IoU	812076	0.581	0.618	0.661	0.720	0.351	0.350
Yandun	0.7	Greedy	353527	0.763	0.778	0.867	0.886	0.157	0.139
Yandun	0.7	Max. IoU	353527	0.763	0.777	0.867	0.886	0.157	0.139

Max. IoU denotes maximum-total-IoU assignment. NMI and ARI are equal-configuration-pair means over pairs with matched events; W. NMI and W. ARI are weighted by matched-event counts. Left and right coverage are mean matched fractions of the respective detector catalogues across all configuration pairs.

Table S17. Yandun physical-horizon and sampling sensitivity

Minutes	Horizon (h)	Clock	Eligible anchors	Positive anchors	Positive rate
15	1	common hourly clock	196458	38174	0.194
15	1	native grid	785688	152383	0.194
15	4	common hourly clock	165650	59359	0.358
15	4	native grid	662573	237724	0.359
30	1	common hourly clock	194166	34850	0.179
30	1	native grid	388019	69862	0.180
30	4	common hourly clock	164112	57838	0.352
30	4	native grid	328086	116004	0.354
60	1	common hourly clock	188305	28732	0.153
60	1	native grid	188305	28732	0.153
60	4	common hourly clock	160194	55441	0.346
60	4	native grid	160194	55441	0.346

Table S18. Event-weight sensitivity under the legacy annual-charge assumptions

Site	Factor	Ramp weight	All nMAE (%)	Ramp nMAE (%)	Base cost
pizhou	0	1	7.738	35.449	25098.126
pizhou	1	2	7.842	34.405	25973.479
pizhou	2	3	7.974	33.759	25931.902
pizhou	4	5	8.629	32.570	30883.870
pizhou	8	9	10.284	31.794	36692.052
yandun	0	1	8.364	36.515	12661.780
yandun	1	2	8.830	35.491	14184.366
yandun	2	3	9.154	34.723	15004.582
yandun	4	5	10.084	33.212	17460.744
yandun	8	9	11.705	30.846	21481.776

Table S19. Adjusted weather association model

Records	Blocks	Risk difference	95% interval	Odds ratio	OR interval
76538	31	0.074	[0.023, 0.125]	1.394	[1.103, 1.762]

Table S20. Greek local-training and frozen-transfer comparison

Training population	Config. pairs	NMI	ARI	Agreement
Greek local	62	0.596	0.536	0.668
Pizhou frozen	62	0.679	0.662	0.767

Table S21. Same-encoder Greek frozen and adapted TCN comparison

Mode	Pairs	Equal NMI	Equal ARI	Pooled NMI	Pooled ARI	Weighted NMI	Weighted ARI
frozen	17357	0.625	0.614	0.547	0.596	0.607	0.628
target finetuned	17357	0.629	0.624	0.552	0.609	0.611	0.639

Table S22. 100-epoch upper-budget controlled detection sensitivity

Model	Protocol	Precision	Recall	F1	Optimizer steps
kanad	calibrated	0.706	0.632	0.657	1725.000
kanad	default	0.181	0.740	0.289	1725.000
mean_rule	calibrated	0.742	0.740	0.741	0.000
mean_rule	default	0.255	0.752	0.381	0.000
rate_rule	calibrated	0.439	0.435	0.437	0.000
rate_rule	default	0.056	0.210	0.088	0.000
tcn_ae	calibrated	0.583	0.397	0.465	2425.000
tcn_ae	default	0.068	0.542	0.121	2425.000
timesnet	calibrated	0.760	0.533	0.623	1841.667
timesnet	default	0.135	0.578	0.219	1841.667
transformer_ae	calibrated	0.814	0.363	0.501	1975.000
transformer_ae	default	0.050	0.499	0.091	1975.000

Table S23. Fixed 10%-power and 2-hour storage costs normalized by test-calendar hours

Site	Model	Prices	Tail premium	Calendar h	MSE CNY/MW/h	Event-weighted CNY/MW/h	Difference
pizhou	tcn	base	0.0	2496.000	11.490	13.782	2.292
pizhou	tcn	base	1000.0	2496.000	15.360	17.577	2.217
pizhou	tcn	base	5000.0	2496.000	30.840	32.685	1.845
pizhou	tcn	high	0.0	2496.000	17.521	22.169	4.649
pizhou	tcn	high	1000.0	2496.000	21.391	25.878	4.487
pizhou	tcn	high	5000.0	2496.000	36.871	40.916	4.045
pizhou	tcn	low	0.0	2496.000	8.474	9.588	1.114
pizhou	tcn	low	1000.0	2496.000	12.344	13.335	0.991
pizhou	tcn	low	5000.0	2496.000	27.824	28.390	0.566
pizhou	timesnet	base	0.0	2496.000	11.102	12.181	1.079
pizhou	timesnet	base	1000.0	2496.000	16.022	18.251	2.229
pizhou	timesnet	base	5000.0	2496.000	35.700	42.532	6.832
pizhou	timesnet	high	0.0	2496.000	16.742	18.925	2.183
pizhou	timesnet	high	1000.0	2496.000	21.661	24.995	3.334
pizhou	timesnet	high	5000.0	2496.000	41.339	49.276	7.937
pizhou	timesnet	low	0.0	2496.000	8.282	8.808	0.526
pizhou	timesnet	low	1000.0	2496.000	13.202	14.879	1.677
pizhou	timesnet	low	5000.0	2496.000	32.880	39.160	6.280
yandun	tcn	base	0.0	914.000	14.675	19.834	5.159
yandun	tcn	base	1000.0	914.000	19.627	24.137	4.511
yandun	tcn	base	5000.0	914.000	39.434	41.352	1.918
yandun	tcn	high	0.0	914.000	23.764	34.207	10.443
yandun	tcn	high	1000.0	914.000	28.716	38.511	9.795
yandun	tcn	high	5000.0	914.000	48.523	55.726	7.202
yandun	tcn	low	0.0	914.000	10.130	12.647	2.516
yandun	tcn	low	1000.0	914.000	15.082	16.951	1.868
yandun	tcn	low	5000.0	914.000	34.889	34.165	-0.724
yandun	timesnet	base	0.0	914.000	12.060	14.501	2.441
yandun	timesnet	base	1000.0	914.000	17.453	21.762	4.309
yandun	timesnet	base	5000.0	914.000	39.027	51.347	12.320
yandun	timesnet	high	0.0	914.000	18.469	23.394	4.925
yandun	timesnet	high	1000.0	914.000	23.863	30.655	6.793
yandun	timesnet	high	5000.0	914.000	45.437	60.427	14.991
yandun	timesnet	low	0.0	914.000	8.855	10.055	1.199
yandun	timesnet	low	1000.0	914.000	14.249	17.456	3.208
yandun	timesnet	low	5000.0	914.000	35.823	46.807	10.985

Table S23 compares training objectives within each site. Calendar-hour normalization makes Pizhou and Yandun exposure lengths explicit; all estimated contrasts are within-site.

Table S24. Validation-selected storage designs across model-seed-price configurations

Design regime	Power fraction	Duration (h)	Configurations
at least 10pct 2h 0.1		2.0	154
at least 10pct 2h 0.2		2.0	61
at least 10pct 2h 0.2		4.0	1
unconstrained	0.0	0.0	97
unconstrained	0.05	1.0	13
unconstrained	0.05	2.0	2
unconstrained	0.1	1.0	20
unconstrained	0.1	2.0	9
unconstrained	0.2	1.0	14
unconstrained	0.2	2.0	60
unconstrained	0.2	4.0	1

Table S25. Perfect-information and feasible online operating costs per installed MW

Site	Model	Training	Prices	Oracle bound	Online control
pizhou	tcn	event weighted	base	22113.263	22117.362
pizhou	tcn	event weighted	high	43053.035	43053.035
pizhou	tcn	mse	base	16395.128	16396.626
pizhou	tcn	mse	high	31449.965	31449.965
pizhou	timesnet	event weighted	base	18118.671	18121.488
pizhou	timesnet	event weighted	high	34955.351	34955.351
pizhou	timesnet	mse	base	15428.404	15429.524
pizhou	timesnet	mse	high	29506.511	29506.511
yandun	tcn	event weighted	base	13625.019	13630.816
yandun	tcn	event weighted	high	26768.344	26768.344
yandun	tcn	mse	base	8909.956	8915.752
yandun	tcn	mse	high	17223.279	17223.279
yandun	timesnet	event weighted	base	8751.009	8756.806
yandun	timesnet	event weighted	high	16884.979	16884.979
yandun	timesnet	mse	base	6519.700	6525.497
yandun	timesnet	mse	high	12383.639	12383.639

Table S26. Exploratory test-grid cells with lower event-weighted storage cost

Site	Model	Prices	Tail premium	Power fraction	Duration (h)	Mean difference	Seeds better
pizhou	tcn	low	5000.0	0.0	0.0	-2099.370	2/3
yandun	tcn	low	5000.0	0.0	0.0	-1480.504	3/3
pizhou	tcn	low	5000.0	0.05	1.0	-1477.668	2/3
yandun	tcn	low	5000.0	0.05	1.0	-1092.228	2/3
yandun	tcn	low	5000.0	0.05	2.0	-923.508	2/3
yandun	tcn	low	5000.0	0.1	1.0	-901.205	2/3
yandun	tcn	low	5000.0	0.05	4.0	-779.969	2/3
yandun	tcn	low	5000.0	0.1	2.0	-661.939	2/3
pizhou	tcn	low	5000.0	0.05	2.0	-636.519	2/3
pizhou	tcn	low	5000.0	0.1	1.0	-581.737	2/3
yandun	tcn	low	5000.0	0.2	2.0	-523.860	3/3
yandun	tcn	low	5000.0	0.2	1.0	-486.951	2/3
pizhou	tcn	low	5000.0	0.05	4.0	-327.066	2/3
yandun	tcn	low	5000.0	0.1	4.0	-261.756	1/3

Mean difference is event-weighted minus MSE total cost in CNY per installed MW, averaged over three seeds. These cells are a post-hoc diagnostic over the test grid; validation-selected policy comparisons appear in Table S37.

Table S27. Controlled detection compute budget and model size

Model	Seeds	Parameters	Max steps	Max epochs	Test n	CKPT bytes
Mean rule	1	0	0	0	800	0
TimesNet	3	37297	2200	88	800	479161
KAN-AD	3	9845	2225	89	800	48673
TCN-AE	3	15928	2500	100	800	68229
Transformer-AE	3	59920	2075	83	800	252208

Optimizer steps, epochs and checkpoint sizes are recorded from the upper-budget benchmark; the mean-change rule is analytic and has no fitted parameters.

Table S28. Validation-selected detection hyperparameters and held-out performance

Model	Selected size	Validation F1	Test F1 (SD)	Precision	Recall	Delay	Seeds
TimesNet	size 8	0.681	0.631 (0.024)	0.680	0.589	3.089	3
KAN-AD	size 16	0.824	0.795 (0.097)	0.829	0.765	1.870	3
TCN-AE	width 8, z 8	0.757	0.729 (0.007)	0.716	0.743	1.578	3
Trans.-AE	width 8, z 8	0.718	0.660 (0.019)	0.600	0.735	0.561	3
Mean rule	window 4	0.774	0.741	0.742	0.740	1.162	1

Trans.-AE denotes the positional Transformer autoencoder. Size denotes TimesNet d_model or KAN-AD Fourier order; width and z are encoder width and latent size for the sequence autoencoders. SD is across training seeds. Model selection uses mean validation F1; the simple rule selects its mean window on the same validation population.

Table S29. Fresh post-selection synthetic confirmation

Condition	Model	Seeds	F1 (SD)	Precision	Recall	Delay
higher noise	KAN-AD	3	0.646 (0.098)	0.545	0.796	1.746
higher noise	Mean rule	1	0.367	0.242	0.759	0.264
higher noise	TCN-AE	3	0.577 (0.011)	0.457	0.783	1.376
higher noise	TimesNet	3	0.533 (0.026)	0.547	0.520	3.069
higher noise	Transformer-AE	3	0.474 (0.015)	0.343	0.770	0.528
same family	KAN-AD	3	0.793 (0.089)	0.847	0.747	1.922
same family	Mean rule	1	0.732	0.718	0.748	1.294
same family	TCN-AE	3	0.716 (0.005)	0.702	0.731	1.430
same family	TimesNet	3	0.604 (0.025)	0.661	0.557	3.132
same family	Transformer-AE	3	0.650 (0.003)	0.583	0.735	1.070

Each condition contains 1,600 fresh sequences, including 800 episodes. Delay is in generator steps, and SD summarizes the three frozen training seeds.

Table S30. Paired F1 differences from the validation-selected mean rule

Condition	Model	F1 difference	95% interval
same family	TimesNet	-0.128	[-0.154, -0.101]
same family	KAN-AD	0.061	[0.040, 0.082]
same family	TCN-AE	-0.016	[-0.038, 0.007]
same family	Transformer-AE	-0.083	[-0.105, -0.059]
higher noise	TimesNet	0.165	[0.138, 0.195]
higher noise	KAN-AD	0.279	[0.259, 0.299]
higher noise	TCN-AE	0.210	[0.188, 0.232]
higher noise	Transformer-AE	0.107	[0.089, 0.126]

Intervals use 2,000 shared resamples of complete synthetic sequences, conditional on the frozen trained models. They characterize sampling uncertainty separately from between-seed variation.

Table S31. CPU score-generation cost of the selected configurations

Model	Parameters	Median ms/sequence	Range	Batch
TimesNet	18697	0.089122	[0.082926, 0.102622]	64
KAN-AD	16013	0.017341	[0.017034, 0.017404]	64
TCN-AE	10944	0.003181	[0.003084, 0.003642]	64
Transformer-AE	18856	0.070616	[0.070247, 0.088320]	64
Mean rule	0	0.000402	[0.000384, 0.000453]	800

Timing uses an AMD Ryzen 7 9700X under WSL2, two CPU threads, one warm-up and five runs of 800 sequences. Normalization and score generation are timed; interval postprocessing is excluded. The simple rule uses a NumPy cumulative-sum implementation.

Table S32. Matched and unmatched composition within detector-pair sides

Site	Feature	Sides used / all	Matched mean	Unmatched mean	Difference
Hill of Towie	Abs. change	249/272	0.390	0.360	0.030
Hill of Towie	Signed change	249/272	-0.016	-0.009	-0.008
Hill of Towie	Duration (h)	249/272	2.649	2.717	-0.068
Hill of Towie	Power range	249/272	0.464	0.432	0.032
Hill of Towie	Start power	249/272	0.509	0.506	0.003
Hill of Towie	Upward share	249/272	0.491	0.492	-0.001
Hill of Towie	Wind (m/s)	249/272	9.254	9.195	0.059
La Haute Borne	Abs. change	249/272	0.314	0.282	0.032
La Haute Borne	Signed change	249/272	0.010	-0.001	0.011
La Haute Borne	Duration (h)	249/272	2.598	2.655	-0.056
La Haute Borne	Power range	249/272	0.361	0.323	0.038
La Haute Borne	Start power	249/272	0.420	0.425	-0.005
La Haute Borne	Upward share	249/272	0.512	0.498	0.014
La Haute Borne	Wind (m/s)	249/272	7.748	7.822	-0.073
Pizhou	Abs. change	253/272	0.410	0.379	0.031
Pizhou	Signed change	253/272	-0.022	-0.012	-0.010
Pizhou	Duration (h)	253/272	2.754	2.737	0.017
Pizhou	Power range	253/272	0.477	0.445	0.032
Pizhou	Start power	253/272	0.549	0.545	0.004
Pizhou	Upward share	253/272	0.477	0.483	-0.006
Pizhou	Wind (m/s)	253/272	6.374	6.372	0.001
Suining	Abs. change	249/272	0.380	0.350	0.030
Suining	Signed change	249/272	-0.004	-0.000	-0.003
Suining	Duration (h)	249/272	2.623	2.668	-0.044
Suining	Power range	249/272	0.440	0.412	0.028
Suining	Start power	249/272	0.499	0.493	0.006
Suining	Upward share	249/272	0.496	0.500	-0.005
Suining	Wind (m/s)	249/272	6.308	6.225	0.083
Yandun	Abs. change	257/272	0.429	0.387	0.042
Yandun	Signed change	257/272	-0.023	0.000	-0.023
Yandun	Duration (h)	257/272	2.790	2.810	-0.020
Yandun	Power range	257/272	0.514	0.477	0.038
Yandun	Start power	257/272	0.358	0.347	0.011
Yandun	Upward share	257/272	0.471	0.495	-0.024
Yandun	Wind (m/s)	257/272	5.646	5.649	-0.003

Means give equal weight to supported detector-pair sides with available feature values in both populations. Power and amplitude use the archived normalization; direction is the fraction of upward events. Earlier wind averages four complete half-hour bins strictly before event start. The detailed companion CSV includes population sizes, valid wind counts and quartiles for every side.

Table S33. Conditional raw25 agreement and retained support

Site	Conditioning	Retained / original	Catalogue cov. L	Catalogue cov. R	Scored fraction	W. NMI	W. ARI
Hill of Towie	amplitude_abs	151847/243684	0.206	0.226	0.992	0.814	0.855
Hill of Towie	direction	239933/243684	0.326	0.358	0.999	0.275	0.335
Hill of Towie	duration_hours	131317/243684	0.178	0.196	0.998	0.837	0.853
Hill of Towie	joint_all_four	104335/243684	0.142	0.155	0.600	0.830	0.845
Hill of Towie	power_start	214671/243684	0.291	0.320	0.995	0.633	0.660
La Haute Borne	amplitude_abs	20627/34042	0.192	0.196	0.871	0.877	0.909
La Haute Borne	direction	33848/34042	0.315	0.322	0.787	0.173	0.225
La Haute Borne	duration_hours	20098/34042	0.187	0.191	0.982	0.844	0.851
La Haute Borne	joint_all_four	14783/34042	0.138	0.141	0.097	0.964	0.973
La Haute Borne	power_start	30788/34042	0.286	0.293	0.892	0.688	0.711
Pizhou	amplitude_abs	169991/274570	0.197	0.208	0.992	0.837	0.876
Pizhou	direction	271655/274570	0.315	0.333	0.998	0.289	0.363
Pizhou	duration_hours	151737/274570	0.176	0.186	0.998	0.843	0.865
Pizhou	joint_all_four	117670/274570	0.137	0.144	0.587	0.790	0.809
Pizhou	power_start	238656/274570	0.277	0.293	0.992	0.638	0.673
Suining	amplitude_abs	63990/102248	0.207	0.221	0.977	0.833	0.872
Suining	direction	100804/102248	0.325	0.349	0.994	0.296	0.359
Suining	duration_hours	56265/102248	0.182	0.195	0.995	0.846	0.864
Suining	joint_all_four	44368/102248	0.143	0.154	0.372	0.847	0.859
Suining	power_start	89315/102248	0.288	0.309	0.963	0.655	0.683
Yandun	amplitude_abs	558155/811453	0.235	0.260	0.999	0.783	0.826
Yandun	direction	791342/811453	0.333	0.369	1.000	0.295	0.356
Yandun	duration_hours	438801/811453	0.184	0.205	1.000	0.800	0.813
Yandun	joint_all_four	347316/811453	0.146	0.162	0.786	0.712	0.732
Yandun	power_start	723809/811453	0.304	0.337	0.998	0.631	0.644

Conditional raw25 scores are computed within detector-pair strata. Direction retains most temporal matches and gives lower information agreement. Amplitude and duration conditioning retain higher within-stratum agreement at lower support. Both-constant strata are excluded from informative scores, and one-side-constant strata contribute zero. Scored fractions refer to retained pairs in defined strata with at least 30 matches.

Table S34. Test-label oracle ceilings for adjacent-mean windows

Condition	Window grid	Oracle mode	Window	F1	Label access
same family	original four windows	oracle window only	4	0.732	test-label oracle
same family	original four windows	oracle window and protocol	4	0.732	test-label oracle
same family	expanded twelve windows	oracle window only	4	0.732	test-label oracle
same family	expanded twelve windows	oracle window and protocol	4	0.732	test-label oracle
higher noise	original four windows	oracle window only	16	0.501	test-label oracle
higher noise	original four windows	oracle window and protocol	4	0.588	test-label oracle
higher noise	expanded twelve windows	oracle window only	6	0.642	test-label oracle
higher noise	expanded twelve windows	oracle window and protocol	6	0.648	test-label oracle

Table S35. Equal-validation noise calibration on a fresh high-noise draw

Model	Policy	F1	F1 SD	Precision	Recall	Seeds
KAN-AD	source frozen	0.641	0.109	0.538	0.796	3
KAN-AD	target validation	0.723	0.107	0.755	0.695	3
Mean rule	source frozen	0.380	-	0.251	0.777	1
Mean rule	target validation	0.626	-	0.606	0.646	1
TCN-AE	source frozen	0.579	0.005	0.458	0.789	3
TCN-AE	target validation	0.653	0.003	0.790	0.577	3
TimesNet	source frozen	0.536	0.023	0.555	0.520	3
TimesNet	target validation	0.570	0.011	0.602	0.540	3

Model	Policy	F1	F1 SD	Precision	Recall	Seeds
Transformer-AE	source frozen	0.488	0.021	0.354	0.789	3
Transformer-AE	target validation	0.623	0.022	0.766	0.525	3

Table S36. Paired gains after source or target-noise calibration

Model	Policy	F1 gain	95% interval
TimesNet	source frozen	0.156	[0.127, 0.185]
TimesNet	target validation	-0.056	[-0.083, -0.030]
KAN-AD	source frozen	0.261	[0.241, 0.281]
KAN-AD	target validation	0.098	[0.074, 0.121]
TCN-AE	source frozen	0.199	[0.178, 0.223]
TCN-AE	target validation	0.027	[0.005, 0.049]
Transformer-AE	source frozen	0.108	[0.090, 0.127]
Transformer-AE	target validation	-0.003	[-0.026, 0.021]

Table S37. Validation-selected policy cost comparison

Site	Model	Prices	Tail	Regime	MSE CNY/MW/h	Event- weighted CNY/MW/h	Difference	Lower EW / seeds
pizhou	tcn	base	0.0	at least 10pct 2h	11.490	13.782	2.292	0/3
pizhou	tcn	base	0.0	unconstrained	10.055	11.871	1.816	0/3
pizhou	tcn	base	1000.0	at least 10pct 2h	15.360	17.577	2.217	0/3
pizhou	tcn	base	1000.0	unconstrained	15.659	17.586	1.927	0/3
pizhou	tcn	base	5000.0	at least 10pct 2h	25.529	29.576	4.047	0/3
pizhou	tcn	base	5000.0	unconstrained	25.529	29.576	4.047	0/3
pizhou	tcn	high	0.0	at least 10pct 2h	17.521	22.169	4.649	0/3
pizhou	tcn	high	0.0	unconstrained	17.797	22.057	4.261	0/3
pizhou	tcn	high	1000.0	at least 10pct 2h	21.391	25.878	4.487	0/3
pizhou	tcn	high	1000.0	unconstrained	21.689	26.365	4.676	0/3
pizhou	tcn	high	5000.0	at least 10pct 2h	29.511	35.943	6.431	0/3
pizhou	tcn	high	5000.0	unconstrained	29.511	35.943	6.431	0/3
pizhou	tcn	low	0.0	at least 10pct 2h	8.474	9.588	1.114	0/3
pizhou	tcn	low	0.0	unconstrained	5.028	5.935	0.908	0/3
pizhou	tcn	low	1000.0	at least 10pct 2h	12.344	13.335	0.991	0/3
pizhou	tcn	low	1000.0	unconstrained	11.675	12.233	0.558	0/3
pizhou	tcn	low	5000.0	at least 10pct 2h	23.538	26.807	3.269	0/3
pizhou	tcn	low	5000.0	unconstrained	23.538	28.661	5.124	0/3
pizhou	timesnet	base	0.0	at least 10pct 2h	11.102	12.181	1.079	0/3
pizhou	timesnet	base	0.0	unconstrained	9.700	10.572	0.872	0/3
pizhou	timesnet	base	1000.0	at least 10pct 2h	16.022	18.251	2.229	0/3
pizhou	timesnet	base	1000.0	unconstrained	16.269	18.443	2.174	0/3
pizhou	timesnet	base	5000.0	at least 10pct 2h	31.143	37.745	6.603	0/3
pizhou	timesnet	base	5000.0	unconstrained	31.143	37.745	6.603	0/3
pizhou	timesnet	high	0.0	at least 10pct 2h	16.742	18.925	2.183	0/3
pizhou	timesnet	high	0.0	unconstrained	17.035	19.133	2.098	0/3
pizhou	timesnet	high	1000.0	at least 10pct 2h	21.661	24.995	3.334	0/3
pizhou	timesnet	high	1000.0	unconstrained	22.008	25.372	3.364	0/3
pizhou	timesnet	high	5000.0	at least 10pct 2h	34.746	42.453	7.707	0/3

Site	Model	Prices	Tail	Regime	MSE CNY/MW/h	Event- weighted CNY/MW/h	Difference	Lower EW / seeds
pizhou	timesnet	high	5000.0	unconstrained	34.746	42.453	7.707	0/3
pizhou	timesnet	low	0.0	at least 10pct 2h	8.282	8.808	0.526	0/3
pizhou	timesnet	low	0.0	unconstrained	4.850	5.286	0.436	0/3
pizhou	timesnet	low	1000.0	at least 10pct 2h	13.202	14.879	1.677	0/3
pizhou	timesnet	low	1000.0	unconstrained	12.434	13.984	1.549	0/3
pizhou	timesnet	low	5000.0	at least 10pct 2h	29.341	35.392	6.050	0/3
pizhou	timesnet	low	5000.0	unconstrained	29.341	35.392	6.050	0/3
yandun	tcn	base	0.0	at least 10pct 2h	14.675	19.834	5.159	0/3
yandun	tcn	base	0.0	unconstrained	13.840	18.191	4.352	0/3
yandun	tcn	base	1000.0	at least 10pct 2h	19.627	24.137	4.511	0/3
yandun	tcn	base	1000.0	unconstrained	22.823	26.416	3.593	0/3
yandun	tcn	base	5000.0	at least 10pct 2h	32.400	41.352	8.952	0/3
yandun	tcn	base	5000.0	unconstrained	32.400	41.352	8.952	0/3
yandun	tcn	high	0.0	at least 10pct 2h	23.764	34.207	10.443	0/3
yandun	tcn	high	0.0	unconstrained	27.679	36.383	8.704	0/3
yandun	tcn	high	1000.0	at least 10pct 2h	28.716	38.511	9.795	0/3
yandun	tcn	high	1000.0	unconstrained	32.942	44.607	11.665	0/3
yandun	tcn	high	5000.0	at least 10pct 2h	39.148	49.955	10.807	0/3
yandun	tcn	high	5000.0	unconstrained	39.148	49.955	10.807	0/3
yandun	tcn	low	0.0	at least 10pct 2h	10.130	12.647	2.516	0/3
yandun	tcn	low	0.0	unconstrained	6.920	9.096	2.176	0/3
yandun	tcn	low	1000.0	at least 10pct 2h	15.082	16.951	1.868	0/3
yandun	tcn	low	1000.0	unconstrained	15.903	17.320	1.417	0/3
yandun	tcn	low	5000.0	at least 10pct 2h	30.986	34.165	3.179	0/3
yandun	tcn	low	5000.0	unconstrained	30.986	34.165	3.179	0/3
yandun	timesnet	base	0.0	at least 10pct 2h	12.060	14.501	2.441	0/3
yandun	timesnet	base	0.0	unconstrained	11.639	13.809	2.169	0/3
yandun	timesnet	base	1000.0	at least 10pct 2h	17.453	21.762	4.309	0/3
yandun	timesnet	base	1000.0	unconstrained	20.601	24.891	4.290	0/3
yandun	timesnet	base	5000.0	at least 10pct 2h	30.617	42.579	11.962	0/3
yandun	timesnet	base	5000.0	unconstrained	30.617	42.579	11.962	0/3
yandun	timesnet	high	0.0	at least 10pct 2h	18.469	23.394	4.925	0/3
yandun	timesnet	high	0.0	unconstrained	23.279	27.618	4.339	0/3
yandun	timesnet	high	1000.0	at least 10pct 2h	23.863	30.655	6.793	0/3
yandun	timesnet	high	1000.0	unconstrained	28.470	34.745	6.275	0/3
yandun	timesnet	high	5000.0	at least 10pct 2h	34.641	45.037	10.396	0/3
yandun	timesnet	high	5000.0	unconstrained	34.641	45.037	10.396	0/3
yandun	timesnet	low	0.0	at least 10pct 2h	8.855	10.055	1.199	0/3
yandun	timesnet	low	0.0	unconstrained	5.820	6.904	1.085	0/3
yandun	timesnet	low	1000.0	at least 10pct 2h	14.249	17.456	3.208	0/3
yandun	timesnet	low	1000.0	unconstrained	14.782	17.987	3.205	0/3
yandun	timesnet	low	5000.0	at least 10pct 2h	28.605	42.921	14.315	0/3
yandun	timesnet	low	5000.0	unconstrained	28.605	42.921	14.315	0/3

Table S38. Representative validation-selected cost decomposition

Site	Training	Prices	Tail	Capital	Throughput	Short	Surplus	Tail cost	Inventory	Total
pizhou	mse	low	0.0	4.920	0.539	2.215	0.740	0.000	-0.004	8.410
pizhou	mse	low	5000.0	9.841	0.810	1.438	0.513	11.115	-0.013	23.705
pizhou	mse	base	0.0	4.920	0.539	4.429	1.480	0.000	-0.004	11.365
pizhou	mse	base	5000.0	9.841	0.810	2.876	1.027	11.115	-0.013	25.656
pizhou	mse	high	0.0	4.920	0.539	8.858	2.959	0.000	-0.004	17.273
pizhou	mse	high	5000.0	9.841	0.810	5.753	2.054	11.115	-0.013	29.560
pizhou	event	low	0.0	4.920	0.450	3.748	0.682	0.000	0.010	9.811
pizhou	weighted	low	5000.0	9.841	0.618	3.267	0.543	11.620	0.010	25.898
	event									
pizhou	event	base	0.0	4.920	0.450	7.496	1.365	0.000	0.010	14.242
pizhou	weighted	base	5000.0	9.841	0.618	6.534	1.086	11.620	0.010	29.708
	event									
pizhou	event	high	0.0	4.920	0.450	14.993	2.730	0.000	0.010	23.103
pizhou	weighted	high	5000.0	9.841	0.680	12.352	1.965	10.838	0.011	35.687
	event									
yandun	mse	low	0.0	4.920	0.638	4.045	0.776	0.000	0.026	10.405
yandun	mse	low	5000.0	4.920	0.638	4.045	0.776	26.179	0.026	36.584
yandun	mse	base	0.0	4.920	0.638	8.089	1.551	0.000	0.026	15.225
yandun	mse	base	5000.0	9.841	0.953	6.267	1.030	16.214	0.049	34.354
yandun	mse	high	0.0	4.920	0.638	16.178	3.103	0.000	0.026	24.865
yandun	mse	high	5000.0	9.841	0.953	12.535	2.059	16.214	0.049	41.651
yandun	event	low	0.0	4.920	0.527	6.329	0.747	0.000	0.026	12.550
yandun	weighted	low	5000.0	4.920	0.527	6.329	0.747	24.167	0.026	36.717
	event									
yandun	event	base	0.0	4.920	0.527	12.659	1.494	0.000	0.026	19.627
yandun	weighted	base	5000.0	4.920	0.527	12.659	1.494	24.167	0.026	43.793
	event									
yandun	event	high	0.0	4.920	0.527	25.317	2.988	0.000	0.026	33.779
yandun	event	high	5000.0	4.920	0.527	25.317	2.988	24.167	0.026	57.946

Table S39. Real fixed 10%-power/2-hour online and LP traces

Site	Training	Scenario	Calendar h	Online CNY/MW/h	LP CNY/MW/h	LP minus online	Online severe steps	LP severe steps
pizhou	mse	base	2496.000	11.365	11.364	-0.001	184	188
pizhou	mse	high	2496.000	17.273	17.273	0.000	184	190
pizhou	mse	low_tail5000	2496.000	27.560	21.927	-5.633	184	118
pizhou	event weighted	base	2496.000	14.242	14.239	-0.002	198	220
pizhou	event weighted	high	2496.000	23.103	23.103	0.000	198	220
pizhou	event weighted	low_tail5000	2496.000	26.945	21.215	-5.730	198	107
yandun	mse	base	914.000	15.225	15.219	-0.006	84	84
yandun	mse	high	914.000	24.865	24.865	0.000	84	84
yandun	mse	low_tail5000	914.000	36.584	30.776	-5.808	84	67
yandun	event weighted	base	914.000	19.627	19.620	-0.006	84	90
yandun	event weighted	high	914.000	33.779	33.779	0.000	84	90
yandun	event weighted	low_tail5000	914.000	36.717	31.327	-5.390	84	66

Table S40. Weather association model diagnostics

Observations	Blocks	IRLS iterations	Converged	Exposure estimate	Cluster SE	Block-t p	BH q
76538	31	22	True	0.332	0.115	0.007	0.023

Table S41. All neural model-size validation and test curves

Model	Configuration	Validation F1	Validation SD	Test F1	Test SD
TimesNet	h08_z08	0.681	0.033	0.631	0.024
TimesNet	h16_z16	0.662	0.013	0.609	0.014
TimesNet	h32_z16	0.632	0.008	0.615	0.007
KAN-AD	h08_z08	0.737	0.085	0.710	0.096
KAN-AD	h16_z16	0.824	0.065	0.795	0.097
KAN-AD	h32_z16	0.796	0.018	0.751	0.015
TCN-AE	sequence_h08_z08	0.757	0.009	0.729	0.007
TCN-AE	sequence_h16_z16	0.660	0.006	0.618	0.012
TCN-AE	sequence_h32_z16	0.647	0.022	0.601	0.021
Transformer-AE	sequence_h08_z08	0.718	0.021	0.660	0.019
Transformer-AE	sequence_h16_z16	0.634	0.002	0.587	0.009
Transformer-AE	sequence_h32_z16	0.609	0.015	0.560	0.012

Table S42. All adjacent-mean candidate curves

Condition	Window	Validation F1	Frozen test F1	Oracle test F1
same family	1	0.492	0.448	0.448
same family	2	0.627	0.563	0.566
same family	3	0.678	0.639	0.644
same family	4	0.774	0.732	0.732
same family	5	0.763	0.729	0.729
same family	6	0.719	0.698	0.698
same family	8	0.701	0.694	0.720
same family	10	0.628	0.621	0.628
same family	12	0.596	0.596	0.596
same family	16	0.484	0.522	0.522
same family	24	0.208	0.255	0.255
same family	32	0.130	0.170	0.170
higher noise	1	0.492	0.158	0.295
higher noise	2	0.627	0.243	0.438
higher noise	3	0.678	0.221	0.517
higher noise	4	0.774	0.367	0.588
higher noise	5	0.763	0.558	0.621
higher noise	6	0.719	0.642	0.648
higher noise	8	0.701	0.300	0.563
higher noise	10	0.628	0.425	0.522
higher noise	12	0.596	0.504	0.539
higher noise	16	0.484	0.501	0.501
higher noise	24	0.208	0.282	0.284
higher noise	32	0.130	0.165	0.165

All candidate rows in S41 and S42 remain available in CSV form with their full selection settings. Oracle rows use test labels and provide an upper-bound diagnostic. Deployable selection uses validation labels. S37 uses validation-selected storage designs; S26 remains a post-hoc test-grid diagnostic. Cost components are in CNY per installed MW per test-calendar hour; unrounded values remain in the source CSV.

Table S43. Yandun raw25 agreement across native and aggregated resolutions

Left min	Right min	Left test	Right test	Pairs	Left cov.	Right cov.	NMI	ARI
15	30	19635	12328	10828	0.551	0.878	0.326	0.354
15	60	19635	7169	5126	0.261	0.715	0.257	0.257
30	60	12328	7169	6087	0.494	0.849	0.339	0.368

Each resolution fits its own training standardizer and k=4 prototypes. Test intervals match within turbine and split at IoU 0.5. The source is Yandun’s native 15-min SCADA; 30- and 60-min series use complete arithmetic means and preserve missing-bin boundaries.

Table S44. Received human-reference morphology by archive

Three observers; central two-hour regions displayed with four hours of context. Observer 1 = R_147f4682; Observer 2 = R_29d247dd; Observer 3 = R_2c332025. V and inverted V are listed separately; quiet and sustained-low are presence-negative.

Observer	Archive	Rise	Fall	V	Inv. V	Osc.	Quiet	Low
Observer 1	greece	14	7	2	1	0	11	5
Observer 1	hill	4	5	8	6	2	9	6
Observer 1	lahaute	6	8	3	2	1	10	10
Observer 1	pizhou	12	13	4	0	1	10	0
Observer 1	sdwpg	13	7	2	2	1	9	6
Observer 1	smarteole	8	10	0	2	2	8	10
Observer 1	suining	9	8	2	4	2	13	2
Observer 1	yandun	9	7	0	5	2	5	12
Observer 2	greece	11	5	3	7	6	8	0
Observer 2	hill	1	5	11	9	7	6	1
Observer 2	lahaute	3	4	10	8	10	4	1
Observer 2	pizhou	8	7	8	7	7	3	0
Observer 2	sdwpg	9	5	6	5	11	3	1
Observer 2	smarteole	4	4	5	8	8	7	4
Observer 2	suining	4	6	8	11	8	3	0
Observer 2	yandun	6	3	1	9	9	8	4
Observer 3	greece	14	5	2	1	1	16	1
Observer 3	hill	0	7	9	5	6	12	1
Observer 3	lahaute	5	5	4	2	5	17	2
Observer 3	pizhou	12	10	2	1	6	9	0
Observer 3	sdwpg	10	3	4	1	5	17	0
Observer 3	smarteole	9	9	1	2	3	8	8
Observer 3	suining	5	8	5	6	3	13	0
Observer 3	yandun	9	3	0	3	5	12	8

Table S45. Event-presence performance of all fixed detectors on the three-observer reference

Presence scoring uses 320 selected windows and the same observer-specific labels as the reference packet. Point estimates are shown; block intervals are in Table S46. The prespecified 120-min threshold is retained as the primary detector.

Observer	Detector	Precision	Recall	F1
Observer 1	Adaptive corridor	0.825	0.680	0.746
Observer 1	Endpoint corridor 0.025	0.914	0.546	0.684
Observer 1	Endpoint corridor 0.050	0.858	0.686	0.762
Observer 1	Endpoint corridor 0.100	0.858	0.809	0.833
Observer 1	Financial tail 120 min	0.825	0.268	0.405
Observer 1	Financial tail 240 min	0.713	0.294	0.416
Observer 1	Financial tail 60 min	0.825	0.242	0.375
Observer 1	Mean shift 120 min	0.757	0.691	0.722
Observer 1	Mean shift 240 min	0.716	0.727	0.721
Observer 1	Mean shift 60 min	0.880	0.567	0.690
Observer 1	OpSDA 120 min	0.867	0.603	0.711
Observer 1	OpSDA 240 min	0.804	0.825	0.814
Observer 1	OpSDA 60 min	0.931	0.278	0.429
Observer 1	SDA 0.025	0.877	0.552	0.677
Observer 1	Threshold 120 min	0.781	0.866	0.822
Observer 1	Threshold 240 min	0.700	0.938	0.802
Observer 1	Threshold 60 min	0.897	0.675	0.771
Observer 2	Adaptive corridor	0.963	0.577	0.721
Observer 2	Endpoint corridor 0.025	1.000	0.434	0.606
Observer 2	Endpoint corridor 0.050	0.968	0.562	0.711
Observer 2	Endpoint corridor 0.100	0.967	0.663	0.787
Observer 2	Financial tail 120 min	0.921	0.217	0.352
Observer 2	Financial tail 240 min	0.875	0.262	0.403
Observer 2	Financial tail 60 min	0.965	0.206	0.340
Observer 2	Mean shift 120 min	0.915	0.607	0.730
Observer 2	Mean shift 240 min	0.893	0.659	0.759
Observer 2	Mean shift 60 min	0.984	0.461	0.628
Observer 2	OpSDA 120 min	0.978	0.494	0.657

Observer	Detector	Precision	Recall	F1
Observer 2	OpSDA 240 min	0.950	0.708	0.811
Observer 2	OpSDA 60 min	1.000	0.217	0.357
Observer 2	SDA 0.025	0.967	0.442	0.607
Observer 2	Threshold 120 min	0.940	0.757	0.838
Observer 2	Threshold 240 min	0.919	0.895	0.907
Observer 2	Threshold 60 min	1.000	0.547	0.707
Observer 3	Adaptive corridor	0.844	0.689	0.758
Observer 3	Endpoint corridor 0.025	0.948	0.561	0.705
Observer 3	Endpoint corridor 0.050	0.871	0.689	0.769
Observer 3	Endpoint corridor 0.100	0.891	0.832	0.860
Observer 3	Financial tail 120 min	0.778	0.250	0.378
Observer 3	Financial tail 240 min	0.662	0.270	0.384
Observer 3	Financial tail 60 min	0.877	0.255	0.395
Observer 3	Mean shift 120 min	0.740	0.668	0.702
Observer 3	Mean shift 240 min	0.701	0.704	0.702
Observer 3	Mean shift 60 min	0.880	0.561	0.685
Observer 3	OpSDA 120 min	0.859	0.592	0.701
Observer 3	OpSDA 240 min	0.794	0.806	0.800
Observer 3	OpSDA 60 min	0.966	0.286	0.441
Observer 3	SDA 0.025	0.893	0.556	0.686
Observer 3	Threshold 120 min	0.772	0.847	0.808
Observer 3	Threshold 240 min	0.704	0.934	0.803
Observer 3	Threshold 60 min	0.918	0.684	0.784

Table S46. Three-observer detector F1 sensitivity to calendar-block length

All 17 configurations and all three observers are retained. Each interval uses 2,000 shared calendar-block resamples within farm.

Observer	Detector	Days	Farm-blocks	F1 interval
Observer 1	Adaptive corridor	3	248	0.693–0.796
Observer 1	Adaptive corridor	7	197	0.690–0.794
Observer 1	Adaptive corridor	14	140	0.693–0.793
Observer 1	Endpoint corridor 0.025	3	248	0.623–0.743
Observer 1	Endpoint corridor 0.025	7	197	0.625–0.737
Observer 1	Endpoint corridor 0.025	14	140	0.625–0.741
Observer 1	Endpoint corridor 0.050	3	248	0.708–0.812
Observer 1	Endpoint corridor 0.050	7	197	0.711–0.809
Observer 1	Endpoint corridor 0.050	14	140	0.709–0.809
Observer 1	Endpoint corridor 0.100	3	248	0.791–0.871
Observer 1	Endpoint corridor 0.100	7	197	0.791–0.872
Observer 1	Endpoint corridor 0.100	14	140	0.791–0.873
Observer 1	Financial tail 120 min	3	248	0.318–0.480
Observer 1	Financial tail 120 min	7	197	0.321–0.483
Observer 1	Financial tail 120 min	14	140	0.309–0.498
Observer 1	Financial tail 240 min	3	248	0.335–0.488
Observer 1	Financial tail 240 min	7	197	0.338–0.488
Observer 1	Financial tail 240 min	14	140	0.331–0.493
Observer 1	Financial tail 60 min	3	248	0.288–0.452
Observer 1	Financial tail 60 min	7	197	0.291–0.450
Observer 1	Financial tail 60 min	14	140	0.283–0.455
Observer 1	Mean shift 120 min	3	248	0.667–0.774
Observer 1	Mean shift 120 min	7	197	0.671–0.771
Observer 1	Mean shift 120 min	14	140	0.670–0.772
Observer 1	Mean shift 240 min	3	248	0.672–0.770
Observer 1	Mean shift 240 min	7	197	0.668–0.768
Observer 1	Mean shift 240 min	14	140	0.674–0.768
Observer 1	Mean shift 60 min	3	248	0.633–0.744
Observer 1	Mean shift 60 min	7	197	0.631–0.742
Observer 1	Mean shift 60 min	14	140	0.632–0.743
Observer 1	OpSDA 120 min	3	248	0.659–0.762
Observer 1	OpSDA 120 min	7	197	0.660–0.757
Observer 1	OpSDA 120 min	14	140	0.665–0.754
Observer 1	OpSDA 240 min	3	248	0.773–0.854
Observer 1	OpSDA 240 min	7	197	0.774–0.852
Observer 1	OpSDA 240 min	14	140	0.774–0.849
Observer 1	OpSDA 60 min	3	248	0.356–0.496
Observer 1	OpSDA 60 min	7	197	0.352–0.494

Observer	Detector	Days	Farm-blocks	F1 interval
Observer 1	OpSDA 60 min	14	140	0.351–0.498
Observer 1	SDA 0.025	3	248	0.621–0.732
Observer 1	SDA 0.025	7	197	0.621–0.726
Observer 1	SDA 0.025	14	140	0.622–0.726
Observer 1	Threshold 120 min	3	248	0.779–0.859
Observer 1	Threshold 120 min	7	197	0.781–0.857
Observer 1	Threshold 120 min	14	140	0.784–0.855
Observer 1	Threshold 240 min	3	248	0.762–0.839
Observer 1	Threshold 240 min	7	197	0.763–0.838
Observer 1	Threshold 240 min	14	140	0.764–0.839
Observer 1	Threshold 60 min	3	248	0.723–0.815
Observer 1	Threshold 60 min	7	197	0.720–0.816
Observer 1	Threshold 60 min	14	140	0.720–0.816
Observer 3	Adaptive corridor	3	248	0.706–0.809
Observer 3	Adaptive corridor	7	197	0.704–0.805
Observer 3	Adaptive corridor	14	140	0.706–0.810
Observer 3	Endpoint corridor 0.025	3	248	0.649–0.763
Observer 3	Endpoint corridor 0.025	7	197	0.649–0.756
Observer 3	Endpoint corridor 0.025	14	140	0.642–0.758
Observer 3	Endpoint corridor 0.050	3	248	0.717–0.817
Observer 3	Endpoint corridor 0.050	7	197	0.720–0.813
Observer 3	Endpoint corridor 0.050	14	140	0.715–0.818
Observer 3	Endpoint corridor 0.100	3	248	0.823–0.897
Observer 3	Endpoint corridor 0.100	7	197	0.820–0.896
Observer 3	Endpoint corridor 0.100	14	140	0.821–0.899
Observer 3	Financial tail 120 min	3	248	0.293–0.449
Observer 3	Financial tail 120 min	7	197	0.294–0.450
Observer 3	Financial tail 120 min	14	140	0.285–0.471
Observer 3	Financial tail 240 min	3	248	0.307–0.451
Observer 3	Financial tail 240 min	7	197	0.308–0.454
Observer 3	Financial tail 240 min	14	140	0.298–0.465
Observer 3	Financial tail 60 min	3	248	0.312–0.471
Observer 3	Financial tail 60 min	7	197	0.313–0.464
Observer 3	Financial tail 60 min	14	140	0.308–0.479
Observer 3	Mean shift 120 min	3	248	0.645–0.753
Observer 3	Mean shift 120 min	7	197	0.643–0.754
Observer 3	Mean shift 120 min	14	140	0.646–0.755
Observer 3	Mean shift 240 min	3	248	0.650–0.753
Observer 3	Mean shift 240 min	7	197	0.646–0.752
Observer 3	Mean shift 240 min	14	140	0.649–0.750
Observer 3	Mean shift 60 min	3	248	0.629–0.739
Observer 3	Mean shift 60 min	7	197	0.626–0.740
Observer 3	Mean shift 60 min	14	140	0.623–0.743
Observer 3	OpSDA 120 min	3	248	0.644–0.751
Observer 3	OpSDA 120 min	7	197	0.646–0.751
Observer 3	OpSDA 120 min	14	140	0.642–0.755
Observer 3	OpSDA 240 min	3	248	0.755–0.843
Observer 3	OpSDA 240 min	7	197	0.752–0.840
Observer 3	OpSDA 240 min	14	140	0.753–0.843
Observer 3	OpSDA 60 min	3	248	0.368–0.506
Observer 3	OpSDA 60 min	7	197	0.364–0.513
Observer 3	OpSDA 60 min	14	140	0.360–0.515
Observer 3	SDA 0.025	3	248	0.627–0.738
Observer 3	SDA 0.025	7	197	0.629–0.737
Observer 3	SDA 0.025	14	140	0.627–0.738
Observer 3	Threshold 120 min	3	248	0.766–0.850
Observer 3	Threshold 120 min	7	197	0.765–0.849
Observer 3	Threshold 120 min	14	140	0.767–0.847
Observer 3	Threshold 240 min	3	248	0.759–0.841
Observer 3	Threshold 240 min	7	197	0.759–0.841
Observer 3	Threshold 240 min	14	140	0.764–0.839
Observer 3	Threshold 60 min	3	248	0.734–0.829
Observer 3	Threshold 60 min	7	197	0.734–0.827
Observer 3	Threshold 60 min	14	140	0.732–0.830

Table S47. Prespecified detector by archive

Threshold at 120 min; all forty reviewed windows per archive are retained.

Observer	Archive	N	TP	FP	FN	Precision	Recall	F1
Observer 1	greece	40	22	3	2	0.880	0.917	0.898
Observer 1	hill	40	24	7	1	0.774	0.960	0.857
Observer 1	lahaute	40	15	6	5	0.714	0.750	0.732
Observer 1	pizhou	40	25	5	5	0.833	0.833	0.833
Observer 1	sdwfpf	40	23	10	2	0.697	0.920	0.793
Observer 1	smarteole	40	15	1	7	0.938	0.682	0.789
Observer 1	suining	40	22	8	3	0.733	0.880	0.800
Observer 1	yandun	40	22	7	1	0.759	0.957	0.846
Observer 2	greece	40	23	2	9	0.920	0.719	0.807
Observer 2	hill	40	30	1	3	0.968	0.909	0.938
Observer 2	lahaute	40	20	1	15	0.952	0.571	0.714
Observer 2	pizhou	40	30	0	7	1.000	0.811	0.896
Observer 2	sdwfpf	40	30	3	6	0.909	0.833	0.870
Observer 2	smarteole	40	15	1	14	0.938	0.517	0.667
Observer 2	suining	40	28	2	9	0.933	0.757	0.836
Observer 2	yandun	40	26	3	2	0.897	0.929	0.912
Observer 3	greece	40	21	4	2	0.840	0.913	0.875
Observer 3	hill	40	25	6	2	0.806	0.926	0.862
Observer 3	lahaute	40	16	5	5	0.762	0.762	0.762
Observer 3	pizhou	40	27	3	4	0.900	0.871	0.885
Observer 3	sdwfpf	40	20	13	3	0.606	0.870	0.714
Observer 3	smarteole	40	15	1	9	0.938	0.625	0.750
Observer 3	suining	40	23	7	4	0.767	0.852	0.807
Observer 3	yandun	40	19	10	1	0.655	0.950	0.776

Table S48. Independent partition-metric sensitivity

Pizhou-frozen K-means, k=4, seed 41; three representations on common supported basic-event test pairs. Higher AMI, Fowlkes–Mallows and NMI, and lower VI indicate closer partitions. These are single-seed sensitivity summaries, distinct from the primary three-seed common-support table.

Archive	Representation	Pairs	AMI	FM	VI (nats)	NMI
greece	gaf bit6	66	0.195	0.506	1.745	0.209
greece	gaf pca6	66	0.192	0.503	1.752	0.208
greece	raw25	66	0.862	0.956	0.213	0.864
hill	gaf bit6	105	0.202	0.451	1.976	0.205
hill	gaf pca6	105	0.204	0.451	1.982	0.206
hill	raw25	105	0.751	0.900	0.412	0.751
hill_2021	gaf bit6	112	0.206	0.452	1.970	0.208
hill_2021	gaf pca6	112	0.207	0.451	1.980	0.209
hill_2021	raw25	112	0.763	0.913	0.399	0.763
lahaute	gaf bit6	90	0.208	0.458	1.899	0.222
lahaute	gaf pca6	90	0.210	0.456	1.901	0.218
lahaute	raw25	90	0.886	0.962	0.181	0.887
pizhou	gaf bit6	105	0.216	0.480	1.878	0.217
pizhou	gaf pca6	105	0.216	0.480	1.880	0.218
pizhou	raw25	105	0.828	0.944	0.276	0.828
sdwfpf	gaf bit6	119	0.227	0.468	1.892	0.227
sdwfpf	gaf pca6	119	0.223	0.468	1.903	0.224
sdwfpf	raw25	119	0.809	0.939	0.309	0.810
suining	gaf bit6	98	0.230	0.461	1.903	0.233
suining	gaf pca6	98	0.227	0.462	1.928	0.234
suining	raw25	98	0.808	0.941	0.301	0.810
yandun	gaf bit6	119	0.211	0.445	1.989	0.213
yandun	gaf pca6	119	0.215	0.448	1.995	0.217
yandun	raw25	119	0.734	0.897	0.475	0.734

Table S49. Signed-weather log-loss decomposition

Occurrence and probability-weighted conditional-direction components sum to the three-class log loss. The conditional-direction contribution is averaged over all evaluated events. Lower scores are better.

Archive	Input	Events	Occurrence	Direction	Total
hill	standalone gaf pca6	79436	0.3066	0.0727	0.3793

Archive Input		Events	Occurrence	Direction	Total
hill	standalone gaf bit6	79436	0.3111	0.0340	0.3451
hill	standalone raw pca6	79436	0.3028	0.0231	0.3259
hill	controlled raw25	79436	0.2594	0.0220	0.2813
hill	scalar controls5	79436	0.2601	0.0310	0.2911
lahaute	standalone gaf pca6	12534	0.2870	0.0537	0.3407
lahaute	standalone gaf bit6	12534	0.2856	0.0412	0.3269
lahaute	standalone raw pca6	12534	0.2834	0.0391	0.3225
lahaute	controlled raw25	12534	0.2615	0.0376	0.2991
lahaute	scalar controls5	12534	0.2680	0.0448	0.3128
pizhou	standalone gaf pca6	94534	0.2250	0.0452	0.2702
pizhou	standalone gaf bit6	94534	0.2232	0.0234	0.2466
pizhou	standalone raw pca6	94534	0.2203	0.0207	0.2410
pizhou	controlled raw25	94534	0.1867	0.0203	0.2070
pizhou	scalar controls5	94534	0.1915	0.0214	0.2129
suining	standalone gaf pca6	33617	0.3171	0.0698	0.3869
suining	standalone gaf bit6	33617	0.3199	0.0534	0.3733
suining	standalone raw pca6	33617	0.3138	0.0546	0.3685
suining	controlled raw25	33617	0.2859	0.0543	0.3402
suining	scalar controls5	33617	0.2834	0.0599	0.3433
yandun	standalone gaf pca6	238131	0.5553	0.1775	0.7328
yandun	standalone gaf bit6	238131	0.5567	0.1717	0.7284
yandun	standalone raw pca6	238131	0.5326	0.1680	0.7006
yandun	controlled raw25	238131	0.4879	0.1743	0.6622
yandun	scalar controls5	238131	0.4982	0.1837	0.6820

Table S50. Chronologically calibrated LiDAR direction discrimination

AUROC uses the increase/decrease subset (705 and 285 events); the complete three-class test has 1,064 and 444 events. Intervals use 2,000 seven-day block draws. All three reported feature sets are retained.

Turbine Input		Events	Blocks	AUROC	95% interval
T11	standalone gaf pca6	705	4	0.494	0.468–0.544
T11	standalone gaf bit6	705	4	0.890	0.822–0.974
T11	scalar controls5	705	4	0.858	0.815–0.928
T07	standalone gaf pca6	285	4	0.477	0.321–0.599
T07	standalone gaf bit6	285	4	0.905	0.808–0.947
T07	scalar controls5	285	4	0.887	0.847–0.964

Table S51. All forecast variants under observed Elexon prices

Each forecast supplies its own scheduled position in this exposure comparison. Gross debits sum positive interval costs; signed cashflow costs include credits. Monetary values are GBP; contractual revenue and capital are separate.

Lead (h)	Variant	Intervals	nMAE	Gross debit	Signed cost
1	persistence	2780	11.04%	166,615	10,748
1	power calendar	2780	11.65%	223,715	101,168
1	observed weather	2780	11.69%	223,348	99,634
1	historical events	2780	11.95%	230,987	106,175
1	predicted ramp	2780	11.49%	211,164	80,727
	observed weather				
1	predicted ramp	2780	11.53%	211,837	80,228
	historical events				
2	persistence	2768	13.66%	206,169	14,447
2	power calendar	2768	14.95%	312,174	173,238
2	observed weather	2768	14.61%	294,259	152,092
2	historical events	2768	14.62%	294,141	151,947
2	predicted ramp	2768	14.69%	292,263	144,143
	observed weather				
2	predicted ramp	2768	14.47%	279,178	128,456
	historical events				
4	persistence	2746	17.53%	262,275	19,510
4	power calendar	2746	19.23%	428,284	264,869
4	observed weather	2746	19.28%	427,257	263,841

Lead (h)	Variant	Intervals	nMAE	Gross debit	Signed cost
4	historical events	2746	19.05%	421,364	260,805
4	predicted ramp	2746	19.34%	430,049	267,500
4	observed weather				
	predicted ramp	2746	19.29%	427,108	264,663
	historical events				

Table S52. Common-commitment capacity and operating-cost decomposition

All arms use the same one-hour persistence commitment over 3,504 test half-hours when the issued forecast is available; unavailable-forecast intervals use the controller's zero-correction fallback. Future-outcome completeness is excluded from action readiness. Gain is relative to zero storage; net operating gain subtracts throughput times the stated cycling-cost sensitivity. Capital is excluded. Validation selects zero power for all six information variants and all three cost sensitivities. Negative gains at positive capacities explain that decision under this controller.

Variant	Power fraction	GBP/MWh	Cashflow gain	MWh	Net gain
historical events	0%	0	0	0.0	0
historical events	0%	10	0	0.0	0
historical events	0%	30	0	0.0	0
historical events	5%	0	-2,799	727.5	-2,799
historical events	5%	10	-2,799	727.5	-10,074
historical events	5%	30	-2,799	727.5	-24,625
historical events	10%	0	-4,903	1,226.2	-4,903
historical events	10%	10	-4,903	1,226.2	-17,165
historical events	10%	30	-4,903	1,226.2	-41,689
historical events	20%	0	-7,630	1,952.0	-7,630
historical events	20%	10	-7,630	1,952.0	-27,150
historical events	20%	30	-7,630	1,952.0	-66,191
observed weather	0%	0	0	0.0	0
observed weather	0%	10	0	0.0	0
observed weather	0%	30	0	0.0	0
observed weather	5%	0	-2,696	748.4	-2,696
observed weather	5%	10	-2,696	748.4	-10,180
observed weather	5%	30	-2,696	748.4	-25,147
observed weather	10%	0	-4,067	1,209.1	-4,067
observed weather	10%	10	-4,067	1,209.1	-16,158
observed weather	10%	30	-4,067	1,209.1	-40,341
observed weather	20%	0	-7,292	1,890.5	-7,292
observed weather	20%	10	-7,292	1,890.5	-26,197
observed weather	20%	30	-7,292	1,890.5	-64,008
persistence	0%	0	0	0.0	0
persistence	0%	10	0	0.0	0
persistence	0%	30	0	0.0	0
persistence	5%	0	0	0.0	0
persistence	5%	10	0	0.0	0
persistence	5%	30	0	0.0	0
persistence	10%	0	0	0.0	0
persistence	10%	10	0	0.0	0
persistence	10%	30	0	0.0	0
persistence	20%	0	0	0.0	0
persistence	20%	10	0	0.0	0
persistence	20%	30	0	0.0	0
power calendar	0%	0	0	0.0	0
power calendar	0%	10	0	0.0	0
power calendar	0%	30	0	0.0	0
power calendar	5%	0	-2,637	756.0	-2,637
power calendar	5%	10	-2,637	756.0	-10,197
power calendar	5%	30	-2,637	756.0	-25,316
power calendar	10%	0	-4,590	1,220.6	-4,590
power calendar	10%	10	-4,590	1,220.6	-16,796
power calendar	10%	30	-4,590	1,220.6	-41,209
power calendar	20%	0	-7,697	1,874.9	-7,697
power calendar	20%	10	-7,697	1,874.9	-26,445
power calendar	20%	30	-7,697	1,874.9	-63,942
predicted ramp	0%	0	0	0.0	0
historical events					

Variant	Power fraction	GBP/MWh	Cashflow gain	MWh	Net gain
predicted ramp	0%	10	0	0.0	0
historical events					
predicted ramp	0%	30	0	0.0	0
historical events					
predicted ramp	5%	0	-2,910	754.0	-2,910
historical events					
predicted ramp	5%	10	-2,910	754.0	-10,449
historical events					
predicted ramp	5%	30	-2,910	754.0	-25,529
historical events					
predicted ramp	10%	0	-5,354	1,211.2	-5,354
historical events					
predicted ramp	10%	10	-5,354	1,211.2	-17,467
historical events					
predicted ramp	10%	30	-5,354	1,211.2	-41,692
historical events					
predicted ramp	20%	0	-8,411	1,848.4	-8,411
historical events					
predicted ramp	20%	10	-8,411	1,848.4	-26,894
historical events					
predicted ramp	20%	30	-8,411	1,848.4	-63,862
historical events					
predicted ramp	0%	0	0	0.0	0
observed weather					
predicted ramp	0%	10	0	0.0	0
observed weather					
predicted ramp	0%	30	0	0.0	0
observed weather					
predicted ramp	5%	0	-3,068	745.7	-3,068
observed weather					
predicted ramp	5%	10	-3,068	745.7	-10,526
observed weather					
predicted ramp	5%	30	-3,068	745.7	-25,440
observed weather					
predicted ramp	10%	0	-5,208	1,194.5	-5,208
observed weather					
predicted ramp	10%	10	-5,208	1,194.5	-17,153
observed weather					
predicted ramp	10%	30	-5,208	1,194.5	-41,044
observed weather					
predicted ramp	20%	0	-7,679	1,831.7	-7,679
observed weather					
predicted ramp	20%	10	-7,679	1,831.7	-25,995
observed weather					
predicted ramp	20%	30	-7,679	1,831.7	-62,629
observed weather					

Table S53. Detector-perturbation axes

Every row crosses seven sites (Pizhou, Suining, Yandun, La Haute Borne, Hill of Towie, Greece and SDWPF) and lags of 60, 120 and 240 min. The seven perturbations therefore define 147 comparisons.

Family	Reference	Perturbed values	Comparisons
Amplitude threshold	0.20	0.10, 0.15, 0.25, 0.30	84
Financial tail probability	0.05	0.01, 0.025, 0.10	63

Table S54. Independent human agreement before adjudication

320 paired windows, three actual observers. Nominal Krippendorff alpha and shared-calendar intervals preserve all ratings per window. The earlier 120-region reference is a separate cohort. One further review is ongoing; only received responses enter this table.

Field	Block days	Agreement	Alpha	95% interval
event presence	3	0.796	0.528	0.457–0.597

Field	Block days	Agreement	Alpha	95% interval
event presence	7	0.796	0.528	0.455–0.599
event presence	14	0.796	0.528	0.456–0.602
morphology	3	0.559	0.476	0.426–0.519
morphology	7	0.559	0.476	0.426–0.523
morphology	14	0.559	0.476	0.434–0.518

Table S55. Jurisdiction-specific settlement requirements used to design transfer simulations

These are policy inputs and simulation requirements, not station invoices. Jiangsu specifies 15-min submitted curves, 96-point medium-short-term assessment, a monthly allowance for unqualified points and rated-capacity point charges. RTE specifies half-hourly imbalance settlement by balance-responsible-party imbalance sign, system trend, VWAP and coefficient k. S22 now evaluates the Chinese ultra-short-term accuracy component on native quarter-hour points. EMS/submission records are needed for actual invoices; aligned French historical prices remain a separate requirement.

Jurisdiction	Policy quantity	Requirement used in simulation design	Current evidence
Jiangsu	Medium-short forecast	15-min curve; 96-point assessment; next-day $\geq 90\%$, day-10 $\geq 70\%$	Official Jiangsu No. 53 (2022)
Jiangsu	Assessment allowance	Monthly unqualified-point allowance of 2%; point charge uses rated capacity	Official clause; station EMS missing
Jiangsu	Ultra-short accuracy	97%/87% point accuracy at 15 min/4 h; CNY 4 per 10 MW per failed point	Native rule simulation in S22; no medium/short-term allowance
France	BRP imbalance	Positive/negative sign, trend, VWAP and k determine interval price	RTE official rule; historical price alignment pending

Table S56. Decision-aligned forecast extension at the two-hour horizon

The selected candidates use validation-only model/loss/blend selection. The test calendar is reused from v19 and is therefore exploratory. Gross-debit intervals compare selected event-aware training with the legacy ramp-mixture reference using shared 7-day farm blocks.

Variant	nMAE	Gross debit (GBP)	Ramp nMAE
persistence	13.66%	206,169	34.69%
predicted ramp observed weather	14.69%	292,263	30.20%
predicted ramp historical events	14.47%	279,178	30.34%
selected weather	13.61%	225,532	32.45%
selected weather events	13.69%	233,084	32.33%
selected cost weather	13.61%	225,532	32.45%
selected cost weather events	13.69%	233,084	32.33%

Selected event-aware versus legacy ramp-mixture: gross-debit reduction 16.51% (7-day interval 11.39–22.31%); nMAE reduction 0.79 percentage points (0.40–1.22).

Table S57. Cluster-count sensitivity on frozen test populations

K-means, Pizhou training, three seeds. Each k uses the intersection of supported informative configuration pairs across the available representations and seeds. N denotes that intersection size. Six-representation and raw-only analyses are labelled separately; differing support sets across k must be retained when comparing medians.

Population	k	Representations in support intersection	N	Raw25 ARI	Statistics9 ARI	GAF/PCA6 ARI
greece	2	6	24	0.961	0.456	0.135
greece	4	6	66	0.817	0.599	0.186
greece	6	6	66	0.735	0.835	0.294
hill	2	1	105	0.898	—	—
hill	4	6	104	0.717	0.488	0.209
hill	6	1	105	0.602	—	—
hill_2021	2	6	91	0.904	0.233	0.144
hill_2021	4	6	112	0.711	0.492	0.206
hill_2021	6	6	112	0.587	0.651	0.252
lahaute	2	1	90	0.977	—	—
lahaute	4	6	89	0.785	0.579	0.195
lahaute	6	1	90	0.686	—	—
pizhou	2	6	63	0.930	0.324	0.147
pizhou	4	6	105	0.747	0.501	0.227
pizhou	6	6	105	0.653	0.643	0.284
sdwvf	2	1	119	0.940	—	—
sdwvf	4	6	119	0.745	0.501	0.220
sdwvf	6	1	119	0.605	—	—
suining	2	1	98	0.927	—	—
suining	4	6	98	0.724	0.515	0.223
suining	6	1	98	0.639	—	—
yandun	2	1	119	0.884	—	—
yandun	4	6	119	0.714	0.506	0.214
yandun	6	1	119	0.661	—	—

Table S58. Many-to-many correspondence at IoU 0.5

The hierarchy-inclusive graph contains primitive and composite intervals. A split/merge component has more than one node on at least one side. Coverage pools unique connected nodes over eligible nodes across all configuration pairs. Component counts are summed over configuration pairs, not unique independent weather episodes. Full IoU 0.3/0.5/0.7 and containment configuration tables accompany this summary. The local component ledger retains individual interval clocks.

Population	Components	Split/merge (%)	Both sides multiple (%)	Mixed hierarchy (%)	Left/right coverage (%)
greece	18,327	19.8	8.2	16.4	32.8/32.4
hill	245,457	36.7	21.1	35.8	37.3/41.8
hill_2021	370,583	33.4	17.8	31.9	36.2/40.8
lahaute	38,198	24.7	12.1	22.2	34.2/36.1
pizhou	276,979	29.9	15.8	26.8	35.3/37.5
sdwvf	1,113,510	29.1	15.7	27.1	34.8/37.3
suining	102,670	33.7	18.6	31.6	36.8/39.9
yandun	818,477	33.1	18.8	33.4	38.9/43.9

Figure S1. Split and merge structure in the overlap graph

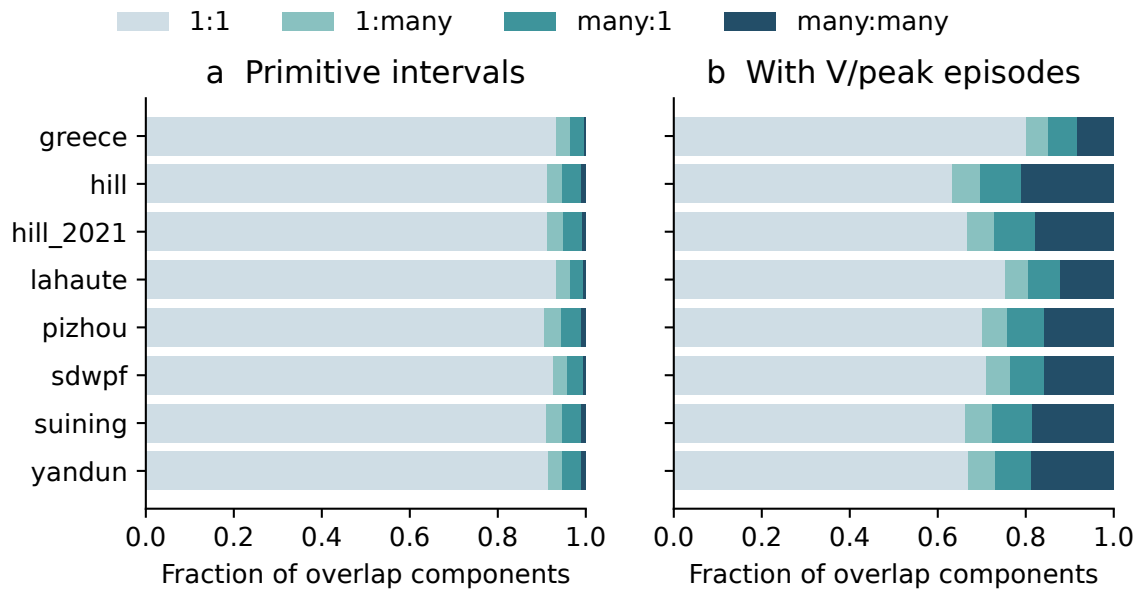


Figure S1: Primitive and hierarchy-inclusive correspondence at IoU 0.5. Each segment gives the fraction of overlap components with one-to-one, one-to-many, many-to-one or many-to-many topology. All 136 configuration pairs are retained in each population.

Table S59. Primary policy-free results and strong controls

Lead	Validation-selected approach	nMAE (%)	Accuracy charge (CNY)	Saving (CNY)
15 min	Persistence	3.039	112,705.56	0.00
15 min	Shape pipeline	2.919	108,472.98	4,232.58
15 min	Scalar/weather control	2.932	108,787.80	3,917.76
15 min	nMAE-selected reference	2.910	106,654.02	6,051.54
4 h	Persistence	17.557	148,804.92	0.00
4 h	Shape pipeline	14.167	121,695.42	27,109.50
4 h	Scalar/weather control	14.179	121,555.50	27,249.42
4 h	nMAE-selected reference	13.589	121,100.76	27,704.16

CNY denotes Chinese yuan. The scalar/weather arm and nMAE reference are selected on validation under the same candidate budget; all methods share the test targets within a horizon. These are simulated accuracy charges, not complete operating costs.

Table S60. Paired calendar-block uncertainty for policy-free comparisons

Positive reductions favour the first named method. Low/high are 95% percentile limits for percentage reduction relative to the named comparator.

comparison	block days	blocks	reduction	pct low	high
15 min: shape/persistence	3	35	3.7554	2.2988	5.0394
15 min: shape/persistence	7	16	3.7554	2.4109	4.9564
15 min: shape/persistence	14	9	3.7554	2.7374	4.7095
15 min: shape/scalar	3	35	0.2894	-0.5303	1.0705
15 min: shape/scalar	7	16	0.2894	-0.5530	0.9175
15 min: shape/scalar	14	9	0.2894	-0.5396	1.0307
4h shape vs persistence	3	35	18.2181	14.0247	22.3006

comparison	block days	blocks	reduction pct	low	high
4h shape vs persistence	7	16	18.2181	15.4317	21.3821
4h shape vs persistence	14	9	18.2181	15.5854	21.5959
4h shape vs scalar	3	35	-0.1151	-1.1399	0.9432
4h shape vs scalar	7	16	-0.1151	-1.0524	0.7287
4h shape vs scalar	14	9	-0.1151	-1.2945	0.8194
4h none band vs mean	3	35	10.5435	5.5009	15.3175
4h none band vs mean	7	16	10.5435	4.5487	16.0544
4h none band vs mean	14	9	10.5435	5.7141	15.1012
4h jma band vs mean	3	35	4.7007	0.1119	9.0097
4h jma band vs mean	7	16	4.7007	-0.3927	9.5338
4h jma band vs mean	14	9	4.7007	2.0677	7.6969
4h jma gfs band vs mean	3	35	4.5753	1.4295	7.4117
4h jma gfs band vs mean	7	16	4.5753	1.1109	7.3343
4h jma gfs band vs mean	14	9	4.5753	2.1051	7.1037

Table S61. Classifier-seed robustness with fixed transformations and selected settings

Lead (min)	seed	Charge (CNY)	Saving (%)	nmae
15	41	108472.9800	3.7554	0.0292
15	42	108438.0000	3.7865	0.0292
15	43	108403.0200	3.8175	0.0292
240	41	121695.4200	18.2181	0.1417
240	42	122430.0000	17.7245	0.1423
240	43	122744.8200	17.5129	0.1421

Table S62. Selection-objective comparison

Positive savings favour fee-selected over nMAE-selected predictions. Both selections use validation; testing does not alter either selection.

Lead (min)	Block days	Blocks	Saved CNY	Reduction %	Low %	High %
15	3	35	-1818.960	-1.705	-2.722	-0.629
15	7	16	-1818.960	-1.705	-2.439	-0.902
15	14	9	-1818.960	-1.705	-2.443	-0.991
240	3	35	-594.660	-0.491	-3.064	1.952
240	7	16	-594.660	-0.491	-3.364	1.671
240	14	9	-594.660	-0.491	-2.954	1.910

Table S63. Full-calendar attribution of rule-based charges

States use absolute actual-minus-persistence change. The first state lies within the horizon-specific tolerance, Moderate changes lie above tolerance and below 20% capacity, and high ramps reach at least 20%. Introduced charges and avoided charges both enter totals.

Lead (min)	state	points	Persistence CNY	Pipeline CNY	Saved CNY
15	Within tolerance	6228	0.0000	7800.5400	-7800.5400
15	Moderate change	3145	110012.1000	97978.9800	12033.1200
15	High ramp	77	2693.4600	2693.4600	0.0000
240	Within tolerance	5091	0.0000	14971.4400	-14971.4400
240	Moderate change	1175	41101.5000	18154.6200	22946.8800
240	High ramp	3079	107703.4200	88569.3600	19134.0600

Table S64. Complete-income British test outcomes

The July–December 2021 test uses frozen validation selections, observed MID execution-price references, imbalance settlement and a simulated trading fee. Negative differences indicate lower total income than passive persistence.

Lead (h)	Arm	Intervals	Revenue GBP	Difference GBP
1	weather	8430	6958536.4060	-16311.5635
1	weather events	8430	6955018.3262	-19829.6432
2	weather	8428	6905633.2851	-22117.4134
2	weather events	8428	6906029.4332	-21721.2653
4	weather	8424	6916025.2306	-25991.0653
4	weather events	8424	6915705.8628	-26310.4331

Figure S2. Preserved British economic comparison

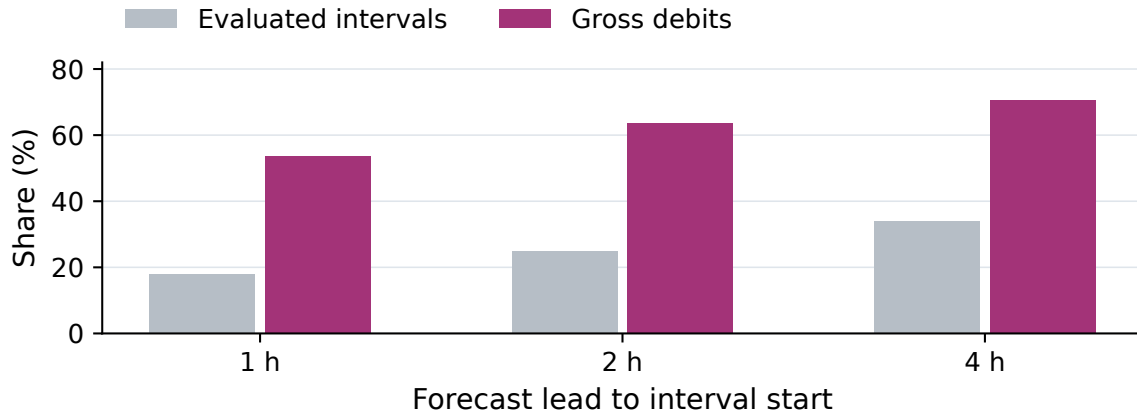


Figure S2: Earlier British large-ramp exposure concentration. Grey bars show interval share and magenta bars the share of persistence gross debits.

Figure S3. Preserved British economic comparison

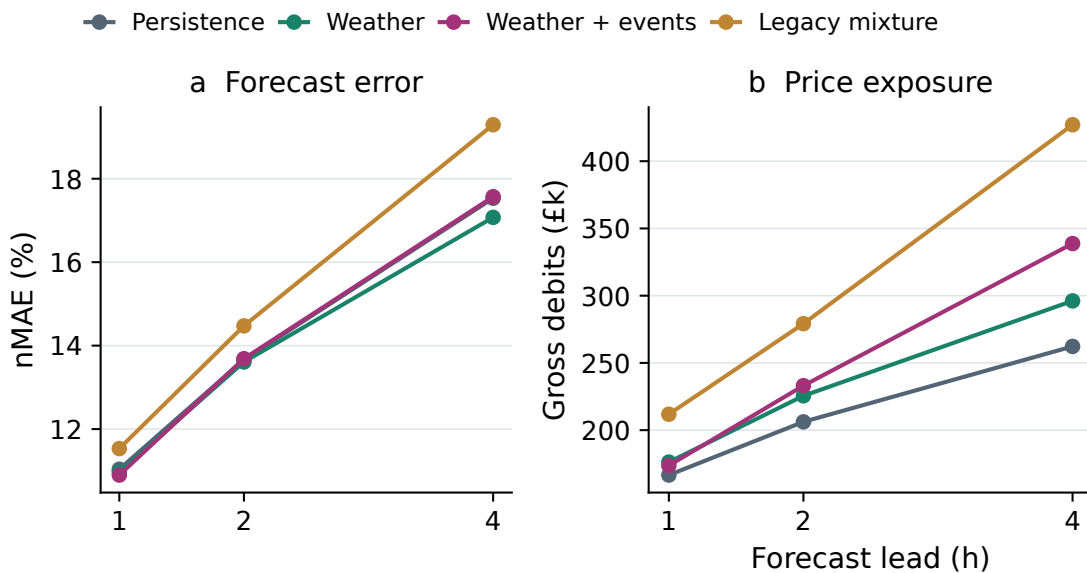


Figure S3: Earlier British forecast-error and gross-debit comparison, on common targets within each horizon. This gross-exposure task is distinct from complete trading income in Table S64.

Figure S4. Preserved British economic comparison

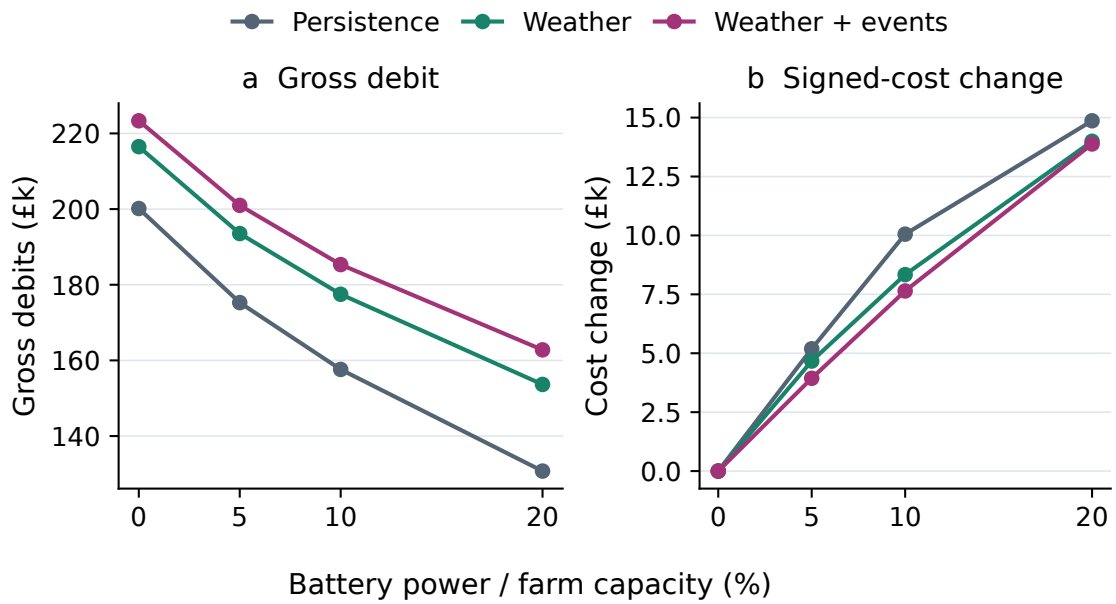


Figure S4: Earlier ex-post metered-correction capability. The event-aware 20% battery reduces gross debits by 27.1% while signed settlement cost rises. Realized half-hour power, inventory restoration and model-specific schedules define this task; it is not an issue-time forecast or net-profit gain.

Table S65. Physical factorial evaluation populations

Population	All events	Directional events	Seven-day blocks
Pizhou ERA5	94534	5862	16
Suining ERA5	33617	3202	11
Yandun ERA5	238131	56960	7
La Haute ERA5	12534	963	21
Hill ERA5	79436	7749	12
T11 LiDAR	1064	705	4
T07 LiDAR	444	285	4
SMARTEOLE	13931	9265	15

Table S66. Direction AUROC in the common-input factorial

S denotes the five scalar descriptors, G5 the common five angular coordinates, and b the protected polarity coordinate. All four columns use the same event set within each row.

Population	S	S+G5	S+b	S+G5+b
Hill ERA5	0.9476	0.9558	0.9485	0.9566
La Haute ERA5	0.7312	0.7769	0.7351	0.7727
Pizhou ERA5	0.9235	0.9211	0.9236	0.9201
Suining ERA5	0.7824	0.7992	0.7849	0.8010
Yandun ERA5	0.6006	0.5971	0.5992	0.5962
T07 LiDAR	0.8870	0.9211	0.8721	0.9156
T11 LiDAR	0.8580	0.8573	0.8522	0.8554
SMARTEOLE	0.8579	0.8748	0.8862	0.8835

Table S67. Paired increments of S+G5+b

Positive gains indicate higher AUROC or lower loss/Brier. The same seven-day blocks are used for both compared models. These are exploratory pointwise intervals; support is in S65.

Population	Reference	Metric	Gain	Lower	Upper
Pizhou ERA5	S	Log loss	0.0048	0.0010	0.0083
Pizhou ERA5	S	Brier	0.0037	0.0014	0.0059
Pizhou ERA5	S	AUROC	-0.0034	-0.0259	0.0104
Pizhou ERA5	S+G5	Log loss	-0.0001	-0.0007	0.0005
Pizhou ERA5	S+G5	Brier	0.0001	-0.0001	0.0004
Pizhou ERA5	S+G5	AUROC	-0.0010	-0.0046	0.0010
Suining ERA5	S	Log loss	-0.0007	-0.0125	0.0103
Suining ERA5	S	Brier	-0.0016	-0.0075	0.0038
Suining ERA5	S	AUROC	0.0185	-0.0044	0.0490
Suining ERA5	S+G5	Log loss	-0.0003	-0.0013	0.0005
Suining ERA5	S+G5	Brier	-0.0001	-0.0003	0.0001
Suining ERA5	S+G5	AUROC	0.0017	-0.0025	0.0085
Yandun ERA5	S	Log loss	0.0080	0.0023	0.0132
Yandun ERA5	S	Brier	0.0071	0.0051	0.0088
Yandun ERA5	S	AUROC	-0.0044	-0.0148	0.0077
Yandun ERA5	S+G5	Log loss	-0.0003	-0.0008	0.0002
Yandun ERA5	S+G5	Brier	-0.0001	-0.0004	0.0001
Yandun ERA5	S+G5	AUROC	-0.0009	-0.0022	0.0003
La Haute ERA5	S	Log loss	0.0079	0.0030	0.0135
La Haute ERA5	S	Brier	0.0029	0.0009	0.0051
La Haute ERA5	S	AUROC	0.0415	0.0181	0.0755
La Haute ERA5	S+G5	Log loss	-0.0003	-0.0017	0.0011
La Haute ERA5	S+G5	Brier	-0.0001	-0.0006	0.0005
La Haute ERA5	S+G5	AUROC	-0.0043	-0.0121	0.0037
Hill ERA5	S	Log loss	0.0054	-0.0036	0.0145
Hill ERA5	S	Brier	0.0038	-0.0011	0.0083
Hill ERA5	S	AUROC	0.0090	0.0013	0.0224
Hill ERA5	S+G5	Log loss	0.0011	0.0000	0.0020
Hill ERA5	S+G5	Brier	0.0004	-0.0000	0.0007
Hill ERA5	S+G5	AUROC	0.0008	-0.0004	0.0024
T11 LiDAR	S	Log loss	0.0039	-0.0251	0.0309
T11 LiDAR	S	Brier	0.0016	-0.0115	0.0119
T11 LiDAR	S	AUROC	-0.0026	-0.0565	0.0198
T11 LiDAR	S+G5	Log loss	-0.0064	-0.0088	0.0001
T11 LiDAR	S+G5	Brier	-0.0045	-0.0049	-0.0032
T11 LiDAR	S+G5	AUROC	-0.0019	-0.0084	0.0056
T07 LiDAR	S	Log loss	0.0223	0.0044	0.1150
T07 LiDAR	S	Brier	0.0115	0.0037	0.0704
T07 LiDAR	S	AUROC	0.0286	-0.0100	0.0752
T07 LiDAR	S+G5	Log loss	-0.0051	-0.0086	0.0139
T07 LiDAR	S+G5	Brier	-0.0027	-0.0044	0.0062
T07 LiDAR	S+G5	AUROC	-0.0055	-0.0150	0.0023
SMARTEOLE	S	Log loss	0.0395	0.0155	0.0677
SMARTEOLE	S	Brier	0.0045	-0.0018	0.0118
SMARTEOLE	S	AUROC	0.0256	0.0120	0.0390
SMARTEOLE	S+G5	Log loss	0.0135	0.0091	0.0186
SMARTEOLE	S+G5	Brier	0.0011	0.0003	0.0019
SMARTEOLE	S+G5	AUROC	0.0087	0.0039	0.0154

Table S68. Paired contrasts for the original LiDAR headline models

This table reuses the original stored predictions without refitting. It distinguishes the large angular-encoding recovery from the smaller difference relative to scalar-only heads. Each instrument supplies four seven-day blocks.

Instrument	Reference	Metric	Gain	Lower	Upper
T11	Scalar only	Log loss	0.0173	-0.0258	0.0795
T11	Scalar only	Brier	0.0029	-0.0160	0.0465
T11	Scalar only	AUROC	0.0320	-0.0207	0.0587
T11	GAF/PCA6	Log loss	0.1831	0.1058	0.2789
T11	GAF/PCA6	Brier	0.1098	0.0650	0.1674
T11	GAF/PCA6	AUROC	0.3956	0.3302	0.4722
T07	Scalar only	Log loss	-0.0043	-0.1723	0.0232
T07	Scalar only	Brier	-0.0139	-0.1045	0.0090
T07	Scalar only	AUROC	0.0178	-0.1561	0.0669
T07	GAF/PCA6	Log loss	0.1769	0.0496	0.2376
T07	GAF/PCA6	Brier	0.0941	0.0435	0.1374
T07	GAF/PCA6	AUROC	0.4281	0.3449	0.5337

Instrument	Reference	Metric	Gain	Lower	Upper
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Table S69. Direction-AUROC controls for dimension, sign and anchor

Positive gain favours the first named model. Reversed-both is information-preserving; test-only sign corruption and all three probability metrics are retained in the machine-readable noise tables.

Population	Comparison	Gain	Lower	Upper
Pizhou ERA5	Protected / G6	-0.0050	-0.0091	0.0002
Pizhou ERA5	Protected / Pre-PCA signed	-0.0035	-0.0067	0.0006
Pizhou ERA5	Protected / Random bit	-0.0028	-0.0062	0.0007
Pizhou ERA5	Reversed both / Protected	0.0000	0.0000	0.0000
Pizhou ERA5	End sign / Protected	0.0010	-0.0010	0.0046
Pizhou ERA5	Pre sign / Protected	0.0040	-0.0009	0.0086
Pizhou ERA5	Post sign / Protected	-0.0003	-0.0086	0.0102
Pizhou ERA5	Internal sign / Protected	-0.0013	-0.0030	0.0000
Suining ERA5	Protected / G6	0.0063	-0.0000	0.0125
Suining ERA5	Protected / Pre-PCA signed	0.0066	-0.0013	0.0155
Suining ERA5	Protected / Random bit	0.0024	-0.0018	0.0090
Suining ERA5	Reversed both / Protected	0.0000	0.0000	0.0000
Suining ERA5	End sign / Protected	-0.0017	-0.0085	0.0025
Suining ERA5	Pre sign / Protected	-0.0016	-0.0094	0.0037
Suining ERA5	Post sign / Protected	0.0038	-0.0053	0.0146
Suining ERA5	Internal sign / Protected	0.0006	-0.0043	0.0055
Yandun ERA5	Protected / G6	-0.0032	-0.0088	0.0015
Yandun ERA5	Protected / Pre-PCA signed	0.0006	-0.0085	0.0084
Yandun ERA5	Protected / Random bit	-0.0017	-0.0031	0.0007
Yandun ERA5	Reversed both / Protected	0.0000	0.0000	0.0000
Yandun ERA5	End sign / Protected	0.0009	-0.0003	0.0022
Yandun ERA5	Pre sign / Protected	-0.0009	-0.0040	0.0017
Yandun ERA5	Post sign / Protected	0.0064	-0.0024	0.0166
Yandun ERA5	Internal sign / Protected	0.0009	-0.0013	0.0025
La Haute ERA5	Protected / G6	-0.0126	-0.0349	0.0152
La Haute ERA5	Protected / Pre-PCA signed	-0.0159	-0.0311	0.0050
La Haute ERA5	Protected / Random bit	-0.0032	-0.0109	0.0048
La Haute ERA5	Reversed both / Protected	0.0000	0.0000	0.0000
La Haute ERA5	End sign / Protected	0.0043	-0.0037	0.0121
La Haute ERA5	Pre sign / Protected	0.0076	-0.0084	0.0204
La Haute ERA5	Post sign / Protected	0.0141	-0.0040	0.0296
La Haute ERA5	Internal sign / Protected	-0.0001	-0.0091	0.0082
Hill ERA5	Protected / G6	0.0016	-0.0018	0.0053
Hill ERA5	Protected / Pre-PCA signed	0.0005	-0.0015	0.0024
Hill ERA5	Protected / Random bit	0.0004	-0.0001	0.0010
Hill ERA5	Reversed both / Protected	0.0000	0.0000	0.0000
Hill ERA5	End sign / Protected	-0.0008	-0.0024	0.0004
Hill ERA5	Pre sign / Protected	0.0037	0.0007	0.0066
Hill ERA5	Post sign / Protected	0.0036	-0.0001	0.0081
Hill ERA5	Internal sign / Protected	-0.0003	-0.0014	0.0004
T11 LiDAR	Protected / G6	0.0366	0.0287	0.0596
T11 LiDAR	Protected / Pre-PCA signed	0.0036	-0.0083	0.0073
T11 LiDAR	Protected / Random bit	0.0029	-0.0024	0.0059
T11 LiDAR	Reversed both / Protected	0.0000	0.0000	0.0000
T11 LiDAR	End sign / Protected	0.0019	-0.0056	0.0084
T11 LiDAR	Pre sign / Protected	-0.0158	-0.0218	-0.0108
T11 LiDAR	Post sign / Protected	-0.0144	-0.0330	-0.0025
T11 LiDAR	Internal sign / Protected	-0.0039	-0.0164	0.0065
T07 LiDAR	Protected / G6	0.0334	0.0209	0.0836
T07 LiDAR	Protected / Pre-PCA signed	-0.0005	-0.0377	0.0339
T07 LiDAR	Protected / Random bit	-0.0013	-0.0184	0.0192
T07 LiDAR	Reversed both / Protected	0.0000	0.0000	0.0000
T07 LiDAR	End sign / Protected	0.0055	-0.0023	0.0150
T07 LiDAR	Pre sign / Protected	-0.0093	-0.0128	0.0029
T07 LiDAR	Post sign / Protected	-0.0141	-0.0333	-0.0069
T07 LiDAR	Internal sign / Protected	-0.0000	-0.0099	0.0084
SMARTEOLE	Protected / G6	0.0063	-0.0005	0.0131
SMARTEOLE	Protected / Pre-PCA signed	0.0068	-0.0040	0.0233
SMARTEOLE	Protected / Random bit	0.0026	-0.0026	0.0089
SMARTEOLE	Reversed both / Protected	0.0000	0.0000	0.0000
SMARTEOLE	End sign / Protected	-0.0087	-0.0154	-0.0039

Population	Comparison	Gain	Lower	Upper
SMARTEOLE	Pre sign / Protected	-0.0269	-0.0394	-0.0163
SMARTEOLE	Post sign / Protected	0.0117	0.0033	0.0216
SMARTEOLE	Internal sign / Protected	-0.0086	-0.0131	-0.0046

Table S70. Anchor flips under relative representation-channel noise

Noise standard deviation is a fraction of the pre-normalization shape amplitude. Values summarize three fixed perturbation seeds, not calendar uncertainty. Scalar descriptors and detectors remain unchanged.

Population	Noise / amplitude	Mean flip rate	Minimum	Maximum
Hill ERA5	0.0000	0.0000	0.0000	0.0000
Hill ERA5	0.0100	0.0029	0.0028	0.0029
Hill ERA5	0.0500	0.0158	0.0156	0.0161
Hill ERA5	0.1000	0.0346	0.0343	0.0348
Hill ERA5	0.2000	0.0857	0.0855	0.0859
Hill ERA5	0.4000	0.2127	0.2108	0.2146
Pizhou ERA5	0.0000	0.0000	0.0000	0.0000
Pizhou ERA5	0.0100	0.0032	0.0030	0.0033
Pizhou ERA5	0.0500	0.0175	0.0172	0.0178
Pizhou ERA5	0.1000	0.0378	0.0371	0.0388
Pizhou ERA5	0.2000	0.0919	0.0911	0.0924
Pizhou ERA5	0.4000	0.2196	0.2177	0.2207
T07 LiDAR	0.0000	0.0000	0.0000	0.0000
T07 LiDAR	0.0100	0.0038	0.0000	0.0068
T07 LiDAR	0.0500	0.0143	0.0090	0.0180
T07 LiDAR	0.1000	0.0293	0.0225	0.0338
T07 LiDAR	0.2000	0.0796	0.0653	0.0968
T07 LiDAR	0.4000	0.2027	0.1959	0.2072
T11 LiDAR	0.0000	0.0000	0.0000	0.0000
T11 LiDAR	0.0100	0.0034	0.0000	0.0056
T11 LiDAR	0.0500	0.0172	0.0150	0.0216
T11 LiDAR	0.1000	0.0420	0.0357	0.0508
T11 LiDAR	0.2000	0.0865	0.0799	0.0902
T11 LiDAR	0.4000	0.2206	0.2124	0.2265
SMARTEOLE	0.0000	0.0000	0.0000	0.0000
SMARTEOLE	0.0100	0.0022	0.0021	0.0024
SMARTEOLE	0.0500	0.0097	0.0090	0.0109
SMARTEOLE	0.1000	0.0201	0.0181	0.0216
SMARTEOLE	0.2000	0.0608	0.0573	0.0635
SMARTEOLE	0.4000	0.1836	0.1789	0.1874

Table S71. Small-amplitude and opposing-extrema strata

Gaussian standard deviation is 0.005 training-production-scale units before shape normalization. Means are over three fixed seeds. NaN denotes an empty stratum, not zero instability. The shape normalizer a and extrema margin m are defined in S24.

Population	Stratum	Events	Mean flip rate
Pizhou ERA5	all	94534	0.0052
Pizhou ERA5	extreme gap ge 0.1	91325	0.0001
Pizhou ERA5	extreme gap lt 0.1	3209	0.1523
Pizhou ERA5	normalizer 0.05 0.2	33	0.1313
Pizhou ERA5	normalizer ge 0.2	94501	0.0052
Pizhou ERA5	normalizer lt 0.05	0	nan
T07 LiDAR	all	444	0.0060
T07 LiDAR	extreme gap ge 0.1	429	0.0008
T07 LiDAR	extreme gap lt 0.1	15	0.1556
T07 LiDAR	normalizer 0.05 0.2	0	nan
T07 LiDAR	normalizer ge 0.2	444	0.0060
T07 LiDAR	normalizer lt 0.05	0	nan
T11 LiDAR	all	1064	0.0060
T11 LiDAR	extreme gap ge 0.1	1025	0.0003
T11 LiDAR	extreme gap lt 0.1	39	0.1538
T11 LiDAR	normalizer 0.05 0.2	0	nan

Population	Stratum	Events	Mean flip rate
T11 LiDAR	normalizer ge 0.2	1064	0.0060
T11 LiDAR	normalizer lt 0.05	0	nan

Table S72. Source-separated reference agreement

All panels assess the same 320 windows. Human-only and AI-only sources remain distinct. The mixed-panel presence point estimate exceeds 0.7, but its interval includes lower values; agreement is not annotation accuracy.

Panel	Target	Reviews	Alpha	Lower	Upper
human3	event presence	3	0.5279	0.4553	0.5994
human3	morphology	3	0.4761	0.4257	0.5228
AI7	event presence	7	0.9582	0.9287	0.9817
AI7	morphology	7	0.9577	0.9413	0.9727
mixed10	event presence	10	0.7067	0.6420	0.7596
mixed10	morphology	10	0.7411	0.7150	0.7657

Table S73. Pairwise agreement by source combination

Each row summarizes observer-pair fractions of identically labelled assessable windows. Min/max describe pairs rather than confidence bounds.

Target	Source pair	Pairs	Mean	Minimum	Maximum
event presence	AI-AI	21	0.9881	0.9812	0.9969
event presence	human-AI	21	0.8289	0.7625	0.9344
event presence	human-human	3	0.7958	0.7656	0.8438
morphology	AI-AI	21	0.9659	0.9531	0.9844
morphology	human-AI	21	0.6403	0.6031	0.6719
morphology	human-human	3	0.5594	0.4656	0.6562

Table S74. Protected anchor sign and scalar endpoint direction

The populations are those included in the representation-noise diagnostic. All retained endpoints in these populations have nonzero direction. The opposing-sign fraction measures coordinate redundancy directly; it is separate from predictive information gain.

Population	Events	Opposing signs (%)
Hill ERA5	79,436	8.154
Pizhou ERA5	94,534	9.224
T07 LiDAR	444	6.757
T11 LiDAR	1,064	7.519
SMARTEOLE	13,931	2.821

Table S75. Unique-interval support for measured-wind reconstruction

Identical turbine/start/end intervals are counted once. Native 10-min and 30-min power tracks share these intervals and wind targets; counts must not be added across tracks. Source periods and context purging are specified in S26.

Reference	Train	Validation	Test	Seven-day test blocks
T11	1108	689	507	4
T07	533	335	170	4
SMARTEOLE	5025	1566	449	4

Table S76. All-event wind-process error reductions

Values are relative RMSE reductions (%) with paired 95% intervals. Same-support comparisons use native 10-min inputs for both arms. The 30-to-10-min contrast includes aggregation and temporal-support changes; it is not a representation-only effect. Counts are given in S75.

Reference	Comparison	Block days	Reduction [interval]	Blocks
T11	Full curve, same support	3	4.50 [-5.93, 12.59]	7
T11	Full curve, same support	7	4.50 [-3.55, 13.69]	4
T11	Full curve, same support	14	4.50 [-1.89, 16.71]	3
T11	Geometry, same support	3	11.09 [5.21, 19.77]	7
T11	Geometry, same support	7	11.09 [6.21, 19.86]	4
T11	Geometry, same support	14	11.09 [2.91, 24.84]	3
T11	Full curve, 30 to 10 min	3	12.14 [4.73, 23.70]	7
T11	Full curve, 30 to 10 min	7	12.14 [4.98, 22.50]	4
T11	Full curve, 30 to 10 min	14	12.14 [1.93, 30.50]	3
T07	Full curve, same support	3	4.47 [-13.63, 23.37]	6
T07	Full curve, same support	7	4.47 [-1.75, 16.20]	4
T07	Full curve, same support	14	4.47 [2.12, 17.48]	3
T07	Geometry, same support	3	26.94 [16.59, 39.25]	6
T07	Geometry, same support	7	26.94 [17.80, 40.66]	4
T07	Geometry, same support	14	26.94 [10.88, 28.45]	3
T07	Full curve, 30 to 10 min	3	23.36 [10.53, 38.65]	6
T07	Full curve, 30 to 10 min	7	23.36 [15.57, 35.18]	4
T07	Full curve, 30 to 10 min	14	23.36 [22.83, 43.50]	3
SMARTEOLE	Full curve, same support	3	1.72 [-0.01, 4.74]	6
SMARTEOLE	Full curve, same support	7	1.72 [0.96, 7.90]	4
SMARTEOLE	Full curve, same support	14	1.72 [0.96, 3.10]	2
SMARTEOLE	Geometry, same support	3	9.58 [8.37, 10.71]	6
SMARTEOLE	Geometry, same support	7	9.58 [9.03, 10.76]	4
SMARTEOLE	Geometry, same support	14	9.58 [9.08, 10.27]	2
SMARTEOLE	Full curve, 30 to 10 min	3	4.22 [-2.20, 11.92]	6
SMARTEOLE	Full curve, 30 to 10 min	7	4.22 [-4.91, 18.29]	4
SMARTEOLE	Full curve, 30 to 10 min	14	4.22 [0.92, 5.84]	2

Table S77. Power-defined hierarchy and full-wind-trajectory recovery

Compound status comes from power-catalogue membership before wind-outcome evaluation. Group values are relative RMSE reductions (%); their difference is in percentage points (pp). Intervals are paired percentile intervals. The difference row includes both groups in its event count.

Reference	Population	Events	Days	Gain [interval]	Valid draws
T11	Compound	165	3	11.67 [3.99, 20.45]	2000
T11	Primitive only	342	3	-0.72 [-13.20, 8.92]	2000
T11	Difference (pp)	507	3	12.39 [2.65, 22.07]	2000
T11	Compound	165	7	11.67 [4.91, 21.43]	2000
T11	Primitive only	342	7	-0.72 [-9.04, 9.09]	2000
T11	Difference (pp)	507	7	12.39 [4.80, 20.44]	2000
T11	Compound	165	14	11.67 [5.55, 27.01]	2000
T11	Primitive only	342	14	-0.72 [-6.03, 10.57]	2000
T11	Difference (pp)	507	14	12.39 [1.72, 22.99]	2000
T07	Compound	38	3	24.45 [12.51, 35.85]	2000
T07	Primitive only	132	3	-1.30 [-17.15, 22.29]	2000
T07	Difference (pp)	170	3	25.75 [3.79, 43.57]	2000
T07	Compound	38	7	24.45 [16.78, 41.87]	2000
T07	Primitive only	132	7	-1.30 [-5.90, 5.22]	2000
T07	Difference (pp)	170	7	25.75 [20.01, 38.04]	2000
T07	Compound	38	14	24.45 [18.94, 43.02]	2000
T07	Primitive only	132	14	-1.30 [-22.35, 5.82]	2000
T07	Difference (pp)	170	14	25.75 [21.10, 55.77]	2000
SMARTEOLE	Compound	42	3	5.81 [-11.80, 7.32]	1999
SMARTEOLE	Primitive only	407	3	1.40 [-0.47, 5.07]	2000
SMARTEOLE	Difference (pp)	449	3	4.41 [-16.93, 7.63]	1999
SMARTEOLE	Compound	42	7	5.81 [-9.94, 7.27]	1990
SMARTEOLE	Primitive only	407	7	1.40 [0.43, 9.59]	2000
SMARTEOLE	Difference (pp)	449	7	4.41 [-19.53, 6.84]	1990
SMARTEOLE	Compound	42	14	5.81 [5.51, 6.91]	2000
SMARTEOLE	Primitive only	407	14	1.40 [0.81, 2.64]	2000
SMARTEOLE	Difference (pp)	449	14	4.41 [2.87, 6.09]	2000

Table S78. Common-duration standardization of the full-trajectory contrast

Identical duration-stratum mass is assigned to both groups using power-only support. Values are relative reductions (%) or compound-minus-primitive percentage points. Shared-duration counts and all geometry contrasts are retained in the accompanying CSVs.

Reference	Population	Days	Gain [interval]
T11	Compound	3	11.66 [3.96, 20.47]
T11	Primitive only	3	1.01 [-12.94, 12.21]
T11	Difference (pp)	3	10.65 [-0.50, 20.91]
T11	Compound	7	11.66 [4.84, 21.52]
T11	Primitive only	7	1.01 [-8.58, 12.11]
T11	Difference (pp)	7	10.65 [1.49, 17.98]
T11	Compound	14	11.66 [5.36, 26.96]
T11	Primitive only	14	1.01 [-5.45, 14.12]
T11	Difference (pp)	14	10.65 [-1.62, 19.06]
T07	Compound	3	24.45 [12.51, 35.85]
T07	Primitive only	3	-3.71 [-23.50, 25.57]
T07	Difference (pp)	3	28.16 [0.91, 46.27]
T07	Compound	7	24.45 [16.78, 41.87]
T07	Primitive only	7	-3.71 [-12.27, 20.71]
T07	Difference (pp)	7	28.16 [19.74, 33.88]
T07	Compound	14	24.45 [18.94, 43.02]
T07	Primitive only	14	-3.71 [-6.16, 22.48]
T07	Difference (pp)	14	28.16 [20.54, 39.58]
SMARTEOLE	Compound	3	5.81 [-11.80, 7.32]
SMARTEOLE	Primitive only	3	0.40 [-2.43, 4.92]
SMARTEOLE	Difference (pp)	3	5.41 [-16.47, 9.21]
SMARTEOLE	Compound	7	5.81 [-9.94, 7.27]
SMARTEOLE	Primitive only	7	0.40 [-0.58, 10.85]
SMARTEOLE	Difference (pp)	7	5.41 [-20.79, 7.57]
SMARTEOLE	Compound	14	5.81 [5.51, 6.91]
SMARTEOLE	Primitive only	14	0.40 [-0.58, 2.41]
SMARTEOLE	Difference (pp)	14	5.41 [3.10, 7.49]

Table S79. Ordered-correspondence controls for internal wind geometry

Errors are in m/s. Clean and perturbed inputs share the frozen model, scalar descriptors and endpoints. The perturbed value is the mean of seeds 101/202/303; the range describes these seeds, not sampling uncertainty.

Reference	Control	Clean RMSE	Perturbed RMSE	Seed range
T11	Interior order shuffled	1.0518	1.6119	1.6100–1.6153
T11	Residual reassigned	1.0518	1.8362	1.7942–1.8699
T07	Interior order shuffled	0.6147	1.4367	1.3929–1.4777
T07	Residual reassigned	0.6147	1.5024	1.4939–1.5157
SMARTEOLE	Interior order shuffled	0.7838	1.1628	1.1466–1.1715
SMARTEOLE	Residual reassigned	0.7838	1.0972	1.0897–1.1100

Table S80. Auxiliary recovery of measured return processes

Classes are derived from reconstructed curves at the fixed 1.5-m/s threshold, not fitted as a separate classifier. Actual return counts are 330/507 at T11, 87/170 at T07 and 194/449 at SMARTEOLE. The always-return comparator illustrates F1's prevalence dependence; precision and recall are therefore shown together. Paired intervals, AUROC and average precision are retained in process_class_intervals.csv and primary_results.csv.

Reference	Prediction	Events	Precision	Recall	F1
T11	Scalar control	507	0.947	0.379	0.541
T11	Ordered trajectory	507	0.937	0.633	0.756
T11	Always return	507	0.651	1.000	0.789
T07	Scalar control	170	0.969	0.356	0.521
T07	Ordered trajectory	170	0.968	0.690	0.805
T07	Always return	170	0.512	1.000	0.677
SMARTEOLE	Scalar control	449	1.000	0.082	0.152
SMARTEOLE	Ordered trajectory	449	0.909	0.206	0.336
SMARTEOLE	Always return	449	0.432	1.000	0.603

Table S81. Conditional information with the implemented pre-window-power control

The joint analysis uses direction, absolute amplitude, duration and the pair-side mean of pre-window power. All three Pizhou representations use the same retained pair set. Mutual information is in nats; normalized mutual information and remaining entropy are dimensionless.

Representation	Pairs	Retention	MI above permutation	Normalized MI	Remaining entropy
raw25	268,569/274,569	97.8%	0.022	0.221	0.130
gaf_pca6	268,569/274,569	97.8%	0.436	0.404	0.862
gaf_bit6	268,569/274,569	97.8%	0.436	0.406	0.862

Table S82. Seven-day calendar-block contributions and leave-one-block-out diagnostics

Each row uses the frozen validation-selected models and the same event targets. Ordinary rows give one seven-day block; leave_out rows remove that block. These rows describe temporal concentration in the evaluated period and are not additional independent weather replications.

Reference	Block	Events	Compound	Scalar RMSE	Ordered RMSE	Gain
T11	1	68	31	2.219	1.966	11.41%
T11	2	213	68	1.403	1.486	-5.95%
T11	3	178	44	1.080	1.006	6.83%
T11	4	48	22	1.864	1.553	16.71%
T11	leave_out_1	439	134	1.344	1.321	1.72%
T11	leave_out_2	294	97	1.553	1.379	11.24%
T11	leave_out_3	329	121	1.673	1.606	3.98%
T11	leave_out_4	459	143	1.447	1.411	2.53%
T07	1	3	1	0.935	0.909	2.81%
T07	2	68	15	1.183	1.205	-1.87%
T07	3	82	16	0.813	0.736	9.55%
T07	4	17	6	1.322	1.091	17.48%
T07	leave_out_1	167	37	1.036	0.989	4.50%
T07	leave_out_2	102	23	0.921	0.811	11.97%
T07	leave_out_3	88	22	1.204	1.175	2.39%
T07	leave_out_4	153	32	0.997	0.976	2.13%
SMARTEOLE	1	4	0	0.490	0.641	-30.73%
SMARTEOLE	2	190	11	1.658	1.641	1.03%
SMARTEOLE	3	172	25	1.165	1.144	1.80%

Reference	Block	Events	Compound	Scalar RMSE	Ordered RMSE	Gain
SMARTEOLE	4	83	6	0.823	0.751	8.70%
SMARTEOLE	leave_out_1	445	42	1.350	1.327	1.77%
SMARTEOLE	leave_out_2	259	31	1.059	1.028	2.97%
SMARTEOLE	leave_out_3	277	17	1.446	1.422	1.69%
SMARTEOLE	leave_out_4	366	36	1.438	1.420	1.22%

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